

‘Pushing’ versus ‘holding’ isometric muscle actions; what we know and where to go: A scoping and systematic review with meta-analyses

Running head: Pushing vs holding isometric muscle actions

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ABSTRACT

Background: Pushing isometric muscle actions (PIMA) are regularly utilized to evaluate strength, fatigue, and neuromechanical aspects. Holding isometric muscle actions (HIMA) are largely unknown, although practitioners prescribe them in rehabilitation and performance contexts. The lack of knowledge and consensus in research on the distinction between two isometric types and limited scientific backing makes appropriate use in clinical and performance contexts difficult.

Objective: To gather research directly comparing PIMA and HIMA, and to summarize and synthesize findings. We also aimed to identify potential practical applications for both tasks. Lastly, we highlight existing gaps in the literature and propose directions for future research.

Methods: CINAHL, Embase, MEDLINE, PubMed and Web of Science databases were searched for peer-reviewed articles comparing PIMA and HIMA in humans. Risk-of-bias and study quality were assessed via established assessments for quasi-experimental studies and funnel plots, respectively. Findings were synthesized where possible, with meta-analyses and meta-regressions performed on time-to-task-failure (TTF), ratings of perceived exertion (RPE), heart rate (HR), and mean arterial pressure (MAP).

Results: Fifty-four studies (publication year=2012.9±6.9; 1995-2024) were identified (N=856 participants; ~29.5±10.1 years). Thirty-five included performance parameters (e.g., TTF), 45 examined neurological outputs (e.g., electromyography (EMG), electroencephalography), and 14 explored cardiovascular or metabolic (e.g., glucose uptake, oxygenation) variables. Meta-analysis of 23 studies revealed consistently longer TTF for PIMA vs HIMA at the same absolute intensity (n=407; $g=-0.74$, $p<0.001$), except for two studies examining axial muscles ($g=1.78-3.59$, $p<0.001$). Meta-analyses of 6-11 studies detected no absolute differences in HR, MAP, or RPE (n=136-194; $g=-0.11$ to 0.18 , $p=0.07-0.96$), except for RPE at 50% of TTF being greater during PIMA (n=164; $g=-0.31$, $p=0.01$). PIMA mostly showed higher force fluctuations, discharge rates, D1-inhibition and peak torque, while HIMA indicated higher heteronymous facilitation, EMG burst rates, interspike interval variation, muscular glucose uptake, and faster increases in force/position fluctuations, EMG amplitude, RPE, HR, and MAP. Findings on muscle activation were mixed. HIMAs showed fewer neurological alterations during experimental joint pain.

Conclusions: Evidence suggests distinguishing two types of isometric muscle action indicating more complex control strategies for HIMA than PIMA. Findings revealed similarities to anisometric actions, suggesting that the control strategies of HIMA and PIMA resemble the ones

for muscle lengthening and shortening, respectively. HIMAs could provide a time-efficient approach for inducing musculoskeletal, neural, and cardiovascular adaptations in rehabilitation. PIMA may be beneficial for prolonged activation and agonist neuromuscular adaptations. Methods varied widely across studies, making additional meta-analyses impossible. More consistent methodology and data reporting are recommended. Randomized controlled trials are required to confirm the use of PIMA vs HIMA in clinical or performance contexts. The knowledge of both isometric types should be implemented in research and education.

Registration: The original protocol was prospectively registered at the National Institute of Health Research PROSPERO (CRD42024530386).

Key Points

- Sports medicine practitioners often use pushing (PIMA) and holding (HIMA) isometric muscle actions in rehabilitation and sports performance, yet limited evidence supports their differentiated use.
- HIMA generally reduces time-to-task-failure, except in cases involving axial muscles. Minimal differences were found in absolute heart rate, mean arterial pressure, or ratings of perceived exertion, although the rate of increase was higher during HIMA.
- HIMA typically increases antagonist and synergist muscle activation, glucose uptake, and electroencephalography coherence. Trends in fatigue and agonist activation varied, while fewer neurological alterations were observed during experimental joint pain.
- While randomized control trials are needed, PIMA seems more beneficial for agonist neuromuscular adaptations, while HIMA appears suitable for time-efficient musculoskeletal and cardiovascular rehabilitation.

1. INTRODUCTION

Different types of muscle actions (i.e., concentric, isometric, eccentric) are commonly used in research to examine acute physiological responses and are also essential in clinical and sports performance settings to elicit specific morphological, architectural, and neuromuscular adaptations through tailored movements and workloads [1, 2]. While concentric and eccentric actions are often highlighted for their distinct metabolic costs and adaptation profiles, such as the differential effects of eccentric actions on neuromuscular cross-education, muscle fascicle length, and region-specific hypertrophy [2, 3], the importance of differentiating between two distinct types of isometric muscle actions has often been underemphasized. Emerging research (e.g., [4, 5]) underscores the need to recognize and distinguish these two forms, as they exhibit unique characteristics across multiple neurological and neuromuscular parameters potentially, influencing outcomes and application [1, 6-9].

Research examining individual contraction or muscle action types has existed since at least 1895, with Adolph Fick examining cardiac muscle under isometric conditions [10]. However, isometrics is commonly known as one single form of isometric muscle action. There has been a recent surge in studies comparing ‘two types’ of isometric muscle actions [4, 5] and hybrid muscle actions such as ‘quasi-isometrics’ [11-17] and ‘adaptive force’ [18-29]. Two types of isometrics are known by several names, including pushing (PIMA) and holding (HIMA) isometric muscle actions [4], force and position (control, matching, target) tasks [30], and overcoming and yielding isometrics [31, 32]. Regardless of the semantics, PIMA involves pushing or pulling against an immovable object (e.g., strain-gauge, wall, power rack). At the same time, HIMA is defined as maintaining a set position while resisting an external force (e.g., gravity, inertially loaded weight [dumbbells, barbells], resistance cable stack, training partner, examiner) (Figure 1).

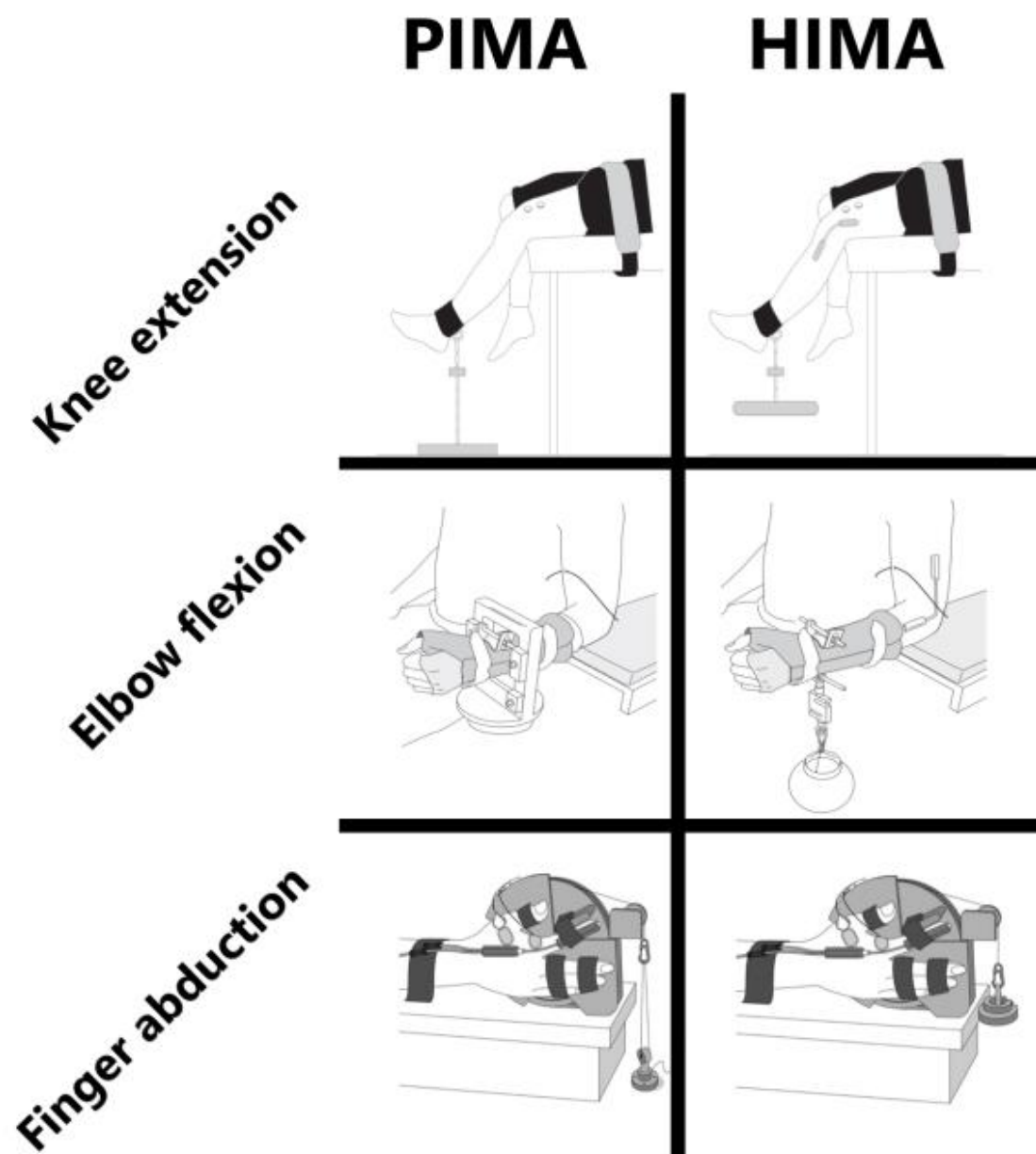


Figure 1. Examples of single-joint pushing (PIMA) and holding (HIMA) isometric muscle actions in the current literature (images adapted from Rudroff et al. [33]; Pasco et al. [34]; Yunoki et al. [35])

While PIMAs are far more common in sports medicine and physiological research [1], typically as a means to assess maximal force production capacity [36, 37], several recent studies have utilized HIMA in rehabilitative and performance contexts [38-42]. Additionally, the recent output of investigations on HIMA and PIMA anecdotally correlates to a surge in online resources and courses outlining practical applications of HIMA and PIMAs [43-47], despite limited scientific

evidence supporting their delineated use. Finally, the authors of this article have noticed inconsistent methods and substantial gaps across much of the PIMA and HIMA literature. Therefore, the purpose of this review was to systematically gather and evaluate research directly by comparing PIMA and HIMA, to assess whether objective measures can differentiate those types, and to utilize relevant literature to assist in providing recommendations to researchers and practitioners.

2. METHODS

The systematic review conformed to the "Preferred Reporting Items for Systematic Reviews and Meta-Analyses" (PRISMA) guidelines [48]. Therefore, no Institutional Review Board approval was necessary.

2.1 Inclusion and exclusion criteria

All included studies met the inclusion criteria of: 1) published in English; 2) peer-reviewed journal publications or doctoral dissertations; 3) performed with human participants; and 4) evaluated the mechanical, neurological, perceptual, or physiological effects of PIMA and HIMA. Studies were excluded if they: 1) were conference proceedings (papers, posters, presentations); 2) did not include both PIMA and HIMA conditions.

2.2 Literature search strategy

The initial electronic search utilized CINAHL, Embase, MEDLINE, PubMed and Web of Science databases from inception to January 2023. Key terms were searched for within the article title, abstract, and keywords using conjunctions 'OR' and 'AND' with truncation '*'. The following Boolean phrases comprised the search terms: (isometric* OR static) AND (contraction* OR action* OR task*) AND (holding OR position) AND (pushing OR force). No limiters or filters were applied. Secondary searches included: a) screening the reference lists of included studies; b) examining studies that cited the included studies (forward citation tracking through Google Scholar); c) search alerts to monitor any new search results; and d) contacting the most common authors of the included outputs. The database search was re-run regularly with the final search performed in October 2024.

2.3 Study selection

Database results were downloaded and uploaded to the EndNote reference manager (version X9.0.3; Clarivate Analytics, Philadelphia, PA, USA). The reference manager was utilized to remove 3080 duplicates, while 22 duplicates were manually identified and removed. The remaining publications were uploaded to the Rayyan systematic review software [49], where titles and abstracts were independently screened for eligibility by two authors blinded to each other's decisions. Three authors then independently screened the full texts to finalize inclusion eligibility. All disagreements were resolved through discussion, with the lead author making the final decision.

2.4 Data extraction

A Google Sheets spreadsheet was used to record extracted data, including 1) publication details including authors and year; 2) participant information (e.g., sample size, sex, age, strength, training status); 3) key equipment utilized (e.g., strain-gauge, electrodes, transcranial magnetic stimulation); 4) details of each condition (e.g., intensity, number of contractions, taken to failure); 5) means, standard deviation and where available, raw data (e.g., pre-, post-, during condition); and 6) reported statistical outputs (p -values, effect sizes, percent changes, ranges). Corresponding author(s) were contacted via email if insufficient data were reported. Three authors independently completed data extraction, with the other three cross-checking the spreadsheet. Any differences were resolved via discussion, with the lead author finalizing each decision. Web Plot Digitizer software (Version 4.1; <https://automeris.io/WebPlotDigitizer/>) extracted data from figures when the authors did not report raw data.

2.5 Risk-of-bias and study quality

Studies that met the inclusion criteria were assessed to determine their risk-of-bias and methodological and statistical reporting quality based on established scales. Although several risk-of-bias tools are available [50-52], most tools are designed to assess intervention studies. Therefore, these assessments contain criteria (e.g., control group, participant allocation, participants lost before follow-up) irrelevant to the observational and acute studies included in the present review. Thus, we created unique risk-of-bias and quality assessments by combining the

most relevant criteria from standard tools while adding other qualities deemed valuable by the present authorship.

The classification of the research study design was based on that presented by Manjali and Gupta [53] and used questions to assessors adapted from the Centre for Evidence-Based Medicine [54]. The degree of internal validity was assessed by incorporating the Joanna Briggs Institute checklist for quasi-experimental studies [55], revised with an update of the critical appraisal tool in March 2024, into the study selection scoring questions posed to assessors. Additional publication reporting-quality questions were posed based on recommended guidelines for statistics [56], participant characteristics (e.g., demographics, evaluated physical characteristics) and training status stratification [57].

Hence, the six risk-of-bias domains were 1) ‘The cause-and-effect relationships were clearly stated’, 2) ‘Detailed study population and participant details’, 3) ‘Reliable exposure’, 4) ‘Reliable outcome measures’, 5) ‘Selective reporting’, and 6) ‘Overall’. The methodological and reporting quality domains were 1) ‘Repeated condition and outcome measurements’, 2) ‘Appropriate statistical analysis’, 3) ‘Exact p -values’, 4) ‘Effect sizes (e.g., Cohen’s d , Hedges’ g , Pearson’s r)’, 5) ‘Confidence intervals’, and 6) ‘Overall’.

At least two authors independently completed the risk-of-bias and quality assessments, with a third author settling any discrepancies. The two authors who have published widely on PIMA vs HIMA were not involved in the risk-of-bias or quality assessments. The results were uploaded to the Robvis RoB 2.0 visualization tool [58], producing the ‘traffic-light’ and ‘summary’ plots. Paint-3D (Microsoft Corporation, Redmond, WA) was used to edit and combine plots for clarity and aesthetics.

2.6 Statistical analysis

2.6.1 Calculation of effect sizes

Raw data, effect sizes, and p -values were extracted and entered into a GoogleSheets spreadsheet. The extracted raw data were used to calculate standardized mean differences (Cohen’s effect size [d]) between PIMA and HIMA conditions. The standardized mean differences were

interpreted as: small=0.20-0.49, moderate=0.50-0.79, and large>0.80 [59]. Studies already reporting Cohen's d effect sizes were double-checked where possible. Results were eventually distilled to statistically significant ($p<0.05$), or 'potentially meaningful' ($d\geq 0.20$) between condition differences. Standardized mean differences were calculated for all time points and variables using the following equation:

$$\text{Cohen's } d = (\text{MeanPIMA} - \text{MeanHIMA}) / (\text{pooledSD})$$

Percentage differences between PIMA and HIMA conditions were also calculated and reported where appropriate using the following equation:

$$\% \Delta = ((\text{HIMA} - \text{PIMA}) / \text{PIMA}) * 100$$

2.6.2 Meta-analytic synthesis

Meta-analytical procedures were performed using SPSS (Version 29.0, IBM Corp; Armonk, NY, USA). Upon careful examination of the extracted data, the authors agreed that time to task failure (TTF), ratings of perceived exertion (RPE), heart rate (HR), and mean arterial pressure (MAP) data were collected and reported with sufficient detail and consistency to warrant meta-analysis. For example, while several studies examined myoelectrical variables, most reported percent change instead of raw data or standardized effect sizes. Other variables, such as force loss, were not reported in enough studies to warrant meta-analysis. Therefore, the following methods are only for TTF, RPE, HR, and MAP.

Each study examining TTF provided mean and standard deviation data required to calculate standardized mean differences between PIMA and HIMA conditions. RPE, HR, and MAP data were often reported in figure format and extracted as previously detailed. This raw data was entered into the statistical software directly to avoid potential human error. However, standardized mean differences were confirmed by manual calculation. Including multiple effects from a single study violates the assumption of independence as effects from the same study are likely to be more similar than effects from others and would, therefore, influence statistical power. However, in the case of only a small proportion of studies reporting multiple outcomes, the dependence might be ignored if a sensitivity analysis is performed [60, 61]. For TTF, four of 23 studies reported multiple outcomes with overlapping participants at different intensities [62, 63],

forearm positions [64], or muscle groups [65]. Sensitivity analysis revealed only small differences in meta-analysis results with single and combined outcomes (see results). Therefore, the forest plots illustrate single outcomes, while statistical results are provided for single and combined outcomes. For combined analysis, the effect sizes from a single study utilizing multiple loading intensities, age groups, or forearm orientations were averaged and entered into the statistical software as a single comparison. The Hedges' g effect size with the adjusted standard error was selected to correct for studies with small sample sizes. A random-effects model with restricted maximum likelihood was chosen due to large inter-study variability regarding age, sex, muscle groups, and loading intensities. The interpretation of the pooled effect size with 95% confidence intervals (95%CI) was based on the following thresholds: trivial<0.20, small=0.20–0.49, medium=0.50–0.79, and large $\geq$ 0.80 [59]. Where appropriate, bubble plots and subgroup analyses were employed to examine the effect of loading intensity, muscle group, or time relative to TTF. Muscle groups were ordered by typical amounts of muscle mass [66, 67] and coded as: 1-finger abductors, 2-wrist extensors, 3-ankle dorsi-flexors, 4-elbow flexors, 5-elbow extensors, 6-knee extensors, and 7-trunk extensors.

2.6.3 Statistical heterogeneity

The I^2 statistic was used to evaluate study heterogeneity and represents the percentage of total variation in estimated effects across studies due to heterogeneity rather than chance. The I^2 statistic was interpreted as low<25%, moderate=25-50%, and high>50% heterogeneity [68]. Funnel plots were visually examined with excessive asymmetry or multiple studies outside the funnel, suggesting the potential for publication bias [69].

3. RESULTS

The systematic literature search yielded 54 studies that fulfilled the inclusion criteria [4, 5, 18, 20-23, 25, 30, 33-35, 62-65, 70-107]. Forty-two studies were found through database searching [4, 5, 18, 20-23, 25, 34, 35, 62, 63, 65, 71, 72, 75-83, 85-99, 103, 105, 106], while 12 publications were included following reference list searches [30, 33, 64, 70, 73, 74, 100-102, 104, 107]. Nine of the studies not found via database search were from the same laboratory group [33, 64, 70, 73, 74, 100-102, 104]. Therefore, the primary investigator of the relevant laboratory was contacted,

resulting in one additional study being included [84]. The search strategy and inclusion/exclusion results are summarized in **Figure 2**.

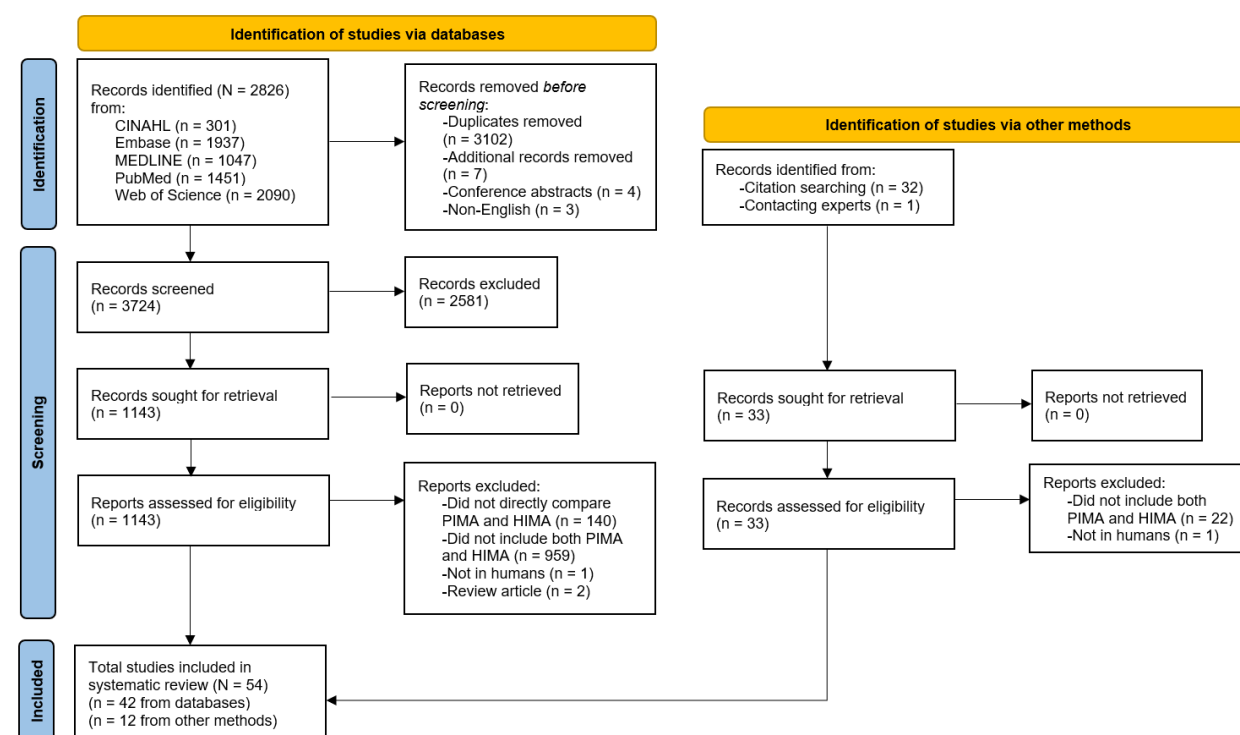


Figure 2. PRISMA diagram

3.1 Study characteristics

The 54 studies included 856 participants (~608 male, ~248 female, ~29.5±10.1 years) [4, 5, 18, 20-23, 25, 30, 33-35, 62-65, 70-107] with publication years ranging from 1995 [106] to 2024 [18], with a mean publication year of 2012.9±6.9. Demographic numbers are approximate, as sex [70, 91, 106] and age were absent [70, 106], or not fully reported [21, 23, 30, 34, 35, 62, 72-75, 82, 87, 88, 104] in several studies. Twelve studies [18, 20, 22, 25, 33, 64, 75, 86, 92, 94, 95, 102] were exclusively male, 39 were mixed sex [4, 5, 21, 23, 30, 34, 35, 62, 63, 65, 71-74, 76-85, 87-90, 93, 96-101, 103-105, 107], with three studies not reporting sex [70, 91, 106]. Forty-one studies examined young (mean: 21-33.5 years) participants [4, 5, 20-23, 25, 30, 35, 62-65, 71, 73-81, 84-105, 107]. In contrast, two examined older (mean: 64-72 years) participants [18, 83], four studies had participants with a mean range of 41.5-52 years [33, 34, 72, 82], and four included young and old sub-groups [33, 34, 72, 82]. No study included participants under 18 years of age. Single studies compared participants with knee osteoarthritis and healthy controls [18], and strength and

endurance athletes [20], while one study utilized soccer players [22]. Two studies injected saline solutions into the patella tendon to induce experimental joint pain [96, 99].

All 54 studies utilized single-joint muscle actions [4, 5, 18, 20-23, 25, 30, 33-35, 62-65, 70-107]. The most common agonist muscle group of interest was the elbow flexors (n=25) [20-23, 30, 34, 62-65, 74, 78, 80, 81, 83, 84, 86, 87, 89, 94, 95, 100, 102, 105, 106], followed by the knee extensors (n=8) [33, 75, 76, 96-99, 101]. Finger abductors (n=6) [35, 71, 88, 91, 92, 104], wrist extensors (n=3) [70, 72, 73] and flexors (n=2) [70, 77], ankle dorsi- (n=3) [82, 85, 93] and plantar-flexors (n=3) [79, 90, 107], elbow extensors (n=4) [4, 5, 25, 106], hip/trunk extensors (n=2) [65, 103], knee flexors (n=1) [18], and hip flexors (n=1) [22] were also agonist muscle groups of interest. Bittmann et al. [22] (elbow and hip flexion) and Thomas et al. [65] (trunk extension and elbow flexion) compared PIMA and HIMA across multiple agonist muscle groups. Two studies compared the elbow flexors with vertical and horizontal forearm positions [95, 100], while another examined elbow flexion with supinated and neutral forearms [64]. The mean contraction intensity was $\sim 34.6 \pm 27.8\%$ of pushing maximal voluntary isometric contraction (MVIC). However, intensity differed widely, ranging from 3.8-100% of MVIC. Most studies (n=39) investigated intensities $\leq 30\%$ MVIC [30, 33-35, 62-65, 70-74, 76, 79-85, 88-92, 94-98, 100-105, 107], 14 studies investigated intensities of 40-90% MVIC [4, 5, 25, 62, 63, 71, 75, 77-79, 86, 87, 92, 93], while five studies examined contractions as high as 100% [18, 20-23]. Seven included multiple intensities [5, 30, 34, 62, 63, 71, 72, 79, 86, 90, 92, 106], and five studies used adaptive muscle action ranging from submaximal to supramaximal intensities in one trial [18, 20-23]. Muscle actions were taken to failure in 28 studies [4, 18, 20-23, 33, 62-65, 73, 74, 76, 78, 80, 82-85, 87, 89, 92, 98, 101-103, 105].

3.2 Risk-of-bias and study quality

The risk-of-bias assessments of each study are illustrated in **Figure 3A**, while **Figure 3B** summarizes the findings across all 54 studies. The overall risk-of-bias was minimal, with 44 studies assessed as ‘low’ [4, 5, 18, 20-23, 25, 33-35, 62, 64, 65, 71-78, 80, 82, 85-100, 102-105] with ten assessed as having ‘some concerns’ [30, 63, 70, 79, 81, 83, 84, 101, 106, 107] and zero studies flagged for ‘high’ overall risk-of-bias. Five studies lacked detail regarding study participants [30, 70, 79, 106, 107] and were substantially older (2005.2 ± 8.1 , 1995-2015) than the

average publication date (2012.9±6.9, 1995-2024) of the 54 studies, suggesting improved reporting and peer-review standards. Neither Buchanan and Lloyd [106] nor Baudry and Enoka [70] reported sex, age, height, mass, or any other demographic information or baseline characteristics. While Garner et al. [79] provided participant sex, age, height, and mass, inclusion and exclusion criteria, physical activity level, and baseline characteristics (e.g., strength, power) were not reported. Finally, neither Magalhães [107] nor Mathis et al. [30] reported height, mass, inclusion or exclusion criteria, or any baseline characteristics. While no other studies or categories received a ‘high’ risk-of-bias classification, the most common concern was the lack of in-house or intra-study reliability statistics for most outcome measures [30, 33-35, 62-65, 70-74, 76, 77, 79-86, 90-107]. Like the participant details category, studies reporting their reliabilities [4, 5, 18, 20-23, 25, 75, 78, 87-89] have substantially more recent publication dates (2020±4.6) than those that did not (2010.7±6.0) [30, 33-35, 62-65, 70-74, 76, 77, 79-86, 90-107].

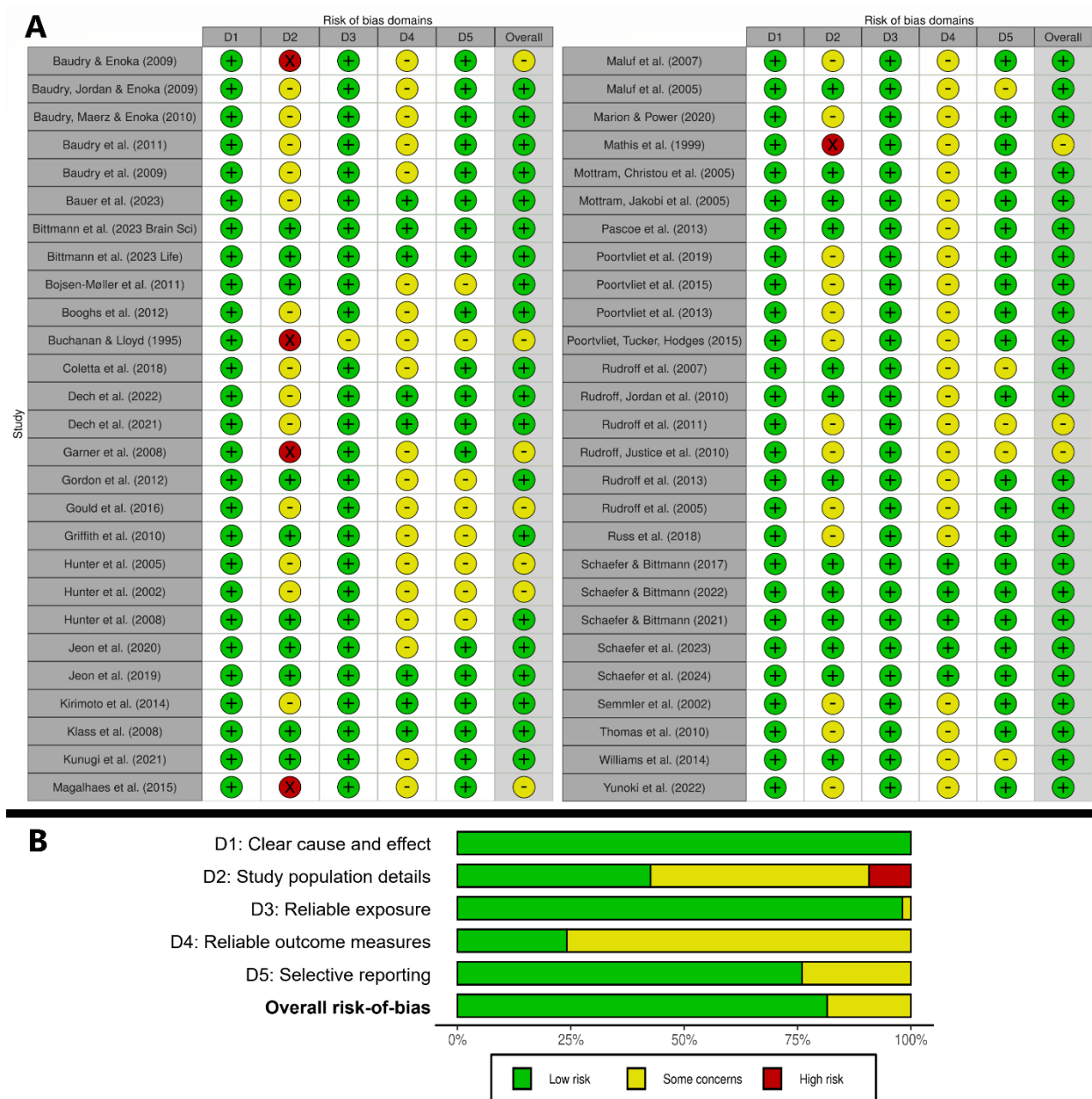


Figure 3. Risk-of-bias assessment illustrations of individual studies (A), and all 54 studies pooled (B)

Figure 4A illustrates each study's methodological and reporting quality assessments, while **Figure 4B** summarizes the findings across all included studies. Overall quality scores were mostly 'medium' [30, 33-35, 62-65, 70-74, 76, 77, 79-87, 89-105] with 12 studies receiving 'high' overall quality scores [4, 5, 18, 20-23, 25, 75, 78, 88, 107]. The oldest study [106] was deemed 'low' quality. While all studies were considered to have utilized appropriate statistical analyses (e.g., t-test vs ANOVA, parametric vs non-parametric), several only sometimes [18, 20, 33, 63, 64, 87,

90, 98, 99, 101, 105] or never [65, 76, 83, 84, 86, 102, 106] reported exact *p*-values. Only 15 studies provided effect sizes for all critical condition and outcome measures [4, 5, 18, 20-23, 25, 65, 75, 78, 87, 88, 105, 107], while confidence interval reporting was only fully present in four studies [18, 20, 25, 107]. Interestingly, only two studies achieving ‘high’ quality were over eight years old [88, 107], with a mean publication year of 2020.6±3.4 [4, 5, 18, 20-23, 25, 75, 78, 88, 107], highlighting the relative improvement in methodological and statistical reporting quality in recent years.

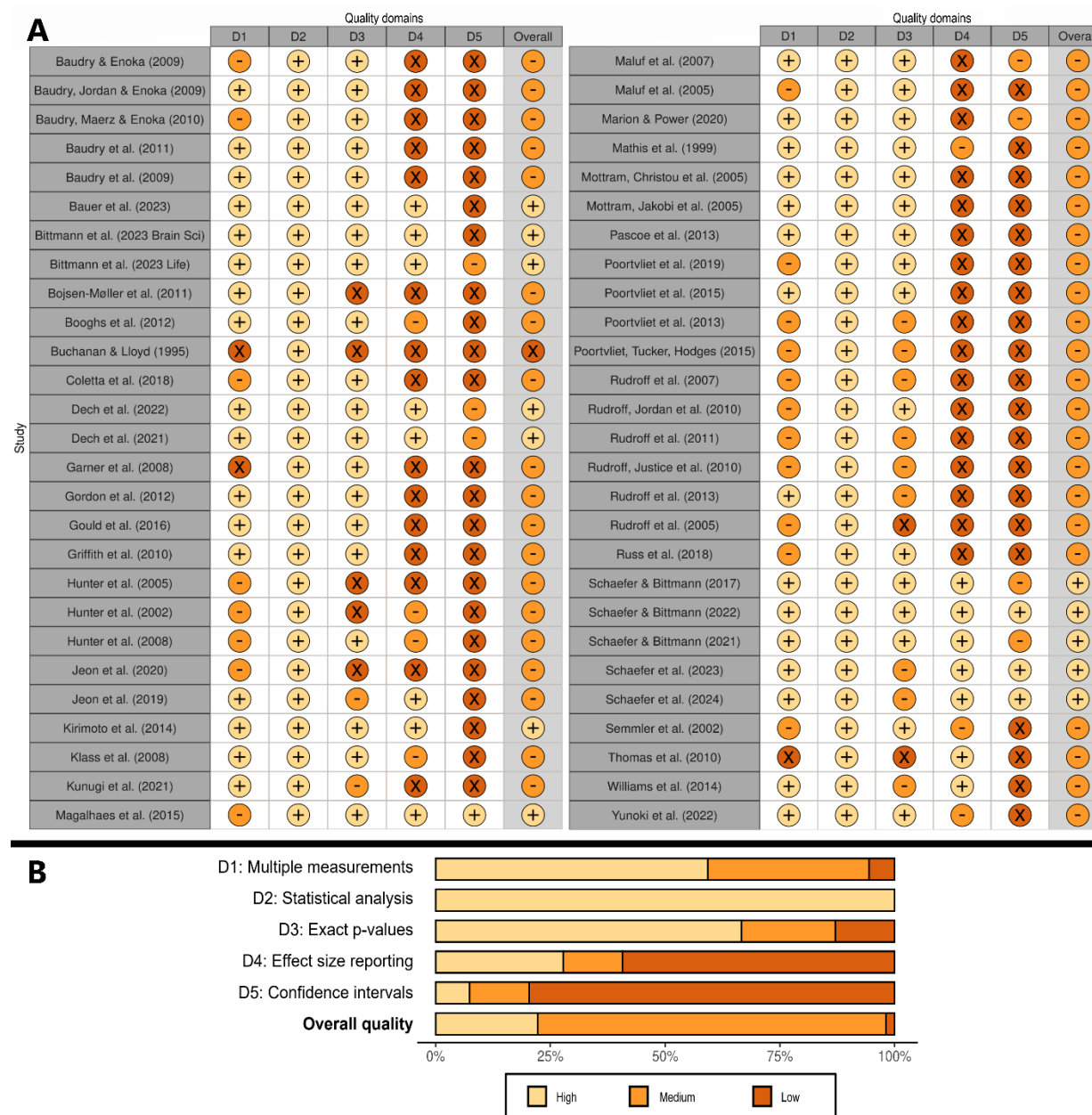


Figure 4. Methodological and reporting quality of individual studies (A), and all 54 studies pooled (B)

3.3 Summary of isometric muscle action differences

An overview of the primary outcomes, excluding meta-analyses, including the total number and type (e.g., TTF, RPE increase, discharge rate) of comparisons and the number of statistically significant and potentially meaningful differences, is provided in **Table 1**. A total of 360 comparisons were identified and included in the results. Of these, 52 comparisons showed statistically significant ($p < 0.05$) higher effects for HIMA than PIMA. In contrast, 93 comparisons showed greater effects for PIMA compared to HIMA, and 201 comparisons were statistically similar between conditions. Notably, an opposite trend emerged when examining standardized effect sizes without considering p-values, with 72 comparisons favoring PIMA with larger effects, 50 favoring HIMA, and 45 holding trivial effects (d or $g < 0.20$).

Table 1. Overview of outcomes including the number of outcomes, and direction of statistical significance ($p < 0.05$) and standardized effect size.

Parameter	Number of outcomes						
	Total	Significant p -value			Effect size		
		PIMA > HIMA	HIMA > PIMA	n.s.	PIMA > HIMA	HIMA > PIMA	Irrelevant effect
TTF	30	16	2	12	24	3	3
Fluctuations	8	6	1	1	6	1	-
Increase fluctuations	18	4	10	4	6	11	1
Max. torque	3	3	0	-	3	0	-
MVIC decline*	29	0	4	25	7	8	9
RPE 50% TTF	14	-	-	-	8	3	3
RPE 100% TTF	16	0	0	16	3	2	11
RPE increase	16	0	9	6	-	-	-
aEMG agonists	48	5	8	35	-	-	-
aEMG synergists	15	2	2	11	-	-	-
aEMG antagonists	17	1	0	17	-	-	-
Increase aEMG agonists	17	0	8	9	-	-	-
Increase aEMG synergist	10	0	5	5	-	-	-
Increase aEMG antagonists	9	0	4	5	-	-	-
Coactivation	13	0	3	10	-	-	-
EMG burst rate	4	0	2	2	0	2	2
Burst rate increase	3	0	3	0	0	2	0
Discharge rate	12	8	0	4	7	0	0
CV of ISI	7	1	5	1	-	-	-
D1 inhibition	7	5	1	1	5	1	-
Heteronymous reflex	11	1	9	1	-	7	-

HR at end	15	0	0	15	3	4	8
HR increase	12	0	8	4	-	-	-
MAP at end	14	0	1	13	0	6	8
MAP increase	12	0	8	4	-	-	-
TOTAL	360	52	93	201	72	50	45

effect size: Cohen's d or Hedges' $g > 0.20$

*irrespective of epoch of TTF (90% or 100% TTF)

In case the sum of results is not equal to the total number of outcomes the information was not available.

Domain-specific (performance and biomechanics, neural and neuromuscular, and cardiovascular and metabolic) results are presented in detail herein.

3.4 Parameters of performance and biomechanics

The 35 studies examining performance and biomechanical differences between PIMA and HIMA are summarized in **Table 2** [4, 5, 18, 20-23, 33, 62-65, 73-76, 78, 80-85, 87, 89, 92, 94-96, 98, 100-103, 105]. Twenty-four studies compared TTF between PIMA and HIMA conditions [4, 25, 33, 62-65, 73, 74, 76, 78, 80, 82-85, 87, 89, 92, 98, 101-103, 105] while two studies investigated the failure rate during personal interaction (similar to TTF) [5, 25]. Twenty-two studies quantified force or position fluctuations or variability throughout each condition [62-64, 73-76, 80-85, 92, 94-96, 98, 100-103], 20 studies measured MVIC reductions following PIMA and HIMA [20, 33, 62, 63, 73, 74, 76, 80-83, 85, 89, 92, 94, 95, 100, 101, 103, 105], five studies compared the maximal force capacity between the two muscle actions [18, 20-23] and 13 studies compared RPE during both tasks [63, 64, 76, 80-85, 92, 95, 101, 105].

Table 2. Summary of studies comparing performance parameters between pushing (PIMA) and holding (HIMA) isometric muscle actions.

↑=sign. larger effect from HIMA vs PIMA; ↓=sign. lower effects from HIMA vs PIMA; ↔=no significant difference between HIMA vs PIMA.

Significance p , effect size Cohen's d and the percentage difference between HIMA and PIMA are given.

Study	Participants	Relevant Measures	Conditions	Performance Results
Baudry, Maerz, Gould & Enoka, 2011 [73]	N=7 (4 m, 3 f) 19–35 yrs MVIC: Pre-PIMA: 123±41 N Pre-HIMA: 124±46 N	Wrist extensors PIMA: servo-controlled torque motor HIMA: inertial load simulated by torque motor, angle TTF Force fluctuations (CV) MVIC decline	Wrist extension 0°, shoulder abduction 74°, elbow flexion 90°, forearm pronated 20% MVIC to failure (1×) with electrical stimulation (4 sets of 10 reflexes) PIMA: >5% force change for >3s HIMA: >11.5° angle change for >3s corrections allowed Visual & verbal feedback	↓ TTF ($p<0.0001$, $d=1.79$, −32.0%) HIMA: 495±119s vs PIMA: 728±140s ↓ Force fluctuations at end ($p=0.002$, $d=1.00$, −49.8%) ↓ Force fluctuation increase ($p<0.001$) ↔ MVIC decline ($p>0.05$, $d=0.16$, 10.3%)
Baudry, Rudroff, Pierpoint & Enoka, 2009 [74]	N=24 (15 m, 9 f; 4 dropouts) 18–37 yrs MVIC: Pre-PIMA: 216±73 N Pre-HIMA: 231±85 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer TTF Force fluctuations (CV) MVIC decline	Elbow flexion 90°, forearm vertical & supinated 15% MVIC to failure (1×) & ~25% MVIC (twice recruitment threshold) for 90% of TTF (1×) PIMA: >5% force change for >3s HIMA: >11.5° angle change for >3s corrections allowed Visual & verbal feedback	↓ TTF ($p=0.0002$, $d=0.83$, −35.7%) HIMA: 1225±666s vs PIMA: 1904±944s ↑ Force fluctuations increase ($p=0.0001$, $d=0.66$, 50.6%) ↔ MVIC decline after failure tasks ($p=0.60$, −5.5%) ↑ MVIC decline after 90% TTF tasks ($p<0.001$, $d=1.74$, −50%)
Bauer, Gomes, Oliveira, Santos, Pezarat-Correia & Vaz, 2023 [75]	N=13 m 18–35 yrs MVIC: -	Knee extensors Isokinetic dynamometer PIMA: isometric mode HIMA: isotonic mode Torque sample entropy Torque fluctuations (CV)	Knee flexion 70° 40% MVIC for 30s (1×) PIMA: - HIMA: >10° angle change PIMA: visual feedback HIMA: with & without visual feedback	↓ Torque sample entropy with feedback ($p<0.001$, $d=2.51$, −26.9%) ↓ Torque sample entropy without feedback ($p<0.001$, $d=2.41$, −28.4%) ↑ Torque fluctuations with feedback ($p<0.001$, $d=4.09$, 123.9%) ↑ Torque fluctuations without feedback ($p<0.001$, $d=3.68$, 123.9%)
Bittmann, Dech & Schaefer, 2023a [21]	N=12 (9 m, 3 f) m=30.9±9.3 yrs f=31.3±6.8 yrs MVIC: 275±47 N	Elbow flexors PIMA: force transducer (handheld device incl. gyrometer) HIMA: adapt to increasing force at given position (Adaptive Force (AF); manual test with handheld device incl. gyrometer) Max force	Elbow flexion 90°, forearm vertical & supinated 100% MVIC (2×) For HIMA: increasing to 100% MVIC without conditioning, with precontraction in lengthened position with passive return (CL) and with CL with 2 nd precontraction in test position (CL-CT) No termination criteria or feedback	↔ Max force without conditioning & after CL-CT ($p>0.05$) ↓ Max force after CL ($p<0.05$, $d=1.90$, −44.7%) HIMA: 152.1±78.6 N; PIMA: 275±47 N

Bittmann, Dech & Schaefer, 2023b [22]	N=13 m soccer players 26.4±4.3 yrs MVIC: Elbow: 314±41 N Hip: 312±44 N	Elbow & hip flexors PIMA: force transducer (handheld device incl. gyrometer) HIMA: adapt to increasing force at given position (AF; manual test with handheld device incl. gyrometer) Max force	Supine position, elbow flexion 90°, forearm vertical & supinated, hip & knee flexion 90° 100% MVIC (3×) For HIMA: increasing to 100% MVIC (1) without conditioning; (2) after manipulation (muscle spindle slack) No termination criteria or feedback	↔ max force without conditioning (elbow: $p=0.359$; hip: $p=0.924$) ↓ max force after manipulation elbow ($p<0.05$, $d=2.63$, -37.1%) HIMA: 197.2±47.7 N; PIMA: 313.8±40.5 N ↓ max force after manipulation hip ($p<0.05$, $d=2.15$, -32.9%) HIMA: 209.2±50.7 N; PIMA: 311.5±44.4 N
Bojsen-Møller, Schwartz & Magnusson, 2011 [76]	N=14 (6 m, 8 f) 29±5 yrs MVIC: 644±47 N	Knee extensors PIMA: force transducer HIMA: inertial load, electrogoniometer & force transducer TTF Force fluctuations (SD) MVIC decline (n=8) RPE	Knee flexion 60° 20% MVIC to failure (1×) PIMA: >1% force drop for >5s HIMA: >1° angle change for >5s corrections allowed Visual & verbal feedback	↔ TTF ($p=0.10$, $d=0.80$, -10.4%) HIMA: 379±180s; PIMA: 423±228s ↓ increase in force fluctuations at middle ($p<0.01$, $d=5.25$, -35.1%)* ↓ increase in force fluctuations at end ($p<0.01$, $d=5.19$, -48.0%)* ↔ MVIC decline ($p>0.05$, 9.0%) ↔ RPE ($p>0.50$)
Booghs, Baudry, Enoka & Duchateau, 2012 [62]	N=15 (8 m, 7 f) 18–36 yrs MVIC: Pre-PIMA: 20% task: 293±82 N 60% task: 299±119 N Pre-HIMA: 20% task: 296±86 N 60% task: 292±111 N	Elbow flexors PIMA: force transducer HIMA: inertial load, potentiometer TTF Force fluctuations (SD) MVIC decline	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVIC to failure (n=12) (1×) 60% MVIC to failure (n=9) (1×) PIMA: >2% or 5% force change for 3s HIMA: >1.5° or 3° angle change for 3s corrections allowed Visual & verbal feedback	↓ TTF 20% task ($p=0.011$, $d=0.73$, -24.2%) HIMA: 404±159s; PIMA: 533±194s ↔ TTF 60% task ($p=0.13$, $d=0.57$, -15.6%) HIMA: 54±19s; PIMA: 64±16s ↓ Force fluctuation 20% task ($p<0.001$, $d=0.53$, -18.2%) ↓ Force fluctuation 60% task ($p<0.001$, $d=0.53$, -35.1%) ↔ MVIC decline 20% task ($p>0.05$, $d=0.23$, 9.6%) ↔ MVIC decline 60% task ($p>0.05$, $d=0.73$, -19.1%)
Dech, Bittmann & Schaefer, 2021 [23]	N=13 (9 m, 4 f) m: 29.4±6.4 yrs f: 32.0±2.9 yrs MVIC: m: 67±17 Nm f: 25±4 Nm	Elbow flexors PIMA: force transducer HIMA: adapt to increasing force at given position (AF; pneumatic device incl ACC) Max torque (M & max of trials) Torque CV between trials	Elbow flexion 90°, forearm vertical & neutral 100% MVIC (4×2 timepoints) For HIMA: increasing to >100% MVIC; max. HIMA=breaking point from isometric to eccentric action (<2° change) No termination criteria or feedback	↓ Max torque (t_1 : $p=0.004$, $d=0.99$, -16.0% ; t_2 : $p=0.009$; $d=0.89$, -16.3%) ↓ Mean of max torques (t_1 : $p<0.001$, $d=1.32$, -26.7% ; t_2 : $p=0.001$; $d=1.23$, -29.8%) ↑ Torque CV (t_1 : $d=1.49$, 438.5%; t_2 : $d=1.53$; 499.0%)
Dech, Bittmann & Schaefer, 2022 [78]	N=10 (8 m, 2 f) 30.7±11.7 yrs MVIC: left: 69±22 Nm right: 70±24 Nm	Elbow flexors (both sides) PIMA: force transducer (seated) or inertial load with intermittent twitches every 7s (standing) HIMA: inertial load (standing) TTF	Elbow flexion 90°, forearm horizontal & supinated 60% MVIC to failure (1×) PIMA: < target force for 2s or twitches impossible HIMA: < target angle for 2s No feedback	↔ TTF ($p=0.394$, $d=0.38$, -11.0%) HIMA: 44.8±18.1s; PIMA: 50.3±9.5s ↓ TTF HIMA vs PIMA with twitches ($p=0.043$, $d=1.03$, -19.2%) HIMA: 42.6±7.6s; PIMA: 52.8±11.6s ↓ TTF HIMA vs both PIMA tasks ($p=0.047$, $d=0.65$, -15.2%) HIMA: 43.7±13.5s; PIMA: 51.6±10.4s

Gordon, Rudroff, Enoka & Enoka, 2012 [80]	N=20 (15 m, 5 f) 21±4 yrs MVIC: Pre-PIMA: 273±90 N Pre-HIMA: 291±93 N	Elbow flexors (both sides) PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer TTF Force fluctuations (CV) MVIC decline RPE	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVIC to failure (1×) PIMA: >5% force change for 5s HIMA: >11.5° angle change for 5s corrections allowed Visual & verbal feedback	↓ TTF ($p<0.001$, $d=0.96$, -31.1%) HIMA: 253±103s; PIMA: 367±133s ↓ Force fluctuations at start ($p<0.001$, $d=0.74$, -52.3%) ↑ Force fluctuation rate of increase ($p<0.001$, $d=1.13$, 3.0%) ↔ MVIC decline ($p=0.73$, $d=0.52$, -23.3%) ↑ RPE rate of increase ($p=0.002$, $d=0.61$, 22.2%)
Gould, Cleland, Mani, Amiridis & Enoka, 2016 [81]	N=21 (13 m, 8 f) 21.9±1.9 yrs MVIC: 252±89 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer Force fluctuations (CV) MVIC decline RPE	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 24.9±10.5% MVIC (~4.7±2% above recruitment threshold) for 152±84s (1×) No termination criteria Visual feedback	↓ Force fluctuations at start, middle & end ($d=0.65$, -41.2%, $d=0.62$, -33.3% & $d=0.63$, -35.0%) ↔ MVIC decline ($p=0.77$, $d=0.04$, 4.5%) ↑ RPE at end ($p=0.006$, $d=0.53$, 14.3%)
Griffith, Yoon & Hunter, 2010 [82]	Young: N=17 (8 m, 9 f) Old: N=12 (7 m, 5 f) 23.6±6.5 yrs 70.0±5.0 yrs MVIC young: pre-PIMA: 38.0±10.2 Nm pre-HIMA: 37.4±9.4 Nm MVIC old: pre-PIMA: 35.5±8.9 Nm pre-HIMA: 34.9±10.4 Nm	Dorsi-flexors PIMA: force transducer HIMA: inertial load, potentiometer, ACC TTF Fluctuations (SD) force or ACC MVIC decline RPE	Dorsi-flexion 0°, hip & knee flexion 90° 30% MVIC to failure (1×) PIMA: >5% force drop for 4s HIMA: >18° angle drop corrections allowed Visual & verbal feedback	↓ TTF for young & old ($p=0.03$, $d=0.45$, -17.3%) HIMA: 516±204s; PIMA: 624±270s ↔ Fluctuations rate of increase for young & old ($p=0.34$) ↔ MVIC decline for young & old ($p=0.78$, $d=0.19$, -6.6%) ↔ RPE at start & end for young & old ($p>0.50$) ↔ RPE rates of increase ($p=0.07$)
Hunter, Rochette, Critchlow & Enoka, 2005 [83]	N=18 (10 m, 8 f) 72±4 yrs MVIC: Pre-PIMA: 180±55 N Pre-HIMA: 178±61 N	Elbow flexors PIMA: force transducers HIMA: inertial load, electrogoniometer, ACC TTF Fluctuations force (CV & product ACC x inertial load) & ACC (SD) MVIC decline RPE	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVIC to failure (1×) PIMA: >10% force drop for >5s HIMA: >26° angle change for >5s corrections allowed Visual & verbal feedback	↓ TTF ($p<0.05$, $d=1.57$, -53.5%) HIMA: 636±366s; PIMA: 1368±546s ↑ Force fluctuations at end ($p<0.05$, $d=0.91$, 75.8%) ↑ Fluctuation increase ACC vs force ($p<0.05$) ↔ MVIC decline after tasks ($p>0.05$, -8.8%) ↔ RPE at start & end ($p>0.05$) ↑ RPE rate of increase ($p<0.05$, 105.5%)
Hunter, Ryan, Ortega & Enoka, 2002 [84]	N=16 (8 m, 8 f) 27±4 yrs MVIC: Pre-PIMA: 308±151 N Pre-HIMA: 307±152 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 15% MVIC to failure (1×)	↓ TTF ($p<0.05$, $d=1.06$, -49.9%) HIMA: 702±582s; PIMA: 1402±728s

		TTF Fluctuations force (CV) or ACC (SD) (vertical & side-to-side) RPE	PIMA: >10% force drop for >5s HIMA: >10° angular change for >5s corrections allowed Visual & verbal feedback	Vertical fluctuations: ↑ Relative increase at end ($p<0.05$, $d=1.96$, 255.4%) ↑ Relative increase at same absolute time ($p<0.05$, $d=2.88$, 1227%) Side-to-side fluctuations: ↑ Relative increase at end ($p<0.05$, $d=1.27$, 85.6%) ↑ Relative increase at same absolute time ($p<0.05$, $d=2.37$, 426.8%) ↔ RPE at start & end ($p>0.05$) ↑ RPE rate of increase ($p<0.05$, 89.7%)
Hunter, Yoon, Farinella, Griffith & Ng, 2008 [85]	N=15 (8 m, 7 f) 21.1±1.4 yrs MVIC: pre-PIMA: 333±71 N pre-HIMA: 334±65 N	Dorsi-flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC TTF Fluctuations force (CV) & ACC (SD) MVIC decline RPE	Dorsi-flexion 0°, hip & knee flexion 90° 20% MVIC to failure (1×) PIMA: >5% force drop for 4s HIMA: >18° angle drop for 4s corrections allowed Visual & verbal feedback	↓ TTF ($p=0.03$, $d=0.85$, -53.1%) HIMA: 600±372s; PIMA: 1278±1068s ↑ Fluctuations at end ($p=0.007$, $d=1.22$, 101.4%) ↑ Fluctuation increase ($p=0.004$, $d=1.12$, 328.9%) ↔ MVIC decline ($p=0.57$, -6.7%) ↔ RPE at start & end ($p>0.05$) ↑ RPE increase ($p=0.006$, 108.2%)
Jeon, Ye & Miller, 2019 [87]	N=20 (12 m, 8 f) m: 24±4 yrs f: 22±3 yrs MVIC: m: 382±102 N f: 189±29 N	Elbow flexors (dominant side) PIMA: force transducer HIMA: inertial load, steel hinge (visual control) TTF	Elbow flexion 135°, forearm horizontal & supinated; non-dominant hand on abdomen 50% MVIC to failure (1×) PIMA: < target force for 3s HIMA: < target position for 3s corrections allowed visual (PIMA) & verbal feedback (both)	↓ TTF sexes combined ($p=0.033$, $d=0.35$, -14.4%) HIMA: 33.9±14.9s; PIMA: 39.6±16.6s ↔ TTF men ($d=0.14$, -5.1%) HIMA: 40.9±13.7s; PIMA: 43.1±18.3s ↓ TTF women ($d=0.93$, -31.4%) HIMA: 23.4±9.9s; PIMA: 34.1±13.0s
Klass, Levenez, Enoka & Duchateau, 2008 [89]	N=11 (6 m, 5 f) 29.4±6 yrs MVIC: 271±99 N	Elbow flexors (dominant side) PIMA: force transducer HIMA: inertial load, electrogoniometer TTF MVIC decline	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVC to failure (1×) incl. magnetic & electrical stimulation to motor cortex & brachial plexus PIMA: < target force for 5-10s HIMA: >10° angle drop for 5-10s corrections allowed Visual & verbal feedback	↓ TTF ($p<0.001$, $d=1.87$, -56.2%) HIMA: 420±165s; PIMA: 958±371s ↔ MVIC decline ($p=0.57$, $d=0.08$, 4.2%)
Maluf, Shinohara, Stephenson & Enoka, 2005 [92]	N=20 m (2 groups, n=10) 23±5 yrs	First dorsal interosseus (abduction) PIMA: force transducer HIMA: inertial load, potentiometer, ACC	Index finger abduction 0° 20% or 60% MVIC to failure (1×)	↓ Mean force at 20% task ($p=0.009$, $d=0.13$, -2.9%) ↔ Mean force at 60% task ($p=0.134$) ↓ TTF at 20% task ($p=0.005$, $d=1.41$, -39.7%) HIMA: 593±212s; PIMA: 938±328s

	MVIC: Low force group: Pre-PIMA: 34.8±7.5 N Pre-HIMA: 33.8±6.7 N High force group: Pre-PIMA: 32.5±4.0 N Pre-HIMA: 32.1±3.9 N	Mean force TTF Fluctuations force (CV) MVIC decline RPE	PIMA: >1.5% force change for 3s HIMA: >10° angle change for 3s corrections allowed Visual & verbal feedback	↔ TTF at 60% task ($p=0.200$, $d=0.19$, -7.5%) HIMA: 86±31s; PIMA: 93±41s ↓ Fluctuations at 20% task ($p=0.001$, $d=2.33$, -75.5%) ↓ Fluctuations at 60% task ($p<0.001$, $d=2.77$, -68.1%) ↔ MVIC decline at 20% and 60% tasks ($p>0.50$, 13.9% and 10.7%) ↑ RPE increase at 20% task ($p=0.007$, 65.3%) ↔ RPE increase at 60% ($p=0.046$, Bonferroni corrected $\alpha=0.013$, 16.8%)
Mottram, Christou, Meyer & Enoka, 2005 [94]	N=15 m 25.5±5.9 yrs MVIC: 267±48 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC Fluctuations (SD) force or ACC MVIC decline	Elbow flexion 90°, forearm horizontal & neutral, shoulder abduction 15° 22.4±14% MVIC (3.6±2.1% above recruitment threshold) for 161±93s (1×) incl. needle EMG Visual feedback	↑ Fluctuation increase ($p<0.001$, $d=1.20$, 373.7%) ↔ MVIC decline ($p=0.09$, $d=0.35$, -40.2%)
Mottram, Jakobi, Semmler & Enoka, 2005 [95]	N=15 m 25.6±5.8 yrs MVIC: 265±50 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC Fluctuations (SD) force or ACC MVIC decline RPE	Elbow flexion 90°, forearm horizontal & neutral, shoulder abduction 15° 22.2±13.4% MVIC (3.5±2.1% above recruitment threshold) for 161±96s (1×) incl. needle EMG Visual feedback	↑ Relative fluctuations (vertical & horizontal) all time points ($p\leq 0.02$) ↑ Vertical fluctuation increase ($p=0.003$, $d=0.83$, 355.2%) ↑ Horizontal fluctuation increase ($p=0.003$, $d=0.88$, 433.3%) ↔ MVIC decline ($p=0.15$, $d=0.32$, -35.6%) ↑ RPE increase ($p=0.023$, 32.6%)
Poortvliet, Tucker, Finnigan, Scott & Hodges, 2019 [96]	N=17 (14 m, 3 f) 33±6 yrs MVIC: Pre-PIMA: 461±148 N Pre-HIMA: 460±149 N	Knee extensors PIMA: force transducer HIMA: inertial load, inclinometer Fluctuations (SD) force or position Experimental pain (hypertonic saline injection to infrapatellar fat pad)	Supine, knee & hip flexion 90° 10% MVIC for 30s (6×) with & without pain No termination criteria Visual & verbal feedback	↓ Fluctuation no pain ($d=3.16$, -80.1%) ↓ Fluctuation with pain ($d=3.54$, -85.6%) Significant changes in fluctuations for no pain vs pain only for PIMA, not for HIMA
Poortvliet, Tucker & Hodges, 2013 [98]	N=17 (9 m, 8 f) 32±7 yrs MVIC: 444±175 N	Knee extensors PIMA: force transducer (strain-gauge) HIMA: inertial load, electric inclinometer TTF Fluctuations (SD) force & position	Supine, knee & hip flexion 90° 20% MVIC to failure (1×) PIMA: >5% force change for 5s HIMA: >5° angle change for 5s corrections allowed Visual & verbal feedback	↓ TTF ($p<0.001$, $d=0.60$, -14.8%) HIMA: 184±51s; PIMA: 216±56s ↔ Fluctuations ($p>0.05$)

Rudroff, Barry, Stone, Barry & Enoka, 2007 [64]	N=20 m 27±5 yrs MVIC horizontal: Pre-PIMA: 309±45 N Pre-HIMA: 307±43 N MVIC vertical: Pre-PIMA: 264±55 N Pre-HIMA: 259±41 N	Elbow flexors in two postures PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC TTF Fluctuations (SD) in force or ACC RPE	Elbow flexion 90°, forearm vertical or horizontal 20% MVIC to failure (each 1×) PIMA: >5% force change for >5s HIMA: >11.5° angle change for >5s corrections allowed Visual & verbal feedback	↓ TTF horizontal forearm ($p=0.003$, $d=1.15$, -40.9%) HIMA: 312±156s, PIMA: 528±216s ↔ TTF vertical forearm ($p=0.99$, -1.3%) HIMA: 468±270s; PIMA: 474±246s ↑ Fluctuation increase (both postures; $p=0.008$, $d=0.98$, 124.5%) ↔ RPE ($p=0.904$) ↑ RPE increase horizontal forearm ($p=0.003$, 72.0%)
Rudroff, Jordan, Enoka, Matthews, Baudry & Enoka, 2010 [100]	N=23 (20 m, 3 f) 21±6 yrs MVIC neutral: Pre-PIMA: 242±57 N Pre-HIMA: 248±26 N MVIC supinated: Pre-PIMA: 249±43 N Pre-HIMA: 244±57 N	Elbow flexors (neutral or supinated) PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC Fluctuations (SD) force or ACC MVIC decline	Elbow flexion 90°, forearm vertical, neutral or supinated) 5% MVIC above recruitment threshold Neutral: 16.4±8% MVIC; 148±47s Supinated: 17.7±12% MVIC; 141±65s (1×) Visual feedback	↑ Fluctuations increase supinated ($p=0.02$, $d=1.34$, 39.3%) ↔ Fluctuations increase neutral ($p>0.05$) ↔ MVIC decline neutral & supinated ($p=0.65$, $d=0.15$, -16.7% & $p=0.73$, $d=0.32$, 31.3%)
Rudroff, Justice, Holmes, Matthews & Enoka, 2011 [63]	N=21 (10 m, 11 f) 23±6 yrs MVIC pre-PIMA: 20%: 165±73 N 30%: 169±86 N 45%: 148±68 N 60%: 142±61 N MVIC pre-HIMA: 20%: 181±74 N 30%: 164±88 N 45%: 157±67 N 60%: 152±56 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer TTF Fluctuations force (CV) MVIC decline RPE	Elbow flexion 90°, forearm horizontal, neutral 20% MVIC (n=10) 30% MVIC (n=10) 45% MVIC (n=10) 60% MVIC (n=10) to failure (1×) PIMA: >5% force change for >5s HIMA: >11.5° angle change for >5s Visual & verbal feedback	↓ TTF for 20% & 30% tasks 20%: HIMA: 299±77s; PIMA: 576±80s ($p<0.02$, $d=3.53$, -48.1%) 30%: HIMA: 168±36s; PIMA: 325±70s ($p<0.02$, $d=2.82$, -48.3%) ↔ TTF for 45% & 60% tasks 45%: HIMA: 132±29s; PIMA: 178±35s ($p>0.05$, $d=1.43$, -25.8%) 60%: HIMA: 87±14s; PIMA: 86±14s ($p>0.05$, $d=0.07$, 1.2%) ↓ Fluctuations (all intensities) ($p<0.001$, $d=0.87$, -40.4%) ↔ MVIC decline for all intensities ($p=0.94$, $d=0.14$ to 0.36, -17.2% to 30%) ↑ RPE increase rate at 20% & 30% tasks ($p=0.006$, $d=2.43$, 200%) ↔ RPE increase rate at 45% & 60% tasks ($p=0.86$)
Rudroff, Justice, Matthews, Zuo & Enoka, 2010 [101]	N=13 (9 m, 4 f) 25±7 yrs MVIC: Pre-PIMA: 189±40 N Pre-HIMA: 179±43 N	Knee extensors PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer TTF Fluctuations force (CV) MVIC decline RPE	Supine, knee & hip flexion 90° 20% MVIC to failure (1×) PIMA: >5% force change for 5s HIMA: >10° angle change for 5s Visual & verbal feedback	↓ TTF ($p=0.0015$, $d=1.35$, -50.9%) HIMA: 110±36s; PIMA: 224±114s ↓ Fluctuation increase at 80% TTF ($p<0.05$, $d=1.11$, -49.6%) ↓ Fluctuation increase at end ($p<0.05$, $d=0.78$, -38.0%) ↔ MVIC decline ($p=0.12$, $d=0.06$, -3.2%) ↔ RPE increase ($p=0.21$, 79.1%)

Rudroff, Kalliokoski, Block, Gould, Klingensmith III & Enoka, 2013 [33]	1. <i>Exp: Endurance</i> n=6 m 3 young: 23±4 yrs 3 old: 72±4 yrs MVIC: young: 500±28 N old: 331±29 N 2. <i>Exp: Muscle activation</i> n=12 m 6 young: 26±6 yrs 6 old: 77±6 yrs MVIC: young: 462±77 N old: 354±91 N	Knee extensors PIMA: force transducer HIMA: inertial load, goniometer, force transducer TTF (1 Exp) MVIC decline (1 & 2 Exp)	Supine, knee flexion 45°, trunk-thigh 180° 1. <i>Exp:</i> 25% MVIC to failure (1×) PIMA: < target force for 5s HIMA: >10° angular change for 5s 2. <i>Exp:</i> 25% MVIC until 90% of TTF of HIMA (1×) (young: 848±137s; old:751±83s) Visual feedback	↓ TTF both groups ($p<0.001$, $d=3.40$, -30.3%) HIMA: 770±94s; PIMA: 1105±103s ↔ MVIC decline both groups ($p=0.94$) ↑ MVIC decline after 90% TTF for young ($p=0.02$, $d=1.48$, 42.6%) ↑ MVIC decline after 90% TTF for old ($p=0.017$, $d=2.60$, 33.2%)
Rudroff, Poston, Shin, Bojsen-Møller & Enoka, 2005 [102]	N=8 m 26±5 yrs MVIC: 304±107 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC TTF Fluctuations (SD) force or ACC	Elbow & shoulder flexion 90°, forearm vertical & supinated 20% MVIC to failure (1×) PIMA: >5% force change for 5s HIMA: >10° angle change for 5s corrections allowed Visual & verbal feedback	↓ TTF ($p<0.05$, $d=0.50$, -21.7%) HIMA: 447±276s; PIMA: 609±250s ↑ Fluctuation increase ($p<0.05$, $d=0.62$, 46.1%)
Russ, Ross, Clark & Thomas, 2018 [103]	N=16 (7 m, 9 f) 23.6±1.4 yrs MVIC: -	Trunk extensors (mod. Sorensen test) PIMA: force transducer, counterbalanced load (100% MVIC) HIMA: potentiometer, force transducer, counterbalanced load (85% MVIC) TTF Fluctuations force (CV) or ACC (SD) MVIC decline	Prone, trunk extension 0° 15% MVIC to failure (1×) PIMA: >20% force change for >3s HIMA: >1° angular change for >3s Visual feedback	↑ TTF ($p<0.034$, $d=3.68$, 61.5%) HIMA: 3498±396s; PIMA: 2166±324s ↔ MVIC decline ($p>0.05$, $d=1.57$, 23.2%) ↔ Fluctuations ($p>0.05$)
Schaefer & Bittmann, 2017 [4]	N=10 (5 m, 5 f) m: 24±5 yrs f: 24.4±2 yrs MVIC: m: 31.2±9.8 Nm f: 18.3±2 Nm	elbow extensors pneumatic device incl. force transducer PIMA: push against push rod HIMA: resist push rod TTF (total, longest isometric phase, relation longest/total)	Elbow extension 90°, forearm vertical 80% MVIC for 15s (3×) & to failure (2×) PIMA & HIMA: >1.3° angular change no correction allowed PIMA: verbal feedback HIMA: no feedback	↓ TTF total ($p=0.029$, $d=0.67$, -15.0%) HIMA: 50.0±10.3s; PIMA: 58.8±15.5s ↓ TTF longest ($p=0.005$, $d=1.21$, -53.9%) HIMA: 19.1±7.9s; PIMA: 41.4±24.9s ↓ TTF relation ($p<0.001$, $d=1.23$, -46.8%) HIMA: 0.32±0.14; PIMA: 0.59±0.29
Schaefer & Bittmann, 2021 [5]	N=20 (10 m, 10 f) m: 22.1±2.4 yrs f: 21.6±2.1 yrs	elbow extensors pairwise interaction, force transducer PIMA: push against partner resistance HIMA: resist partner force	Elbow extension 90°, forearm vertical & neutral 80% MVIC of weaker for 15s (3×) 90% MVIC of weaker to failure (2×)	↑ Failure rate ($p<0.001$, effect size $\Phi=0.75$) HIMA: 85%; PIMA: 12.5%

	MVIC: m: 51.2±22.5 Nm f: 25.8±0.07 Nm	Failure rate	HIMA & PIMA: >7° angular change Visual feedback for pushing partner	
Schaefer, Carnarius, Dech & Bittmann, 2023 [20]	N=12 m athletes (6 endurance, 6 strength) 26.1±3.4 yrs MVIC: endurance: 76.1±10.3 Nm strength: 92.7±18.4 Nm	Elbow flexors PIMA: force transducer HIMA: adapt to increasing force at given position (AF; pneumatic device incl ACC) Max torque (total, start, end) Torque decline	Elbow flexion 90°, forearm vertical & neutral 100% MVIC PIMA: 3×pre-HIMA; 2×post-HIMA HIMA (30×): increasing to >100% MVIC; max. HIMA=breaking point from iso to ecc action (<2° change) No termination criteria or feedback	↓ Max torque total ($p<0.001$, $d=1.67$, -32.6%) HIMA: 56.86±16.21 Nm; PIMA: 84.39±16.68 Nm ↓ Max torque start ($p<0.001$, $d=1.96$, -37.5%) ↓ Max torque end ($p<0.001$, $d=2.37$, -49.2%) ↑ Torque decline ($p=0.041$, $d=0.88$, 132.7%)
Schaefer, Dech, Carnarius, Rönnert, Bittmann & Becker, 2024 [18]	N= 9 m Patients knee osteoarthritis: (n=20): 66±9 yrs Controls (n=19): 62±6 yrs MVIC patients: ~both limbs: 0.96±0.3 Nm/kg MVIC controls: ~both limbs: 1.36±0.26 Nm/kg	Knee flexors PIMA: force transducer HIMA: adapt to increasing force at given position (AF; pneumatic device incl ACC) Max torque	Knee flexion 92°, hip flexion 90° 100% MVIC PIMA: 3×pre-HIMA, 2x post-HIMA HIMA (5×): increasing to >100% MVIC; max. HIMA=breaking point from iso to ecc action (<2° change) No termination criteria or feedback	↓ Max torque patients & controls: Patients less affected knee ($p<0.001$, $d=1.19$, -37.5%) Patients more affected knee ($p<0.001$, $d=1.09$, -38.5%) Controls knee 1 ($p<0.001$, $d=2.61$, -42.8%) Controls knee 2 ($p<0.001$, $d=1.08$, -24.6%)
Thomas, Ross, Russ & Clark, 2010 [65]	N=18 (9 m, 9 f) 22.8±0.92 yrs Elbow flexor subgroup: N=4 (2 m, 2 f) MVIC: -	Trunk extensors & elbow flexors PIMA: force transducer HIMA: inertial load, potentiometer, force transducer TTF	Vertical trunk, flexion knee 55° & hip 85° Elbow & shoulder flexion 90° 15% MVIC to failure (1×) PIMA: >10% force change for >3s HIMA: >1° angle change for >3s Visual & verbal feedback	↑ TTF trunk extensors ($p<0.05$, $d=1.82$, 38.1%) HIMA: 1956±336s; PIMA: 1416±252s ↓ TTF elbow flexors ($p<0.05$, $d=2.68$, -35.1%) HIMA: 1122±150s; PIMA: 1728±282s
Williams, Hoffman & Clark, 2014 [105]	N=10 (5 m, 5 f) 24.5±3.1 yrs MVIC: session 1: 276.4±101.7 N session 2: 272.0±102.9 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer TTF MVIC decline RPE	Elbow flexion 90°, forearm horizontal & neutral, shoulder abduction 10-15° 15% MVIC to failure (1×) incl. magnetic & electrical stimuli to motor cortex, cervicomedullary junction & brachial plexus PIMA: >5% force change for >5s HIMA: >10° angle change for >5s corrections allowed Visual & verbal feedback	↑ TTF ($p<0.01$, $d=0.78$, 53.7%) HIMA: 1614±907s; PIMA: 1050±474s ↔ MVIC decline ($p=0.59$, $d=0.001$, 0.07%) ↔ RPE ($p>0.05$, 2.1%)
Abbreviations (alphabetical): ACC=accelerometer/accelerations, AF=Adaptive Force, CV=coefficient of variation, d =Cohen's d effect size, f=female, HIMA=holding isometric muscle action, m=male, min=minutes, MU=motor unit, MVIC=maximal voluntary isometric contraction, N=newtons, Nm=newton meters, PIMA=pushing isometric muscle action, RPE=rating of perceived exertion (scale 0-10), s=seconds, TTF=time to task failure, yrs=years old, °=angular degrees. Numbers are reported as mean±standard deviation. Effect sizes are pairwise. *values were extracted from figures using Plot Digitizer Online App				

3.4.1 Time to task failure

The 23 studies investigating TTF (n=407) showed almost exclusively longer TTFs for PIMA (718±252 s) than HIMA (588±202 s) with a mean percentage difference of $-22.4 \pm 30.2\%$ (range: -56.2 to 61.5%) [4, 33, 62-65, 73, 74, 76, 78, 80, 82-85, 87, 89, 92, 98, 101-103, 105]. As mentioned in methods, five studies reported multiple outcomes [62-65, 92]. Although the single outcomes were based on partially overlapping or similar participants, the methods (high/low intensities, different forearm positions, different muscles) and results differed largely. Thus, a combination of outcomes would result in considerable information loss, including the levelling of the single outcomes. According to Becker [61], and Park and Beretvas [60] dependence can be ignored if only a few studies report multiple outcomes. The sensitivity analysis showed only small differences in Hedges' g of meta-analyses of single vs combined outcomes ($g = -0.74$ vs -0.70). To avoid loss of information, the forest plot (**Figure 5A**) illustrates the 30 single outcomes of 23 studies. The TTF was significantly longer during PIMA vs HIMA with moderate to large effect ($g = -0.74$, 95%CI: -1.18 to -0.30 , $p < 0.001$, $n = 407$). Combining the outcomes gives a similar result ($g = -0.70$, 95%CI: -1.13 to -0.27 , $p < 0.001$, $n = 358$).

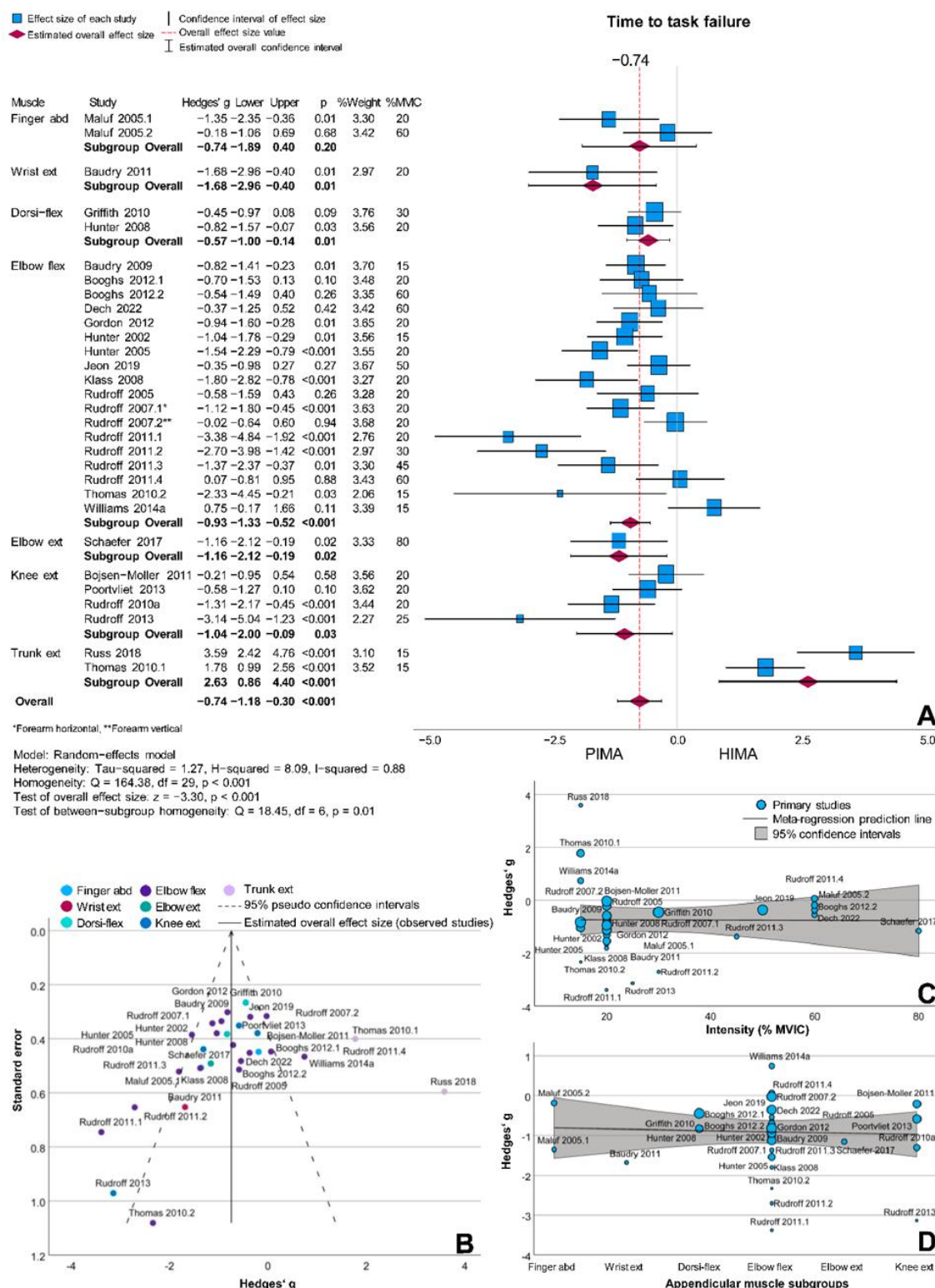


Figure 5. Meta-analysis forest plot (A) comparing time to task failure (TTF) between PIMA and HIMA with funnel plot (B) and bubble chart regressions for loading intensity (C) and appendicular muscle subgroups (D).

Twenty-six of 30 outcomes demonstrated shorter TTF for HIMA vs PIMA with percentage differences ranging from -56.2% to -1.3% ($g=-3.38$ to -0.02). Twenty-four of them showed relevant effect sizes ($g\geq 0.20$), and two were negligible $g<0.20$. The latter occurred for the first dorsal interosseous with intensities of 60% MVIC [92] and for elbow flexors with vertical forearm position at 20% of MVIC [64]. The remaining four of 30 outcomes showed outcomes in the opposite direction, thus longer TTF for HIMA vs PIMA with percentage differences of 1-61% ($g=0.07-3.59$). The main factor regarding TTF is the muscles considered: for appendicular muscles, 26 of 28 outcomes showed higher TTF for PIMA vs HIMA. Only Williams et al. [105] revealed a clear difference with a high effect favoring HIMA over PIMA (54%, $g=0.75$, 15% MVIC, elbow flexors). Rudroff et al. [63] found similar TTFs for HIMA and PIMA at 60% MVIC (%diff: 1%, $g=0.07$, elbow flexors) [105]. Meta-analysis of TTF with only appendicular muscles, including all outcomes, demonstrated a large effect favoring PIMA vs HIMA ($n=373$; $g=-0.90$, 95%CI: -1.18 to -0.62 , $p<0.001$). The two studies [65, 103] examining axial muscles (trunk extensors) had large effects in the opposite direction, thus showing longer TTF for HIMA vs PIMA ($n=34$; $g=2.63$, 95%CI: $0.86-4.40$, $p<0.001$).

The above meta-analyses were highly heterogeneous ($I^2=66\%$ (appendicular), 84% (axial), 85% (combined), 88% (all outcomes)), as confirmed by the funnel plot (**Figure 5B**), supporting the use of the random-effects model. Subgroup analysis and meta-regression bubble plots determined negligible effects of loading intensity (**Figure 5C**) and appendicular muscle size (**Figure 5D**) on TTF. It must be noted that most investigations (23 of 30) were performed at intensities below 40% MVIC [33, 62-65, 73, 76, 80, 83, 84, 89, 92, 98, 101-103, 105]. Five of the remaining seven performed at higher intensities showed moderate to high effect sizes ($g=-1.37$ to -0.35) [4, 62, 63, 78, 87], while the other two were negligible [63, 92].

Related to the TTF is one study by Schaefer and Bittmann [5] in which the ‘failure rate’ during personal interaction with elbow extensors at 90% MVIC was investigated (one partner performed PIMA and the other one HIMA; tasks changed so that each partner performed both HIMA and PIMA). In most trials (85%, $p<0.001$), the partner performing HIMA left the isometric position first. This indicates that the pushing partner could maintain the position for longer, supporting the above results for higher intensities.

3.4.2 Fluctuations of force and position

Twenty-two studies (25 outcomes) investigated the fluctuations of force or position [62-64, 73-76, 80-85, 92, 94-96, 98, 100-103]. The outcome parameters are very heterogeneous. The CV or SD was provided as an indicator of fluctuations. Sixteen studies (18 outcomes) compared the fluctuation increase between HIMA and PIMA [64, 73, 74, 76, 81-85, 94, 95, 100-103]. Of these, ten outcomes showed a significantly higher increase for HIMA vs PIMA ($d=0.62-1.96$), four showed a lower one for HIMA vs PIMA ($d=-5.19$ to -0.63) and four outcomes were non-significant [82, 100]. The other six studies (8 outcomes) examined the fluctuations (CV or SD) over the entire trial period [62, 63, 75, 92, 96, 98]. Of these, six outcomes showed significantly lower fluctuations for HIMA vs PIMA ($d=-3.16$ to -0.53), one revealed significantly higher fluctuations for HIMA vs PIMA ($d=4.05$) [75], and one outcome was non-significant [98].

3.4.3 Maximal torques and strength declines

Only one research group has compared the maximal torques of HIMA (max-HIMA) and PIMA (max-PIMA) [18, 20-23]. Three studies showed clearly significantly lower values in healthy participants for max-HIMA vs max-PIMA (MVIC) for elbow extensors ($d=-0.74$) [23], elbow flexors ($d=-2.37$) [20], and knee flexors ($d=-2.62$) [18] (**Table 2**). Max-HIMA was measured by a pneumatically driven system by which an increasing load was applied on the participant's extremity from low to supramaximal intensities in one trial, whereby their task was to maintain an isometric position for as long as possible. The highest force value under static position stands for max-HIMA. A standard MVIC test measured PIMA by pushing against a stable resistance. Two further studies of the same group investigated max-HIMA during an objectified manual muscle test after different conditionings (similar task as above, but external force was applied by an examiner from low to maximal intensities, not supramaximal) [21, 22]. The maximal torques between HIMA and PIMA were similar at baseline. However, max-HIMA was significantly reduced directly after specific manipulations to shorten muscle spindles. It was not investigated whether max-PIMA would have reacted similarly.

Nineteen studies investigated MVIC decline pre-post fatiguing HIMA or PIMA tasks [33, 62, 63, 73, 74, 76, 80-82, 84, 85, 89, 92, 94, 95, 100, 101, 103, 105]. All 25 outcomes were non-

significant, though wide-ranging ($d=-0.73-1.57$, $-40.2-31.3\%$). Three outcomes (2 studies) reported significance following 90%TTF tasks with larger MVIC declines following HIMA vs PIMA ($p\leq 0.02$, $d=1.48-2.60$, $33.2-50\%$) [33, 74]. Considering the values of relative decline irrespective of the significance and epoch of TTF, 15 reported higher magnitudes of decline for HIMA vs PIMA ($d=0.01-1.57$, $13\pm 10\%$) [62, 63, 73, 76, 81, 89, 92, 100, 103, 105], where 13 found lower declines following HIMA vs PIMA ($d=0.06-0.73$, $-17\pm 12\%$) [62, 63, 74, 80, 82, 83, 85, 94, 95, 101]. One study considered the decline in max-PIMA (=MVIC) and max-HIMA after 30 repeated adaptive HIMA trials (supramaximal), where HIMA showed a 133% higher decline than PIMA [20].

3.4.4 Rating of perceived exertion

Twelve studies investigated the RPE during TTF trials [63, 64, 76, 80, 82-85, 92, 95, 101, 105], with a single study reporting RPE throughout a trial not taken to failure [81]. RPE data was extracted from the figures of several studies [63, 64, 76, 80, 82-85, 92, 101, 105]. Meta-analysis revealed that at the start and the end of TTF the RPE did not differ significantly ($n=194$, $g=-0.02$ to 0.07 , $p=0.43-0.84$) (**Figure 6A**) [63, 64, 76, 80, 82-85, 92, 101, 105]. However, at 50% of TTF the RPE was significantly higher for PIMA than HIMA ($n=164$, $g=-0.31$, 95%CI: -0.55 to -0.07 , $p=0.01$) (**Figure 6A**) [63, 64, 76, 82-85, 92, 101]. The respective bubble plots indicate a positive effect of intensity (**Figure 6B**) and a negative effect of appendicular muscle size (**Figure 6C**). This was vice versa for 0% (clear effect) and 100% TTF (low effect). Considering the significantly shorter TTF for HIMA vs PIMA, the RPE increase rate was higher for HIMA for most studies, indicated by the relative difference (**Figure 7**). This conclusion is further supported by Gould et al. [81], who reported faster increases in RPE for HIMA vs PIMA during contractions over a set time.

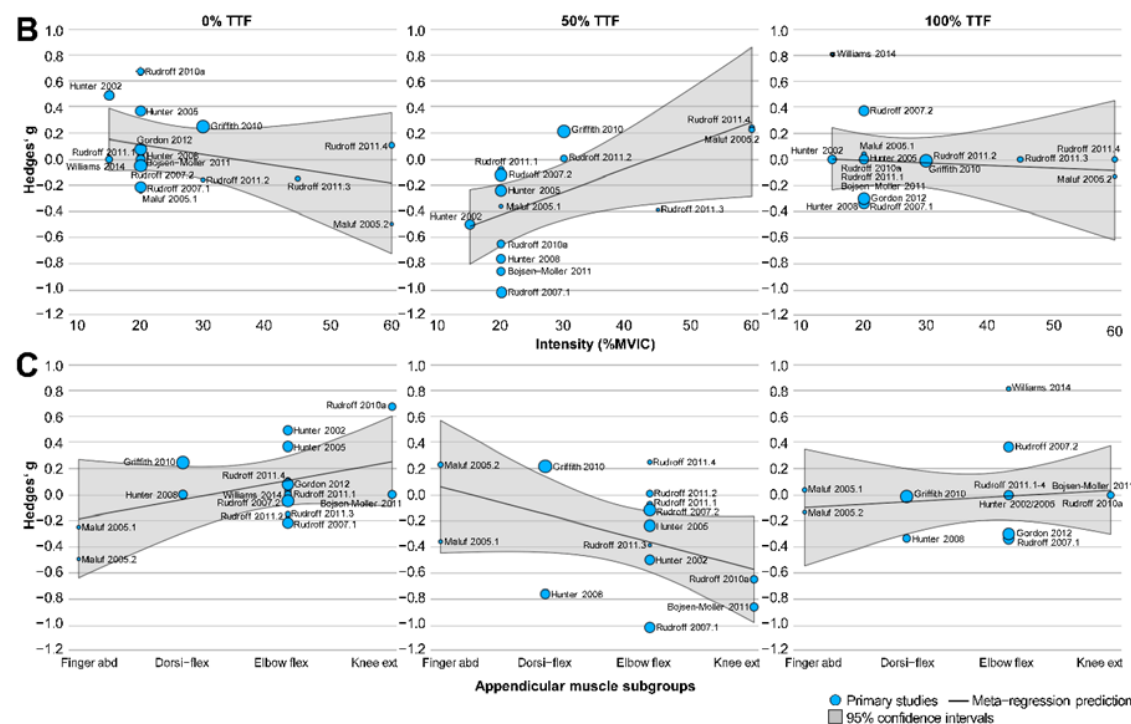
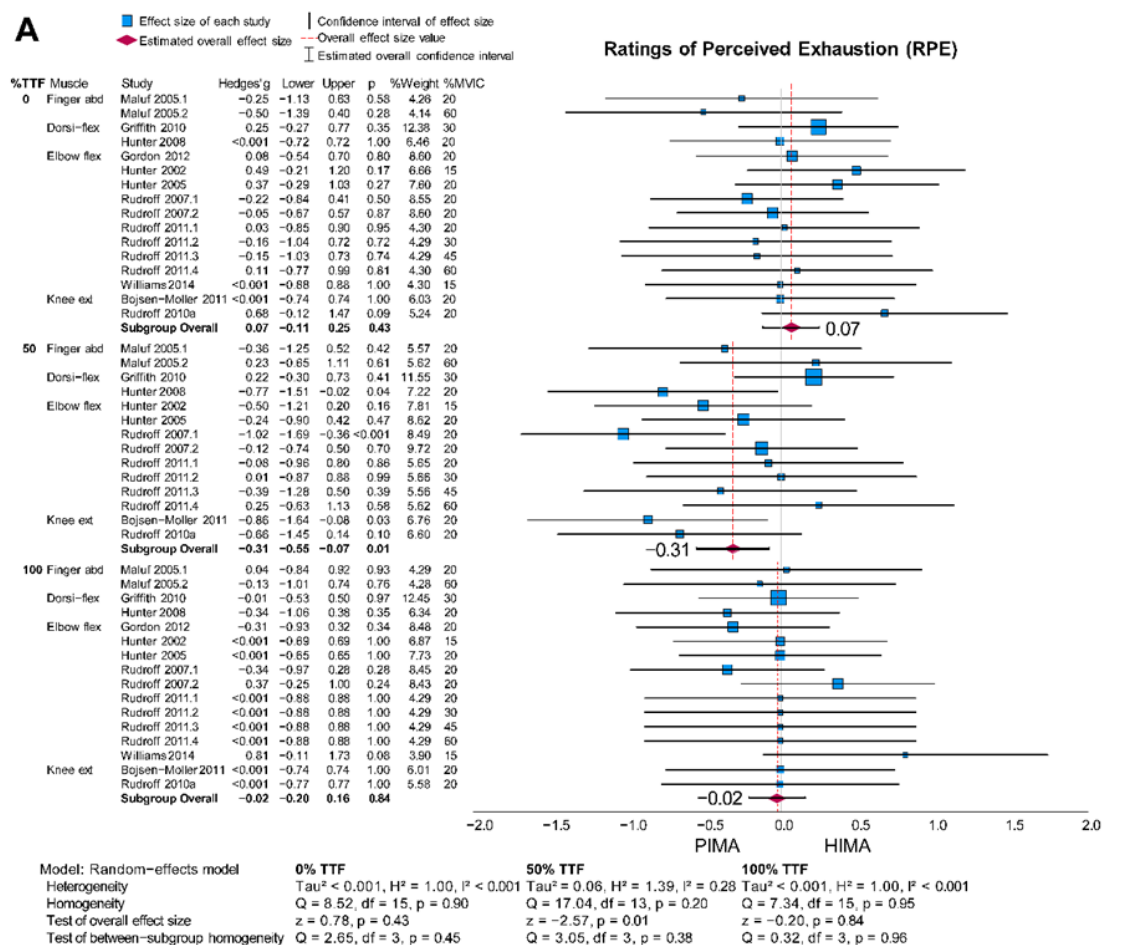


Figure 6. Meta-analysis forest plot (A) comparing ratings of perceived exhaustion (RPE) between PIMA and HIMA with bubble chart regressions for loading intensity (B) and appendicular muscle subgroups (C).

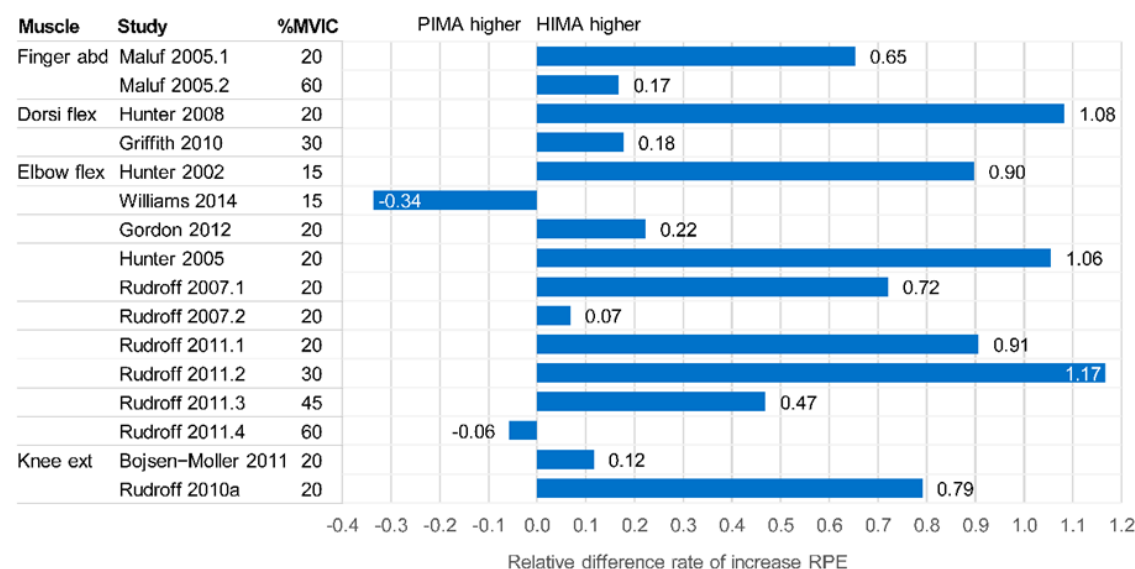


Figure 7. The relative difference in the rate of increased ratings of perceived exertion (RPE) between HIMA and PIMA. The rate of increase was calculated from the start, end and TTF values for PIMA and HIMA, respectively. Relative differences=(HIMA-PIMA)/PIMA.

3.5 Neural and neuromuscular parameters

The 45 studies examining neural or neuromuscular differences between PIMA and HIMA are summarized in **Table 3** [4, 5, 25, 30, 33-35, 62-64, 70-74, 77, 79-105]. Unlike the mechanical (above) and metabolic (below) variables, more than minimal pooling of findings via meta-analyses or otherwise was impossible due to the exceptional variety of collection methods and data processing. Nearly all studies (n=41) employed surface electromyography (sEMG) [30, 33-35, 62-64, 70-74, 77, 79-93, 95-107], while five included needle EMG [34, 94, 95, 100, 104], 12 employed intramuscular EMG [64, 71, 80, 81, 83, 84, 91, 92, 95, 99, 102, 106], three studies used a combination of mechanomyography (MMG) and mechanotendography (MTG), which are biomechanical analyses but reflect oscillatory behavior of motor control [4, 5, 25]. At the same time, three assessed neural function via electroencephalography [25, 96, 97]. Twelve studies included peripheral nerve stimulation [30, 35, 70-73, 77, 88, 89, 91, 105, 107], two used mechanical stimulation [71, 91], while transcranial magnetic stimulation was employed in four

studies [30, 88, 89, 105]. Furthermore, there was a diversity of processing methods and outcomes, including EMG or MMG amplitudes [4, 5, 33, 62-64, 70-74, 79-81, 83, 84, 89, 91, 95, 96, 100-102, 106], EMG root-mean-square (RMS) [77, 82, 85-87, 97-99, 103, 105], parameters of sEMG or MMG frequency [4, 5, 25, 77, 87, 98, 103], burst rates [82-85], burst durations [83, 84], discharge rates [34, 74, 81, 94, 95, 99, 100, 104], interspike interval coefficients of variation (CV) [34, 74, 81, 100], recruitment rates and thresholds [34, 74, 95], H-reflex change [89], heteronymous facilitation [70-73, 88, 91, 107], D1-inhibition [70, 72, 73, 107], D2-inhibition [107], T-reflex amplitude [71], silent periods [30, 88, 89, 105], and coactivation ratios [33, 62, 63, 72, 73, 80, 82, 85, 89, 91, 93, 100, 101, 107], among others. Three studies investigated the coherence of brain activity [25, 96, 97]. Thirty-one studies examined antagonist muscle groups [30, 33, 34, 62-64, 71-74, 80-83, 85-87, 89, 91-93, 95-102, 106, 107], while 34 assessed synergist muscles [4, 5, 25, 30, 33, 62-64, 70-74, 80-85, 88, 91, 92, 95-103, 105]. Finally, four studies had specific focuses, including saline injection-induced experimental joint pain [96, 99], residual force enhancement [93], and skin and core temperature [77].

Table 3. Summary of studies comparing neural and neuromuscular parameters between pushing (PIMA) and holding (HIMA) isometric muscle actions.

↑=sign. larger effect from HIMA vs PIMA; ↓=sign. lower effects from HIMA vs PIMA; ↔=no significant difference between HIMA vs PIMA.

Significance p, effect size Cohen's d and the percentage difference between HIMA and PIMA are given.

Study	Participants	Relevant Measures	Conditions	Results
Baudry & Enoka, 2009 [70]	N=6 Age: - MVIC: -	Wrist adductors (radial deviation) PIMA: - ; HIMA: inertial load sEMG: FCR & ECR Electrical stimulation: radial & median nerves D1 inhibition, heteronymous facilitation aEMG	Position: - 10% MVIC for 35s Termination criteria: - Visual feedback	↓ D1 inhibition FCR & ECR ($p=0.018$, $d=1.15$, -28.45% & $p=0.028$, $d=3.6$, -34.35%) ↑ heteronymous facilitation FCR & ECR ($p=0.03$, $d=0.42$, 11.4% & $p=0.048$, $d=0.21$, 5.2%) ↔ aEMG FCR & ECR ($p>0.05$)
Baudry, Jordan & Enoka, 2009 [71]	<i>H-reflex experiment:</i> N=19 (11 m, 8 f) 25±5.7 yrs <i>T-reflex experiment:</i> N=18 (10 m, 8 f) 25.2±5.8 yrs MVIC: 1.36±0.5 Nm	First dorsal interosseus FDI PIMA: torque transducer, electrical torque motor HIMA: inertial load simulated by torque motor sEMG: FDI & APB imEMG: SPI Electrical stimulation median nerve FDI (n=19) Mechanical stimulation tendon APB (n=18) aEMG Heteronymous reflexes of H-reflex & T-reflex (tendon reflex): SLR & LLR (short & long latency)	Index finger full extension & abduction 0°, shoulder abduction 20°, elbow flexion 95°, forearm horizontal & neutral, thumb abduction 45° 20%, 40% & 60% MVIC (6×) Termination criteria: - Visual feedback	↔ aEMG FDI, APB & SPI ($p>0.45$) ↔ SLR & LLR latency & duration FDI & APB ($p>0.05$) ↑ Heteronymous H-reflex amplitude all intensities ($p<0.01$) ↔ T-reflex amplitude all intensities ($p>0.05$)
Baudry, Maerz & Enoka, 2010 [72]	Young: N=12 (7 m, 5 f) Old: N=12 (5 m, 7 f) [72] 25.9±4.8 yrs 74.0±2.8 yrs MVIC: young: 102.0±26.5 N old: 83.4±20.6 N	Wrist extensors PIMA: servo-controlled torque motor HIMA: inertial load simulated by torque motor, angle sEMG: ECR, FCR, brachioradialis & APB Electrical stimulation: radial & median nerves aEMG Coactivation ratio FCR to ECR D1 inhibition (n=12 young/old), heteronymous Ia facilitation of ECR (n=8 young/old)	Wrist extension 0°, shoulder abduction 74°, elbow flexion 90°, forearm neutral 5%, 10%, 15% MVIC for ~35s Termination criteria: - Feedback: -	↔ aEMG ECR & FCR young & old all intensities ($p>0.05$) ↔ Coactivation ratio young ($p=0.96$) ↑ Coactivation old ($p=0.02$, $d=0.15$, 10.0%) ↓ D1 inhibition young ($p=0.04$, $d=0.71$, -14.3%) ↔ D1 inhibition old ($p=0.63$) ↑ heteronymous Ia facilitation young ($p=0.02$, $d=0.55$, 14.68%) ↔ heteronymous Ia facilitation old ($p=0.89$, $d=0.05$, 0.9%)
Baudry, Maerz, Gould & Enoka, 2011 [73]	N=7 (4 m, 3 f) 19–35 yrs MVIC: Pre-PIMA: 123±41 N Pre-HIMA: 124±46 N	Wrist extensors PIMA: servo-controlled torque motor HIMA: inertial load simulated by torque motor, angle sEMG: ECR, FCR, brachioradialis & APB Electrical stimulation of radial & median nerves aEMG Coactivation ratio FCR to ECR H-reflex, D1 inhibition & heteronymous Ia facilitation of ECR (n=8 young & old)	Wrist extension 0°, shoulder abduction 74°, elbow flexion 90°, forearm pronated 20% MVIC to failure (1×) HIMA: 495±119s vs PIMA: 728±140s* PIMA: >5% force change for >3s HIMA: >11.5° angle change for >3s corrections allowed Visual & verbal feedback	↔ aEMG ECR & FCR at start & end ($p=0.31$ & $p=0.34$) ↔ Coactivation ratio at start & end ($p>0.05$) ↔ aEMG increase brachioradialis ($p>0.05$) ↔ test H-reflex latency, duration & amplitude ($p>0.31$) ↓ D1 inhibition at 2 min ($p=0.042$, $d=1.59$, -33%) ↑ D1 inhibition at end ($p<0.01$, $d=1.57$, 33.4%) ↑ Change of D1 inhibition ($p=0.01$) ↔ Heteronymous Ia facilitation at 2min ($p>0.05$) ↓ Heteronymous Ia facilitation at end ($p=0.049$) ↑ Change of heteronymous Ia facilitation ($p=0.02$)
Baudry, Rudroff, Pierpoint &	N=24 (15 m, 9 f; 4 dropouts) 18–37 yrs	Elbow flexors	Elbow flexion 90°, forearm vertical & supinated	1) TTF: ↔ aEMG elbow flexors start & end ($p>0.05$) ↑ aEMG increase elbow flexors ($p=0.0001$, $d=0.90$, 200%)

Enoka, 2009 [74]	MVIC: Pre-PIMA: 216±73 N Pre-HIMA: 231±85 N	PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer sEMG: elbow flexors (BB, brachioradialis), triceps brachii, deltoid muscle subcutaneous EMG: BB aEMG For BB: RT (recruitment threshold), DT (derecruitment threshold), discharge rate & CV of ISI (interspike intervals)	1) 15% MVIC to failure (1×) HIMA: 1225±666s vs PIMA: 1904±944s* 2) ~25% MVIC (twice RT) for 90% of TTF (1×) PIMA: >5% force change for >3s HIMA: >11.5° angle change for >3s corrections allowed Visual & verbal feedback	↔ aEMG triceps start, end & increase ($p=0.45$) ↔ aEMG deltoid start, end & increase ($p=0.68$) 2) 90% TTF: ↑ aEMG end & increase elbow flexors ($p=0.002$ & $p=0.005$) ↔ RT & DT before tasks ($p=0.97$) ↑ relative decline RT & DT ($p<0.05$) ↓ discharge rate at end ($p=0.02$) ↔ derecruitment all parameters ($p>0.05$) ↑ increase CV ISI ($d=2.31$, 210.9%)
Booghs, Baudry, Enoka & Duchateau, 2012 [62]	N=15 (8 m, 7 f) 18–36 yrs MVIC: Pre-PIMA: 20% task: 293±82 N 60% task: 299±119 N Pre-HIMA: 20% task: 296±86 N 60% task: 292±111 N	Elbow flexors PIMA: force transducer HIMA: inertial load, potentiometer sEMG: BB, brachioradialis, triceps brachii, trapezius aEMG Coactivation ratio elbow flexors to elbow extensor	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVIC to failure (n=12) (1×) HIMA: 404±159s; PIMA: 533±194s* 60% MVIC to failure (n=9) (1×) HIMA: 54±19s; PIMA: 64±16s PIMA: >2% or 5% force change for 3s HIMA: >1.5° or 3° angle change for 3s corrections allowed Visual & verbal feedback	↔ aEMG all muscles 20% & 60% tasks ($p>0.05$) ↔ Coactivation ratio 20% & 60% tasks ($p\geq 0.50$)
Buchanan & Lloyd, 1995 [106]	N=9 Age: - MVIC: -	Elbow flexors & extensors PIMA: force transducer HIMA: inertial load on a pulley sEMG: BB, brachioradialis, triceps brachii (medial & lateral) imEMG: brachialis aEMG	Elbow flexion & extension 90°, neutral forearm, shoulder abduction 90° 5, 10, or 15 lbs (highest amount = ~32±10% of MVIC for elbow flexors & ~39±11% of MVIC for elbow extensors) (5x per intensity & tasks) Termination criteria: - Visual feedback	Significant differences in activation of elbow flexors ($p<0.05$): In 6 of 9 subjects (no p -values): ↑ aEMG BB ↓ aEMG brachialis & brachioradialis In 3 of 9 subjects (no p -values): ↓ aEMG BB ↑ aEMG brachialis & brachioradialis Elbow extensors: ↑ aEMG triceps brachii ($p<0.05$)
Coletta, Mallette, Gabriel, Tyler & Cheung, 2018 [77]	N=20 (13 m, 7 f) 23.8±2.1 yrs MVIC: 28.5±11.2 Nm	Wrist flexors PIMA: force transducer, potentiometer HIMA: isoinertial pulley & potentiometer passive heating & cooling procedure (thermistor for body (rectal) & skin temperature) sEMG: FCR Electrical stimulation: median nerve RMS MPF MDF	Wrist flexion 0°, elbow extension 135° 60% MVIC 1) 3s per 0.5 °C body temp change (several sets) 2) 1min for pre, hot, cool, post (4×, n=18) Termination criteria: - Visual feedback	3s-contractions: ↑ RMS pre, +0.5 °C, -0.5 °C, hot body/cold skin, post ($p<0.05$) ↔ RMS at +1°C, hot body/hot skin ↔ MPF & MDF all conditions ($p>0.05$) 1min-contractions: ↑ RMS at pre ($p=0.038$) ↔ RMS at hot, cool & post ($p>0.05$)
Garner, Blackburn, Wiemar & Campbell, 2008 [79]	N=20 (10 m, 10 f) 22.5±2.7 yrs MVIC: -	Plantar flexors PIMA: force plate HIMA: force plate & inertial load sEMG: soleus aEMG	Plantar flexion 90°, hip & knee flexion 90° 20%, 30%, 40%, 50% MVIC for 3s (5×) Termination criteria: - Visual & verbal feedback	↔ aEMG all intensities ($p=0.386$)

Gordon, Rudroff, Enoka & Enoka, 2012 [80]	N=20 (15 m, 5 f) 23±4 yrs MVIC: Pre-PIMA: 273±90 N Pre-HIMA: 291±93 N	Elbow flexors (both sides) PIMA: force transducer HIMA: nertial load, electrogoniometer, force transducer sEMG: BB, triceps brachii, brachioradialis, deltoid imEMG: brachialis aEMG Coactivation ratios elbow extensors to flexors	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVIC to failure (1×) HIMA: 253±103s; PIMA: 367±133s* PIMA: >5% force change for 5s HIMA: >11.5° angle change for 5s corrections allowed Visual & verbal feedback	↔ aEMG all muscles ($p>0.05$) ↔ Coactivation ratio ($p>0.05$)
Gould, Cleland, Mani, Amiridis & Enoka, 2016 [81]	N=21 (13 m, 8 f) 21.9±1.9 yrs MVIC: 252±89 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer sEMG: BB, brachioradialis & triceps brachii imEMG: brachialis, BB aEMG MU discharge rate CV of ISI	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 24.9±10.5% MVIC (~4.7±2% above recruitment threshold) for 152±84s (1×) No termination criteria Visual feedback	↔ aEMG increase for agonists & antagonists ($p>0.05$) ↔ MU discharge rate at start ($p>0.05$) ↓ MU discharge rate at end ($p<0.05$, $d=0.75$, -15.9%) ↑ Decline MU discharge rate ($p=0.004$, $d=0.55$, 60%) ↔ CV ISI at start ($p>0.05$, $d=0.14$, -4.5) ↑ CV ISI at end ($p<0.05$, $d=0.57$, 23.9%) ↑ Relative change CV ISI ($p=0.008$, $d=0.65$, 625%)
Griffith, Yoon & Hunter, 2010 [82]	Young: n=17 (8 m, 9 f) Old: n=12 (7 m, 5 f) 23.6±6.5 yrs 70.0±5.0 yrs MVIC young: pre-PIMA: 38.0±10.2 Nm pre-HIMA: 37.4±9.4 Nm MVIC old: pre-PIMA: 35.5±8.9 Nm pre-HIMA: 34.9±10.4 Nm	Dorsi-flexors PIMA: force transducer HIMA: inertial load, potentiometer, ACC sEMG: tibialis anterior, gastrocnemius, soleus, RF EMG burst rate RMS Coactivation ratio	Dorsi-flexion 0°, hip & knee flexion 90° 30% MVIC to failure (1×) HIMA: 561±204s; PIMA: 624±270s* PIMA: >5% force drop for 4s HIMA: >18° angle drop corrections allowed Visual & verbal feedback	↑ RMS tibialis ($p<0.001$, $d=0.58$, 25%) ↑ RMS increase rate tibialis ($p<0.01$, $d=0.46$, 47.4%) ↑ RMS increase rate gastrocnemius ($p=0.004$, $d=0.53$, 82.4%) ↑ RMS increase rate soleus ($p=0.002$) ↑ RMS increase rate RF ($p=0.014$) ↑ EMG burst rate increase tibialis ($p=0.007$, $d=0.55$, 40.7%) ↑ Coactivation increase gastrocnemius ($p=0.01$, 327.8%) ↑ Coactivation increase soleus ($p=0.01$, 235.4%)
Hunter, Rochette, Critchlow & Enoka, 2005 [83]	N=18 (10 m, 8 f) 72±4 yrs MVIC: Pre-PIMA: 180±55 N Pre-HIMA: 178±61 N	Elbow flexors PIMA: force transducers HIMA: inertial load, electrogoniometer, ACC sEMG: elbow flexors (BB, brachioradialis), triceps brachii imEMG: brachialis aEMG EMG burst rate & duration	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVIC to failure (1×) HIMA: 636±366s; PIMA: 1368±546s* PIMA: >10% force drop for >5s HIMA: >26° angle change for >5s corrections allowed Visual & verbal feedback	↔ aEMG elbow flexors at start ($p>0.05$) ↓ aEMG elbow flexor end ($p<0.05$, $d=0.85$, -38.1%) ↔ aEMG increase rate (overall) elbow flexors ($p>0.05$) ↓ aEMG increase rate at 25 & 50% TTF elbow flexors ($p<0.05$) ↔ aEMG & increase rate triceps ($p>0.05$) ↑ Burst rate elbow flexors ($p<0.05$, $d=0.20$, 44.2%) ↔ Burst duration elbow flexors ($p>0.05$) ↑ Burst rate increase rate elbow flexors ($p<0.05$) ↔ Burst rate triceps ($p>0.05$)
Hunter, Ryan, Ortega & Enoka, 2002 [84]	N=16 (8 m, 8 f) 27±4 yrs MVIC: Pre-PIMA: 308±151 N Pre-HIMA: 307±152 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC sEMG: elbow flexors (BB, brachioradialis), deltoid imEMG: brachialis aEMG EMG burst rate & duration	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 15% MVIC to failure (1×) HIMA: 702±582s; PIMA: 1402±728s* PIMA: >10% force drop for >5s HIMA: >10° angular change for >5s corrections allowed Visual & verbal feedback	↔ aEMG elbow flexors at start ($p>0.05$) ↓ aEMG elbow flexor at end ($p<0.05$, $d=6.79$, -33.5%) ↔ aEMG increase rate elbow flexors ($p>0.05$) ↔ aEMG deltoid at start & end ($p>0.05$) ↑ aEMG increase rate deltoid ($p<0.05$) ↔ Burst rate & duration elbow flexors ($p>0.05$) ↑ Burst rate brachialis ($p<0.05$, $d=0.69$, 210.5%)

Hunter, Yoon, Farinella, Griffith & Ng, 2008 [85]	N=15 (8 m, 7 f) 21.1±1.4 yrs MVIC: pre-PIMA: 333±71 N pre-HIMA: 334±65 N	Dorsi-flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC sEMG: tibialis anterior, gastrocnemius medialis, VL, RF RMS Coactivation ratio gastrocnemius to tibialis Burst rate tibialis anterior	Dorsi-flexion 0°, hip & knee flexion 90° 20% MVIC to failure (1×) HIMA: 600±372s; PIMA: 1278±1068s* PIMA: >5% force drop for 4s HIMA: >18° angle drop for 4s corrections allowed Visual & verbal feedback	↔ RMS tibialis at start & VL ($p>0.05$) ↓ RMS RF ($p=0.002$, $d=0.46$, -21.6%) ↑ RMS increase & change rate tibialis ($p=0.01$, $d=0.99$, 3567% & $p=0.01$, $d=1.09$, 470%) ↑ RMS increase rate gastrocnemius ($p=0.024$, $d=0.82$, 600%) ↑ RMS increase rate VL ($p=0.025$, $d=0.43$, 81%) ↔ RMS increase & increase rate RF ($p>0.05$) ↔ Coactivation ($p=0.16$) ↔ Burst rate tibialis ($p>0.05$) ↑ Burst rate increase rate tibialis ($p=0.024$, $d=0.61$, 35.7%)
Jeon, Miller & Ye, 2020 [86]	N=19 m 23.7±3.9 yrs MVIC: pre-PIMA: 299±134 N pre-HIMA: 299±132 N	Elbow flexors PIMA: force transducer HIMA: inertial load, steel hinge (visual control) sEMG: BB, triceps brachii RMS (n=19) RT Slopes (regr. lines) BB: RT vs MFR (n=17) & RT vs DT (n=14)	Elbow flexion 135°, forearm horizontal & supinated 40% & 70% MVIC (trapezoid: 4s-10s-4s) (2×) Termination criteria: -; corrections allowed visual (PIMA) & verbal feedback (both)	↔ RMS BB & triceps both intensities ($p>0.05$) ↔ RT BB & triceps both intensities ($p>0.05$) ↑ Slope RT vs MFR BB (intensities combined) ($p=0.010$, $d=0.59$, -31.3%) ↑ Slope RT vs DT BB (intensities combined) ($p=0.023$, $d=0.48$, 25%)
Jeon, Ye & Miller, 2019 [87]	N=20 (12 m, 8 f) m: 24±4 yrs f: 22±3 yrs MVIC: m: 382±102 N f: 189±29 N	Elbow flexors (dominant side) PIMA: force transducer HIMA: inertial load, steel hinge (visual control) sEMG: BB, triceps brachii RMS MF slope of amplitude & frequency over time	Elbow flexion 135°, forearm horizontal & supinated, non-dominant hand on abdomen 50% MVIC to failure (1×) HIMA: 33.9±14.9s; PIMA: 39.6±16.6s* PIMA: < target force for 3s HIMA: < target position for 3s corrections allowed Visual (PIMA) & verbal feedback (both)	↔ RMS BB & triceps ($p=0.529$ & $p=0.935$) ↔ slope RMS BB ↑ slope RMS triceps ($d=0.40$, 63.0%) ↔ MF BB ($p=0.169$) ↑ MF triceps at start & middle ($p=0.009$ & $p=0.044$) ↑ slope MF BB ($d=0.71$, 60.1%) ↔ slope MF TB
Kirimoto, Tamaki, Suzuki, Matsumoto, Sugawara, Kojima & Onishi, 2014 [88]	N=10 (9 m, 1 f) 20–38 yrs MVIC: 2.0±0.4 Nm	First dorsal interosseus PIMA: force transducer HIMA: inertial load, wire-type displacement meter sEMG: FDI, APB TMS: left M1 area Electrical stimulation: ulnar & median nerve SEP MEP cSP Heteronymous reflexes (SLR & LLR)	Index finger abduction 10°, full finger extension, shoulder abduction 10–20°, elbow flexion 110°; forearm neutral, thumb 45° abduction 3 blocks of 20% MVIC for 40–50s (2×) Termination criteria: - Visual feedback	↓ SEP amplitude of P45 ulnar stimulation ($p=0.027$, $d=0.80$, -10%) ↔ SEP amplitude of P45 median stimulation ($p=1.0$) ↔ MEP amplitude ($p>0.255$) ↓ cSP ($p=0.013$, $d=0.52$, -10.7%) ↑ Heteronymous SLR ($p<0.001$, $d=1.32$, 34.6%) ↑ Heteronymous LLT ($p=0.018$, $d=0.60$, 9.9%)
Klass, Levenez, Enoka & Duchateau, 2008 [89]	N=11 (6 m, 5 f) 29.4±6 yrs MVIC: pre-PIMA: 257±81 N pre-HIMA: 271±99 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer sEMG: BB, triceps brachii Electrical stimulation: brachial plexus TMS: left motor cortex aEMG MEP	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVC to failure (1×) HIMA: 420±165s; PIMA: 958±371s* PIMA: < target force for 5–10s HIMA: >10° angle drop for 5–10s corrections allowed	↔ aEMG BB & triceps at start ($p=0.30$ & $p=0.96$) ↓ aEMG BB & triceps at end ($p<0.001$, $d=0.73$, -38.5% & $p<0.05$) ↔ aEMG increase rate BB & triceps ($p>0.05$) ↔ MEP BB & triceps ($p=0.21$ & $p=0.90$) ↓ MEP increase BB ($p>0.001$, $d=1.17$, -35.6%) ↔ MEP increase triceps ($p>0.05$)

		cSP M _{max} H-reflex (n=6)	Visual & verbal feedback	↔ MEP increase rate BB ($p>0.05$) ↑ MEP increase rate triceps ($p<0.05$, $d=3.02$, 86.1%) ↔ cSP start, end & increase ($p>0.05$) ↓ M _{max} decline BB ($d=0.38$, -41.9%) ↔ H-reflex at start ($p=0.65$) ↑ H-reflex change ($p<0.01$, $d=1.35$, 72%)
Kunugi, Holobar, Koder, Toyoda & Watanabe, 2021 [90]	N=12 (10 m, 2 f) 24.8±6.9 yrs MVIC: 75.6±17.5 Nm	Plantar flexors PIMA: torque transducer, angle sensor HIMA: inertial load pully system, angle sensor sEMG: gastrocnemius MFR changes between 20% & 30% MVIC	Plantarflexion 20°, full knee extension, back flexion 20° 20% & 30% MVIC for 15s (2×); ramp contractions (15s-15s) Termination criteria: - Visual feedback	↔ MFR change & CV MFR change ($p=0.59$ & $p=0.26$)
Magalhães, Elias, da Silva, de Lima, de Toledo & Kohn, 2015 [107]	N=10 (5 m, 5 f) 27.9±7.8 yrs MVIC: -	Plantar flexors PIMA: force transducer HIMA: inertial load, force transducer, ACC sEMG: soleus, gastrocnemius, tibialis, VL, semitendinosus Electrical stimulation: posterior tibial nerve (test stimuli), peroneal nerve (conditioning stimuli D1&D2), femoral nerve (heteronymous reflex) D1 & D2 inhibition (amount of inhibition) (n=10)	Plantar flexion (90°), full knee extension, hip ~120° PIMA: 10% MVIC HIMA: EMG level of PIMA 7x57s Termination criteria: EMG RMS 2x higher than in rest Visual feedback: PIMA: >±1% target force HIMA: as close as possible to 90°	↓ amount of D1 inhibition ($p=0.001$, $d=1.56$) ↔ amount of D2 inhibition ($p=0.078$)
Maluf, Barry, Riley & Enoka, 2007 [91]	N=12 27±9 yrs MVIC: pre-PIMA: 30.2±6.9 N pre-HIMA: 29.6±1.8 N	First dorsal interosseus PIMA: force transducer HIMA: inertial load, potentiometer, ACC sEMG: FDI, APB, extensor digitorum, BB imEMG: SPI Electrical stimulation: median nerve (n=10) Mechanical stimulation (stretch reflex, n=12) Tonic activation (relative) heteronymous SLR & LLR	Index finger abduction 0°, shoulder abduction 45°, elbow flexion 90°, forearm neutral, thumb full extension 20% MVIC for 40s (6×) Stimulation only when: PIMA: < ±5% target force HIMA: < ±2° target angle Visual & verbal feedback	↑ SLR electrical stimulation ($p=0.04$, $d=0.50$, 26.7%) ↑ LLR electrical stimulation ($p=0.02$, $d=0.58$, 28.1%) ↔ SLR & LLR mechanical stimulation ($p=0.69$) ↔ Tonic activation all muscles ($p>0.27$)
Maluf, Shinohara, Stephenson & Enoka, 2005 [92]	N=20 m (2×n=10) 23±5 yrs MVIC: Low force group: Pre-PIMA: 34.8±7.5 N Pre-HIMA: 33.8±6.7 N High force group: Pre-PIMA: 32.5±4.0 N Pre-HIMA: 32.1±3.9 N	First dorsal interosseus PIMA: force transducer HIMA: inertial load, potentiometer, ACC sEMG: FDI, extrinsic finger flexors & extensors, BB imEMG: SPI (n=5 per group) EMG activity, EMG increase EMG slope (rate of increase in RMS amplitude)	Index finger abduction 0°, shoulder abduction 45°, elbow flexion 90°, forearm neutral, thumb full extension 20% or 60% MVIC to failure (1×) 20%: HIMA: 593±212s; PIMA: 938±328s* 60%: HIMA: 86±31s; PIMA: 93±41s PIMA: >1.5% force change for 3s HIMA: >10° angle change for 3s corrections allowed Visual & verbal feedback	↑ EMG increase FDI 20% task ($p<0.05$) ↔ EMG increase SPI 20% task ($p>0.05$) ↑ EMG slope FDI 20% task ($p=0.002$), not for other muscles ↔ EMG increase for FDI & SPI 60% task ($p>0.05$) ↔ EMG slope all muscles 60% task ($p>0.05$)

Marion & Power, 2020 [93]	N=12 (6 m, 6 f) 22.8±1.1 yrs MVIC: m: 24.6±7.5 Nm f: 17.3±2 Nm	Dorsi-flexors PIMA: isometric dynamometer HIMA: isoinertial load, angle sEMG: tibialis anterior & soleus Activation reduction Neuromuscular economy (torque per unit RMS during residual force enhancement (rTE) task) Coactivation soleus	Dorsi-flexions 130°, hip flexion 110°, knee flexion 130° 60% MVIC PIMA & HIMA: 10s (1×) followed by rTE task: active lengthening for 3s (90° to 130°) & isometric for 5s < ±5% of target force or angle Visual feedback	↔ Activation reduction ($p=0.743$) ↔ Neuromuscular economy after lengthening ($p=0.971$) ↔ Coactivation ($p=0.591$)
Mathis, de Quervain & Hess, 1999 [30]	N=10 (8 m, 2 f) 23-39 yrs MVIC: -	Elbow flexors PIMA: force transducer, oscilloscope HIMA: inertial load, potentiometer & oscilloscope sEMG: BB, brachioradialis, triceps brachii TMS: 3% above threshold, 50%, 60%, 80% & 100% of max stimulator output (3× every 10s) Peripheral magnetic nerve stimulation (BB, n=4) sP CT MEP amplitude & latency	Elbow flexion 90°, slight shoulder abduction, forearm semipronated 5%, 10% & 20% MVIC duration: - Termination criteria: - Visual feedback	↑ SP BB 50% TMS intensity (5% MVIC: $p<0.01$, $d=1.00$, 24.9%; 10% MVIC: $p=0.002$, $d=0.49$, 13.8%; 20% MVIC: $p<0.01$, $d=0.28$, 8.2%) ↑ SP BB 60% TMS intensity (5% MVIC: $p<0.01$, $d=0.71$, 17.3%; 10% MVIC: $p=0.004$, $d=0.43$, 10.7%; 20% MVIC: $p<0.01$, $d=0.30$, 8.9%) ↑ SP brachioradialis 50% TMS intensity (5% MVIC: $p\leq 0.05$, $d=0.59$, 21.1%; 10% MVIC: $p\leq 0.05$, $d=0.50$, 15.8%; 20% MVIC: $p\leq 0.05$, $d=0.26$, 8.3%) ↑ SP brachioradialis 60% TMS intensity (5% MVIC: $p\leq 0.05$, $d=0.65$, 16.8%; 10% MVIC: $p\leq 0.05$, $d=0.77$, 21.3%; 20% MVIC: $p\leq 0.05$, $d=0.28$, 8.1%) ↔ SP elbow flexors TMS 3% >threshold, 80% & 100% ($p>0.05$) ↑ CT all forces & TMS intensities ($p<0.05$, $d=2.34$ -16.2, 68.7-121.9%) ↔ SP elbow flexors after peripheral stimulation ($p>0.05$) ↔ MEP amplitude & latency ($p>0.05$)
Mottram, Christou, Meyer & Enoka, 2005 [94]	N=15 m 25.5±5.9 yrs MVIC: 267±48 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC nEMG: BB Discharge rate: CV PSD (power spectral density)	Elbow flexion 90°, forearm horizontal & neutral, shoulder abduction 15° 22.4±14% MVIC (3.6±2.1% above RT for 161±93s (1×)) Termination criteria: - Visual feedback	↔ CV discharge rate ($p=0.10$) ↓ PSD at start ($p=0.05$) ↔ PSD at end ($p>0.05$) ↑ PSD %-change over time ($p<0.03$)
Mottram, Jakobi, Semmler & Enoka, 2005 [95]	N=15 m 25.6±5.8 yrs MVIC: 265±50 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC sEMG: BB, triceps brachii, upper trapezius imEMG: brachialis nEMG: BB aEMG all muscles For BB: discharge rate (M, CV), MU recruitment (number of newly recruited MUs), time of recruitment & derecruitment	Elbow flexion 90°, forearm horizontal & neutral, shoulder abduction 15° low threshold MU: 13.4±7.6% MVIC, 222±66s moderate thresh. MU: 37.0±5.4% MVIC, 59±4s (1×) Termination criteria: - Visual feedback	↔ aEMG all muscles ($p>0.05$) ↔ Discharge rate all MUs at start ($p=0.56$) ↓ Discharge rate all MUs at middle & end ($p=0.02$, $d=0.52$, -10.5% & $p=0.001$, $d=0.54$, -11.7%) ↑ Decline discharge rate all MUs ($p<0.03$) ↔ Discharge rate low MUs at start ($p=0.29$) ↓ Discharge rate low MUs at end ($p=0.03$, $d=0.37$, -9.7%) ↓ Discharge rate mod. MUs overall ($p=0.02$, $d=0.74$, -13.0%) ↔ CV all MUs, low & moderate MUs at start ($p>0.05$) ↑ CV all MUs at end ($p=0.01$, $d=0.49$, 19.2%)

				↑ CV low MUs at end ($p=0.01$, $d=0.54$, 18.0%) ↑ CV moderate MUs overall ($p=0.02$, $d=0.44$, 16.5%) ↑ MU recruitment ($p=0.01$, $d=0.53$, 40%) ↔ Recruitment & derecruitment times ($p=0.87$)
Pascoe, Gould, Enoka, 2013 [34]	Young: N=16 (13 m, 3 f) Old: N=14 (12 m, 2 f) 28.0±3.8 yrs 75.1±3.9 yrs MVIC young: 280±91 N old: 200±67 N Comparison HIMA with previous data of PIMA: Riley et al. (2008): Young: n=18 (16 m, 2 f) 25.5 ± 6.2 yrs Pascoe et al. (2011): Old: n=11 (8 m, 3 f) 78.8 ± 5.9 yrs	Elbow flexors PIMA: force transducer HIMA: inertial load, force transducer, electrogoniometer sEMG: BB, triceps brachii nEMG: BB short head time to recruitment ISI (discharge rate & amp; CV)	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction Less than recruitment threshold (1×) HIMA young: large target force diff: 11.6±5.1% MVIC for 295±195s small target force diff: 17.5±6.7% MVIC for 138±21s PIMA young: large: 22.3±10% MVIC (no duration given) small: 27.1±10% MVIC (no duration given) HIMA old: large: 9.24±7.2% MVIC for 325±266s small: 15.5±9.3% MVIC for 185±219s PIMA old: large: 13.5±7% MVIC for 223±147s small: 18.4±7.9% MVIC for 84.4±29s Termination: discharged action potentials for ~120s Riley et al: discharged action potentials for ~60s or force fluctuations >4% Pascoe et al. 2011: discharged action potentials for ~60s Visual feedback (not stated in Riley et al.)	↑ time to recruitment overall ($p=0.049$, $d=0.31$, 48.1%) ↓ mean discharge rate young ($p<0.05$, $d=1.1$, -28%) ↔ mean discharge rate old ($p>0.05$, 4.8%) ↓ CV ISI for young with large target force difference ($p=0.002$, $d=0.64$, -20.3%) ↑ CV ISI for young with small target force difference ($d=0.80$, 46%) ↔ CV ISI old for both target force differences ($p>0.05$, large: $d=0.03$, -1.3%, small: $d=0.42$, -20.2%)
Poortvliet, Tucker, Finnigan, Scott & Hodges, 2019 [96]	N=17 (14 m, 3 f) 33±6 yrs MVIC: Pre-PIMA: 461±148 N Pre-HIMA: 460±149 N	Knee extensors (right) PIMA: force transducer HIMA: inertial load, electronic inclinometer sEMG: RF, VL, VM, semitendinosus & BF EEG CMC EEG power spectra aEMG Experimental pain (hypertonic saline injection to infrapatellar fat pad)	Supine, knee & hip flexion 90° 10% MVIC for 30s (3× without & 3× with pain) PIMA/HIMA: as close as possible to target value Visual & verbal feedback	↔ CMC beta & gamma band ($p=0.292$ & $p=0.867$) ↔ EEG power without pain (beta: $p=0.68$, gamma: $p=0.77$) ↔ aEMG ($p=0.067$) Applies only for PIMA, not for HIMA (no statistics): Significantly lower CMC in beta band for pain vs no pain & EEG power decrease with pain in beta & increase in gamma band
Poortvliet, Tucker, Finnigan, Scott, Sowman & Hodges, 2015 [97]	N=17 (14 m, 3 f) 33±6 yrs MVIC: ~463 N	Knee extensors PIMA: force transducer HIMA: inertial load & electronic inclinometer sEMG: RF, VL, VM, semitendinosus & BF EEG CMC CCC	Supine knee extension 90°, hip flexion 90° 10% MVIC for 30s (6×) PIMA/HIMA: as close as possible to target value Visual & verbal feedback	↔ RMS all muscles ($p=0.84$) ↔ CMC ($p=0.27$) ↓ CCC left hemisphere beta band ($p<0.001$, $d=0.09$, -5%) ↔ CCC left hemisphere gamma band ($p=0.106$) ↓ CCC right hemisphere beta ($p<0.001$, $d=0.26$, -13.8%) ↓ CCC right hemisphere gamma ($p<0.001$, $d=0.24$, -14.2%) ↓ CCC inter-hemispheres beta & gamma ($p<0.001$; beta: $d=0.27$, -14.5%; gamma: $d=0.20$, -11.6%)

		EEG power spectra RMS		↑ EEG power left hemisphere beta ($p<0.05$, $d=0.11$, 0.68%) ↔ EEG power left hemisphere gamma ($p=0.17$) ↔ EEG power right hemi beta & gamma ($p=0.49$ & $p=0.15$)
Poortvliet, Tucker & Hodges, 2013 [98]	N=17 (9 m, 8 f) 32±7 yrs MVIC: 444±175 N	Knee extensors PIMA: force transducer (strain-gauge) HIMA: inertial load, electric inclinometer sEMG: prime mover (VL & VM), auxiliary muscles (tensor fascia latae, BF, semitendinosus) RMS & MDF at start, end & shortest (final 10s pre-failure for task with shortest TTF)	Supine, knee & hip flexion 90° 20% MVIC to failure (1×) PIMA: >5% force change for 5s HIMA: >5° angle change for 5s corrections allowed Visual & verbal feedback	↔ RMS & MDF prime mover ($p>0.05$) ↓ RMS of auxiliary muscles ($p<0.03$) ↔ MDF auxiliary muscles at start ($p=0.75$) ↓ MDF auxiliary muscles at shortest ($p=0.001$, $d=0.72$, -19.3%) ↓ MDF auxiliary muscles at end ($p<0.01$, $d=0.60$, -17.3%)
Poortvliet, Tucker & Hodges, 2015 [99]	N=13 (8 m, 5 f) 31±6 yrs MVIC: -	Knee extensors PIMA: force transducer HIMA: inertial load & electronic inclinometer sEMG: tensor fasciae latae, BF & semitendinosus imEMG: VL & VM RMS, discharge rate (M & SD) & proportion of MU with discharge rate change >10% with pain for (1) 35 SMU discharged across all conditions (n=5) (2) 189 overall identified SMUs (n=11) Experimental pain (hypertonic saline injection to infrapatellar fat pad)	Supine knee extension 90°, hip flexion 90° Intensity to activate 4-7 SMUs (~11N) for 30s (3× without & 3× with pain; n=5 in one session, n=6 two separate sessions) PIMA/HIMA: as close as possible to target value Visual & verbal feedback	(1) 35 SMU discharged across all conditions (n=5) ↔ Discharge rate without pain ($p=0.90$) ↓ Decline discharge rate with pain ($p<0.01$) ↓ Proportion SMU discharge rate change pain ($p<0.05$, -62.6%) ↓ SD discharge rate without pain ($p=0.05$, $d=0.20$, -10%); Note: SD discharge rate with vs no pain sign. lower for PIMA, did not change for HIMA. ↔ RMS EMG VL & VM without pain ($p>0.086$, n=5) (2) 189 SMU (all n=11): ↑ Discharge rate in general ($p=0.032$) ↑ Discharge rate in pain (post hoc: $p = 0.011$) ↔ Decline discharge rate no-pain to pain ($p = 0.052$) ↔ %change discharge rate no-pain to pain ($p = 0.096$)
Rudroff, Barry, Stone, Barry & Enoka, 2007 [64]	N=20 m 27±5 yrs MVIC horizontal: Pre-PIMA: 309±45 N Pre-HIMA: 307±43 N MVIC vertical: Pre-PIMA: 264±55 N Pre-HIMA: 259±41 N	Elbow flexors in two postures PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC sEMG: BB, brachioradialis, triceps brachii, deltoid imEMG: supraspinatus, infraspinatus, teres minor aEMG	Elbow flexion 90°, forearm vertical or horizontal 20% MVIC to failure (1×) horiz. HIMA: 312±156s, PIMA: 528±216s* vertical: HIMA: 468±270s; PIMA: 474±246s PIMA: >5% force change for >5s HIMA: >11.5° angle change for >5s corrections allowed Visual & verbal feedback	↔ aEMG elbow flexors, triceps, deltoid overall ($p>0.163$) ↓ aEMG elbow flexors forearm horizontal final 40% TTF ($p=0.0007$, $d=0.66$, -25.5%) ↑ aEMG supraspinatus both postures ($p<0.007$, $d=0.40$, 35.1%) ↑ aEMG teres minor both postures ($p<0.007$, $d=0.57$, 49.7%) ↑ aEMG infraspinatus both postures ($p<0.007$, $d=0.79$, 76.5%) ↑ EMG increase rate supraspinatus, infraspinatus & teres minor forearm horizontal ($p=0.05$, $p=0.004$ & $p=0.002$) ↔ EMG increase rate elbow flexors both postures ($p>0.05$)
Rudroff, Jordan, Enoka, Matthews, Baudry & Enoka, 2010 [100]	N=23 (20 m, 3 f) 21±6 yrs MVIC neutral: Pre-PIMA: 242±57 N Pre-HIMA: 248±26 N MVIC supinated: Pre-PIMA: 249±43 N Pre-HIMA: 244±57 N	Elbow flexors (neutral or supinated) PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC sEMG: BB, brachioradialis, triceps brachii nEMG: BB aEMG Coactivation ratio elbow extensors to flexors SMU mean discharge rate CV of ISI	Elbow flexion 90°, forearm vertical, neutral or supinated) 5% MVIC above recruitment threshold Neutral: 16.4±8% MVIC; 148±47s Supinated: 17.7±12% MVIC; 141±65s (1×) Visual feedback	↔ aEMG BB, brachioradialis & triceps ($p>0.05$) ↔ Coactivation ratio both postures ($p=0.60$) ↓ SMU discharge rate at end forearm supinated ($p<0.001$, $d=4.56$, -20.2%) ↔ SMU discharge rate forearm neutral ($p>0.05$) ↔ CV ISI at start both postures ($p=0.89$) Applies only for HIMA with supinated forearm, no statistics for neutral & PIMA: significant increase of CV ISI
Rudroff, Justice, Holmes, Matthews &	N=21 (10 m, 11 f) 23±6 yrs MVIC pre-PIMA: 20%: 165±73 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer	Elbow flexion 90°, forearm horizontal, neutral 20%, 30%, 45% & 60% MVIC to failure (1×) 20%: HIMA: 299±77s; PIMA: 576±80s*(n=10) 30%: HIMA: 168±36s; PIMA: 325±70s*(n=11)	↔ aEMG elbow flexors for 20, 30, 45% task ($p>0.05$) ↓ aEMG elbow flexors 60% task all timepoints ($p<0.0001$) ↔ aEMG triceps all tasks ($p>0.05$) ↔ aEMG trapezius 20% & 45% tasks ($p>0.05$)

Enoka, 2011 [63]	30%: 169±86 N 45%: 148±68 N 60%: 142±61 N MVIC pre-HIMA: 20%: 181±74 N 30%: 164±88 N 45%: 157±67 N 60%: 152±56 N	sEMG: BB, brachioradialis, triceps brachii, trapezius aEMG Coactivation ratio (elbow extensors to flexors) EMG power	45%: HIMA: 132±29s; PIMA: 178±35s (n=10) 60%: HIMA: 87±14s; PIMA: 86±15s (n=9) PIMA: >5% force change for >5s HIMA: >11.5° angle change for >5s Visual & verbal feedback	↑ aEMG trapezius 30% & 60% tasks ($p<0.0005$; $d=0.72$, 41.4% & $d=0.42$, 22.8%) ↑ aEMG increase rate elbow flexors 20% & 30% tasks ($p=0.01$, $d=0.78$; 0.10% & $p=0.047$, $d=0.52$, 0.10%) ↔ aEMG increase rate elbow flexors 45 & 60% task ($p>0.05$) ↑ aEMG increase rate trapezius at 20% & 45% tasks ($p<0.05$ & $d=3.16$, 0.50%) ↔ aEMG increase rate trapezius at 30% & 60% tasks ($p>0.05$) ↔ Coactivation ratio all tasks & timepoints ($p=0.90$) ↓ Power in 10-29 Hz all tasks towards end ($p<0.02$) ↔ Power in 0-9 Hz & 30-60 Hz all tasks ($p>0.30$)
Rudroff, Justice, Matthews, Zuo & Enoka, 2010 [101]	N=13 (9 m, 4 f) 25±7 yrs MVIC: Pre-PIMA: 189±40 N Pre-HIMA: 179±43 N	Knee extensors PIMA: force transducer HIMA: inertial load, electrogoniometer, force transd. sEMG: vastus medialis oblique, VM, VL, RF, BF aEMG Coactivation ratio (BF to knee extensors) EMG power	Supine, knee & hip flexion 90° 20% MVIC to failure (1×) HIMA: 110±36s; PIMA: 224±114s* PIMA: >5% force change for 5s HIMA: >10° angle change for 5s Visual & verbal feedback	↔ aEMG knee extensors & flexor ($p=0.50$ & $p=0.90$) ↑ EMG increase rate knee extensors ($p=0.01$, $d=0.48$, 116.7%) ↔ Coactivation ratio ($p>0.50$) ↑ Power in 1-10 Hz knee extensors each time point ($p<0.04$) ↔ Power in 10-29 Hz & 30-60 Hz knee extensors ($p>0.30$)
Rudroff, Kalliokoski, Block, Gould, Klingensmith III & Enoka, 2013 [33]	n=12 m 6 young: 26±6 yrs 6 old: 77±6 yrs MVIC: young: 462±77 N old: 354±91 N	Knee extensors PIMA: force transducer HIMA: inertial load, goniometer, force transducer sEMG: RF, VM, VL, BF aEMG Coactivation ratio	Supine, knee flexion 45°, trunk-thigh 180° 25% MVIC until 90% of TTF of HIMA (1×) (young: 848±137s; old: 751±83s) No termination criteria Visual feedback	↔ aEMG knee extensors for young & old ($p>0.05$) ↔ aEMG knee flexor for young & old ($p=0.59$) ↔ Coactivation ratio for young & old ($p=0.47$)
Rudroff, Poston, Shin, Bojsen-Møller & Enoka, 2005 [102]	N=8 m 26±5 yrs MVIC: 304±107 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC sEMG: BB, brachioradialis, triceps brachii, deltoid imEMG: brachialis aEMG	Elbow & shoulder flexion 90°, forearm vertical & supinated 20% MVIC to failure (1×) HIMA: 447±276s; PIMA: 609±250s* PIMA: >5% force change for 5s HIMA: >10° angle change for 5s corrections allowed Visual & verbal feedback	↔ aEMG elbow flexors start, short BB & triceps overall ($p>0.05$) ↑ aEMG elbow flexors entire task ($p<0.05$, $d=0.22$, 11.8%) ↑ aEMG elbow flexors at 80% TTF ($p<0.05$, $d=0.45$, 26.3%) ↑ aEMG elbow flexors at end ($p<0.05$, $d=0.40$, 23.8%) ↑ aEMG long BB & brachioradialis at 20-100% TTF ($p<0.05$) ↑ aEMG brachioradialis at 80% & 100% of TTF ($p<0.05$)
Russ, Ross, Clark & Thomas, 2018 [103]	N=16 (7 m, 9 f) 23.6±1.4 yrs MVIC: -	Trunk extensors (mod. Sorensen test) PIMA: force transducer, counterbalanced load HIMA: potentiometer, force transducer, counterbalanced load (85% MVIC) sEMG: erector spinae, multifidus, gluteus max, BF RMS MF of EMG power spectrum	Prone, trunk extension 0° 15% MVIC to failure (1×) PIMA: >20% force change for >3s HIMA: >1° angular change for >3s Visual feedback	↔ RMS erector spinae ($p>0.05$) ↔ RMS multifidus ($p=0.062$) ↔ RMS gluteus max ($p=0.078$) ↔ RMS BF ($p=0.073$) ↔ MF decrease for all muscles ($p=0.171$ -0.663)
Schaefer & Bittmann, 2017 [4]	N=10 (5 m, 5 f)	Elbow extensors	Elbow extension 90°, forearm vertical	↔ Amplitude MMGs/MTGtri both tasks ($p=0.069$ -0.765) ↓ Amplitude MMGtri at end ($p=0.012$)

	m: 24±5 yrs f: 24.4±2 yrs MVIC: m: 31.2±9.8 Nm f: 18.3±2 Nm	pneumatic device incl. force transducer PIMA: push against push rod HIMA: resist push rod MMGtri & MMGobl MTGtri Mean amplitude, CV of amplitude between trials MF & power	80% MVIC for 15s (3×) & to failure (2×) HIMA: 19.1±7.9s; PIMA: 41.4±24.9s* PIMA & HIMA: >1.3° angular change no correction allowed PIMA: verbal feedback HIMA: no feedback	↓ CV amplitude MMGobl between 15s trials ($p=0.017$) ↔ MF ($p>0.05$) ↑ Power MTGtri in 8-15 Hz & 10-29 Hz ($p=0.037$ & $p=0.048$) ↔ Power MMGtri & MMGobl ($p>0.05$)
Schaefer & Bittmann, 2021 [5]	N=20 (10 m, 10 f) m: 22.1±2.4 yrs f: 21.6±2.1 yrs MVIC: m: 51.2±22.5 Nm f: 25.8±0.07 Nm	Elbow extensors pairwise interaction, force transducer PIMA: push against partner resistance HIMA: resist partner force MMGtri & MMGobl MTGtri Mean amplitude & amplitude variation MF & power in 8-15 Hz Power ratio: power in 3-7 Hz related to sum of power in 3-7 Hz & 7-12 Hz MQ _{rel}	Elbow extension 90°, forearm vertical & neutral 80% MVIC of weaker for 15s (3×) 90% MVIC of weaker to failure of one partner (2×) HIMA & PIMA: >7° angular change or decline in force Visual feedback for pushing partner	↔ Amplitude MMGs/MTGtri both tasks ($p=0.055$ to 0.573) ↔ Amplitude variation MMGtri & MTGtri ($p=0.219$ - 0.863) ↑ Amplitude variation MMGobl 15s & fatiguing tasks ($p=0.013$, $d=0.71$, 13.8% & $p=0.007$, $d=0.58$, 11.6%) ↔ MF MMGs/MTGs both tasks ($p>0.05$) ↑ CV of MF between fatiguing trials ($p=0.01$, $d=0.44$, 57.4%) ↓ Power MMGobl in 8-15 Hz 15s & fatiguing trials ($p=0.001$, $d=0.39$, -50% & $p=0.011$, $d=0.28$, -33.3%) ↔ Power MMGtri & MMGobl ($p>0.05$) ↑ MQ _{rel} MMGobl 15s & fatiguing tasks ($p=0.04$, $d=0.36$, 24.7% & $p=0.002$, $d=0.66$, 49.1%) ↔ MQ _{rel} MMGtri & MTGtri both tasks ($p=0.053$ - 0.717)
Schaefer & Bittmann, 2022 [25]	N=2 m Partner A: 28 yrs MVIC: 186 N Partner B: 22 yrs 142 N	Elbow extensors Force transducer, accelerometer (ACC) PIMA: push against partner resistance HIMA: hold against partner resistance MMGtri & MMGobl MTGtru EEGleft, EEGright & EEGcentral Wavelet Coherence Analysis: Coh: Coherence (%) in 8-15 Hz within one subject (intra) & between subjects (inter) WF: Weighted frequency of Coh in 8-15 & 3-25 Hz	Elbow, shoulder, hip & knee flexion 90° 70% MVIC of weaker to failure of one partner (6×); PIMA & HIMA alternating HIMA & PIMA: >7° angular change Visual feedback for pushing partner	↔ Coh intra-EEGright-MMGs, intra-EEGcentral-MMGs ($p>0.05$) ↑ Coh intra-EEGleft-MMGs partner B ($p<0.001$, $d=1.79$, 56.8%) ↑ Coh inter-EEGcentral-MMGs & inter-EEGleft-MMGs ($p=0.047$, $d=0.36$, 11.8% & $p=0.007$, $d=0.60$, 21.2%) ↑ Coh force-EEGcentral & force-EEGright ($p=0.017$, $d=1.05$, 44.8% & $p=0.013$, $d=2.41$, 67.6%) ↔ Coh inter-MMGs, inter-EEGs, inter-EEGright-MMGs, force- MMGs, force-EEGleft, ACC-MMGs, ACC-EEGs ($p=0.058$ - 1.00) ↑ WF 8-15 Hz inter-EEGright-MMGs ($p=0.032$, $d=1.28$, 5.1%) ↑ WF 8-15 Hz force-EEGleft & force-EEGright ($p=0.040$, $d=1.91$, 9.0% & $p=0.005$, $d=2.67$, 9.1%) ↑ WF 8-15 Hz ACC-EEGright ($p=0.012$, $d=1.20$, 6.0%) ↔ WF 8-15 Hz inter-EEGcentral-MMGs, inter-EEGleft-MMGs, force-EEGcentral, ACC-EEGcentral & ACC-EEGleft ($p>0.05$) ↑ WF 3-25 Hz inter-EEGcentral-EEGle & inter-EEGcentral- EEGright ($p<0.001$, $d=1.71$, 12% & $p=0.043$, $d=1.73$, 13%)
Semmler, Kornatz, Dinunno, Shi & Enoka, 2002 [104]	N=10 subgroup for HIMA-PIMA from n=17 (12 m, 5 f) 22–45 yrs ~39 N	First dorsal interosseus PIMA: force transducer HIMA: inertial load, displacement transducer sEMG & nEMG: FDI MU discharge rate (mean & CV) MU synchronisation (cross-correlogramm)	Index finger abduction 5°, full index finger extension, 3 rd -5 th finger flexed, elbow flexion 90° Force to sustain discharge of SMU for 2-5 min PIMA: 4.4% MVIC, HIMA: 3.8% MVIC Termination criteria: none for HIMA; PIMA: at least one MU detectable, occasionally target force adjustment	↔ MU discharge rate mean & CV ($p>0.05$) ↔ MU synchronisation ($p>0.05$)

			Visual & audio feedback	
Williams, Hoffman & Clark, 2014 [105]	N=10 (5 m, 5 f) 24.5±3.1 yrs MVIC: session 1: 276.4±101.7 N session 2: 272.0±102.9 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer sEMG: BB, brachioradialis TMS: right motor cortex Electrical stim.: brachial plexus & cervicomedullary junction RMS MEP amplitude & SP duration CMEP SICI: short-interval intracortical inhibition (ratio) ICF: intracortical facilitation (ratio) LICI: long-interval intracortical inhibition (ratio) LII: long-interval inhibition (ratio)	Elbow flexion 90°, forearm horizontal & neutral, shoulder abduction 10-15° 15% MVIC to failure (1×) HIMA: 1614±907s; PIMA: 1050±474s* PIMA: >5% force change for >5s HIMA: >10° angle change for >5s corrections allowed Visual & verbal feedback	↑ SP duration at baseline for BB & brachioradialis ($p<0.001$; $d=0.88$, 14.5% & $d=0.82$, 17.4%) ↔ RMS, $M_{\max}$, MEP, CMEP, SICI, ICF, LII baseline ($p>0.05$) ↔ RMS, RMS increase & $M_{\max}$ both muscles overall ($p\geq 0.07$) ↔ MEP, increase & increase rate both muscles ($p\geq 0.38$) ↔ SP change & rate change rate at end both muscles ($p\geq 0.07$) ↔ MEP in SP change & change rate both muscles ($p\geq 0.53$) ↔ CMEP in SP change & change rate both muscles ($p\geq 0.36$) ↔ SICI change & change rate both muscles ($p\geq 0.26$) ↑ ICF brachioradialis overall ($p=0.02$, $d=1.20$, 20.2%) ↔ ICF BB overall, change & change rate both muscles ($p\geq 0.21$) ↔ LII overall both muscles ($p\geq 0.55$)
Yunoki, Wtanabe, Matsumoto, Kuwabara, Horinouchi, Ito, Ishida & Kirimoto, 2022 [35]	N=18 (15 m, 3 f) 21-35 yrs MVIC: -	First dorsal interosseus PIMA: force transducer HIMA: inertial load, electrogoniometer sEMG: FDI silver-silver chloride electrode: C3 Electrical stimulation: digital nerve CMR amplitude (cutaneomuscular reflex) from EMG (3 components: E1, I1, E2) SEP amplitude from C3 (N20, P25, N33, P45)	Index finger abduction 10°, full finger extension, thumb abduction 45°, forearm neutral, elbow flexion 110°, shoulder abduction 10-20° 20% MVIC for 50-60s (3× for SEP, 1× for CMR) Termination criteria: - Visual feedback	↔ CMR E1 & I1 ($p=0.306$ & $p=0.107$) ↓ CMR E2 ($p<0.001$, $d=4.18$, -32.7%) ↔ SEP N20 ($p=0.073$) ↑ SEP N33 ($p=0.008$, $d=2.54$, 25%)
Abbreviations (alphabetical): °=angular degree, ACC=accelerometer, APB=abductor pollicis brevis, aEMG=EMG amplitude, BB=biceps brachii, BF=biceps femoris, CCC=cortico-cortical coherence, CMC=corticomuscular coherence, cSP=cortical silent period, CV=coefficient of variation, d =Cohen's d effect size, DT=derecruitment threshold, ECR=extensor carpi radialis, EEG=electroencephalography, EMG=electromyography, FCR=flexor carpi radialis, FDI=first dorsal interosseus, imEMG=intramuscular EMG, ISI=interspike interval, LLR=long latency reflex, MDF=median power frequency, MEP=motor evoked potential, MF=mean frequency, MFR=mean firing rate, M=arithmetic mean, $M_{\max}$ =maximal M wave, MMG=Mechanomyography, MTG=Mechanotendography, MPF=mean power frequency, MU=motor unit, MVIC=maximal isometric voluntary contraction, nEMG=needle EMG, PSD=power spectral density, RF=rectus femoris, RMS=root-mean-square, RT=recruitment threshold, s=seconds, SD=standard deviation, sEMG=surface EMG, SEP=somatosensory evoked potential, SLR=short latency reflex, SMU=single motor unit, SP=silent period, SPI=second palmar interosseous, TTF=time to task failure, VL=vastus lateralis, VM=vastus medialis.				
Numbers are reported as mean±standard deviation. Effect sizes are pairwise. *indicates a significant difference for TTF between HIMA & PIMA.				

3.5.1 Neuromuscular activity

As previously described, combining magnitude statistics could not be confidently done. However, some general trends can be observed. Three studies examined HIMA and PIMA on, or in response to, special factors, which will be considered separately (see below). Agonist muscle EMG amplitude differences between PIMA and HIMAs were mostly non-significant. For example, only five of 48 relevant outcomes [33, 62-64, 70-74, 79-87, 89, 92, 97, 98, 100-103, 105] resulted in greater agonist activation during PIMA ($d=0.66-6.79$, 25.5-38.1%) [63, 64, 83, 84, 89], whereas eight outcomes were greater during HIMA ($d=0.22-0.72$, 11.8-25%) [74, 82, 102, 106]. Eight of 17 outcomes [63, 74, 82-85, 87, 89, 92, 100, 105] showed a significantly greater increase of EMG amplitude in agonists during HIMA vs PIMA ($d=0.46-0.99$, 0.1-3566%) [63, 74, 82, 85, 92, 101], with none favoring PIMA.

Regarding synergist muscle groups, two of the 15 relevant outcomes [62, 63, 71, 74, 80, 84, 85, 103] reported greater EMG amplitude during HIMA ($d=0.42-0.72$, 22.8-41.4%) [63] and two favoring PIMA ($d=0.46$, -21.6%) [85, 106]. In five of 10 outcomes [63, 73, 74, 82, 84, 85], the increase in EMG amplitude was significantly higher during HIMA vs PIMA for synergists ($d=0.43-3.16$, 0.5-81%) [63, 82, 84, 85], the other five showed no significant differences. For antagonist muscle groups, only one of the 17 relevant outcomes [33, 62, 70-74, 80, 81, 86, 87, 89, 97, 98, 100-102] was significant with greater EMG amplitude for PIMA vs HIMA [98]. In four of nine relevant outcomes [74, 82, 83, 85, 87, 89, 92], the increase in EMG was significantly higher during HIMA for antagonists ($d=0.40-0.82$, 63-600%) [82, 85, 87].

The EMG burst rate of agonist muscles was significantly greater during HIMA vs PIMA in five of seven outcomes ($d=0.20-0.69$, 35.7-210.5%) [82-85], while three referred to the increase in burst rate. One study considered the burst rate of antagonist muscles, which did not show a significant difference between both tasks [83]. The discharge rate was significantly lower for HIMA vs PIMA in eight of twelve outcomes ($d=0.52-4.56$, -28 to -10.5%) [34, 74, 81, 95, 100] where the remaining four were non-significant, e.g., in older participants or for neutral forearm position [34, 95, 100, 104]. The discharge rate declined significantly stronger during HIMA vs PIMA in all outcomes [81, 95]. The interspike interval CV was mostly (five of seven outcomes; 3x increase/change of CV) greater during HIMA ($d=0.57-2.31$, 23.9-625%) [34, 74, 81, 100], with a single outcome favoring PIMA for large target force differences in young participants ($d=0.64$, -20.3%) [34].

The H-reflex decreased strongly during HIMA vs PIMA ($d=1.35$, 72%) [89]. Heteronymous facilitation was greater for HIMA vs PIMA in nine of eleven outcomes ($d=0.21-1.32$, 5.22-34.6%) [70-72, 88, 91, 107], while one outcome at TTF was significantly lower ($p=0.049$) for the HIMA condition [73]. For old participants, it was non-significant [72]. D1-inhibition was lower for HIMA than PIMA in five of seven outcomes ($p\leq 0.042$, $d=-0.71$ to -3.6, -14.3 to -34%) [70, 72, 73, 107]. As for discharge rate and heteronymous reflex, old participants did not show a significant difference between the isometric tasks regarding D1-inhibition [72]. Furthermore, one study showed a significantly greater D1-inhibition with HIMA at the end ($d=1.57$, 33%), which was lower after 2 min of TTF [73]. For the coactivation ratio, ten of 13 outcomes [33, 62, 63, 72, 73, 80, 82, 85, 100, 101] reported non-significant differences between muscle actions. The three significant findings reported greater coactivation during HIMA wrist extensions for old participants and ankle dorsiflexion [72, 82]. It has to be mentioned that the latter is considered the increase in coactivation ratio [82].

Schaefer and Bittmann used MMG and MTG to examine the elbow extensors during 80% and 90% MVIC muscle actions in a single-person setting [4] or during the muscular interaction of two participants [5]. PIMA led to significantly greater triceps MMG amplitude [4]. However, HIMA had significantly greater MTG power in the triceps tendon in low-frequency ranges of 8-29 Hz [4]. For the single-person setting, the amplitude variation of oblique muscle was significantly lower during HIMA [4], whereas it was significantly higher for HIMA in the paired setting [5]. Moreover, while HIMA led to a significantly lower MMG power of the obliques in 8-15 Hz ($d=0.28-0.39$, 33-50%), PIMA showed a significantly higher CV of MMG frequency between trials, and the power-frequency ratio of the obliques ($d=0.36-0.66$, 24.7-57.4%) [5].

3.5.2 Brain activity

Three studies, with mixed findings, included direct measures of brain activity via electroencephalography (EEG) [25, 96, 97]. Schaefer and Bittmann primarily examined wavelet coherence between EEG, MMG and MTG variables [25] during partner-matched ($n=2$) elbow extensions at 70% of the weaker partner's MVIC. All reported significant differences (6 of 13 outcomes) demonstrated higher coherence or frequency during HIMA vs PIMA ($p \leq 0.047$, $d=0.36-2.67$, 5.1-67.6%). Conversely, five of eleven outcomes from Poortvliet et al. [96], who investigated the knee extensors at 10% of MVIC, showed significant muscle action differences with lower cortico-cortical coherence for HIMA in four cases ($p < 0.001$, $d=0.09-0.27$, 5-14.5%) and significantly higher beta-band EEG power of the left hemisphere ($p < 0.05$, $d=0.11$, 0.68%).

3.5.3 Special factors

Three studies compared PIMA and HIMA on, or in response to, special factors, including skin and core temperature [77], residual force enhancement [93], and experimental joint pain [96, 99]. Coletta et al. [77] reported that EMG RMS was greater for HIMA at modest ($\pm 0.5^{\circ}\text{C}$) core heating and skin cooling conditions. However, no between-muscle action differences in RMS were detected during more extreme heating (1°C) [77]. It must be noted that the RMS was significantly higher for HIMA vs PIMA already before the heating or cooling protocol (baseline). Marion and Power [93] PIMA and HIMA were compared following active lengthening of the dorsi-flexors to examine residual force enhancement, with no between-muscle action differences for any outcome, including activation reduction, neuromuscular economy, or muscle coactivation. Experiments led by Poortvliet compared the effects of hypertonic saline injections to the infrapatellar fat pat to temporarily simulate knee pain during low intensity ($\sim 10\%$) PIMA and HIMA knee extensions [96, 99]. The first study [99] considered 35 single-motor-units ($n=5$), which were assessed in all conditions (no pain, pain, HIMA, PIMA) in one session, and all measured 189 single-motor-units ($n=11$) were assessed in two sessions, and not all participants were evaluated in each condition. For the 35 single-motor-units, the no-pain state showed similar EMG RMS and discharge rates between HIMA and PIMA, where the discharge rate variability tended to be higher for PIMA ($p=0.05$). Under pain, the decline in discharge rate was greater for PIMA vs HIMA. Furthermore, the discharge rate variability between no-pain vs pain state only changed significantly for PIMA, not HIMA. Considering all 189 single-motor-units, the discharge rate, in general, was significantly higher for HIMA vs PIMA and decreased under pain for both HIMA and PIMA. There was a trend of a lower discharge rate for PIMA under pain. Furthermore, the number of single-motor-units which changed the discharge rate from no-pain to pain-state (by $>10\%$; increase or decrease) was twice as high for PIMA (44.9%) than HIMA (22.2%) [99]. The second study by Poortvliet et al. found no significant muscle action differences for corticomuscular coherence (EEG and sEMG) [96]. However, comparing the no-pain vs pain state only PIMA, not HIMA, showed significant changes: under pain, corticomuscular coherence was lower, and the power in the beta band decreased while it increased in the gamma band. Poortvliet et al. suggest that brain activity is less affected by pain during HIMA [96].

3.6 Cardiovascular and metabolic

The 14 studies examining cardiovascular or metabolic differences between PIMA and HIMA are summarized in **Table 4** [33, 62-64, 78, 80, 82-85, 92, 95, 101, 102]. Eleven studies assessed HR and MAP, allowing for meta-analysis (see below), with mean contraction intensities of 28.1% (range: 15-60% MVIC) [63, 64, 80, 82-85, 92, 95, 101, 102]. Seven of the HR and MAP comparisons utilized the elbow flexors [63, 64, 80, 83, 84, 95, 102], while others employed the dorsi-flexors [82, 85], knee extensors [101], or first dorsal interossei [92] as prime movers. A single study utilized near-infrared spectroscopy, finding no significant differences between PIMA and HIMA for tissue oxygenation index or normalized index of total hemoglobin for the biceps or triceps brachii during elbow flexions at 20% or 60% MVIC [62]. Another isolated study used an O²C spectrophotometer to assess biceps brachii capillary venous oxygen saturation and relative hemoglobin during elbow flexions at 60% MVIC, with no significant difference between PIMA and HIMA [78]. Finally, a single study employed computed tomography with venous glucose injection to assess standardized glucose uptake of the agonist, antagonist, and synergist muscles during PIMA and HIMA knee extensions at 25% MVIC [33]. While all differences were non-significant in the older population ($d=-0.13$ to -0.26 , -7.7 to -4.3%), HIMA resulted in greater glucose uptake into the agonist ($d=0.68$, 32.6%), antagonist ($d=0.47$, 15.4%), and hip ($d=0.19$, 9.7%) muscles of the young group [33]. HP and MAP data were extracted from the figures of nine studies [63, 64, 80, 82-85, 92, 101].

Table 4. Summary of studies comparing cardiovascular and metabolic parameters between pushing (PIMA) and holding (HIMA) isometric muscle actions.

↑=sign. larger effect from HIMA vs PIMA; ↓=sign. lower effects from HIMA vs PIMA; ↔=no significant difference between HIMA vs PIMA.

Significance p, effect size Cohen's d and the percentage difference between HIMA and PIMA are given.

Study	Participants	Relevant Measures	Conditions	Results
Booghs, Baudry, Enoka & Duchateau, 2012 [62]	N=15 (8 m, 7 f) 18–36 yrs MVIC: Pre-PIMA: 20% task: 293±82 N 60% task: 299±119 N Pre-HIMA: 20% task: 296±86 N 60% task: 292±111 N	Elbow flexors PIMA: force transducer HIMA: inertial load, potentiometer NIRS: biceps & triceps brachii TOI (%) nTHI	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% or 60% MVIC to failure (n=12) (1×) 20% (n=12): HIMA: 404±159s; PIMA: 533±194s* 60% (n=9): HIMA: 54±19s; PIMA: 64±16s PIMA: >2% or 5% force change for 3s HIMA: >1.5° or 3° angle change for 3s corrections allowed Visual & verbal feedback	↔ TOI biceps at 20% & 60% ($p>0.05$) ↔ TOI triceps at 20% & 60% ($p=0.13$) ↔ TOI slope biceps ($p=0.36$, $d=0.24$, -12.5%) ↔ nTHI biceps 20% & 60% ($p>0.05$) ↔ nTHI triceps 20% & 60% ($p>0.05$, $d=0.08$, -4.6%)
Dech, Bittmann & Schaefer, 2022 [78]	N=10 (8 m, 2 f) 30.7±11.7 yrs MVIC: left: 69±22 Nm right: 70±24 Nm	Elbow flexors (both sides) PIMA: force transducer (seated) or inertial load with intermittent twitches every 7s (standing) HIMA: inertial load (standing) Spectrophotometer (O2C): biceps brachii SvO ₂ TSS	Elbow flexion 90°, forearm horizontal & supinated 60% MVIC to failure (1×) HIMA: 44±14s; PIMA: 52±10s* PIMA: < target force for 2s or twitches impossible HIMA: < target angle for 2s No feedback	↔ SvO ₂ decrease ($p=0.121$ -0.909) ↔ SvO ₂ slope ($p=0.373$ -0.913) ↔ TTS ($p=0.309$ -0.630)
Gordon, Rudroff, Enoka & Enoka, 2012 [80]	N=20 (15 m, 5 f) 21±4 yrs MVIC: Pre-PIMA: 273±90 N Pre-HIMA: 291±93 N	Elbow flexors (both sides) PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer automated blood pressure monitor HR MAP	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVIC to failure (1×) HIMA: 253±103s; PIMA: 367±133s* PIMA: >5% force change for 5s HIMA: >11.5° angle change for 5s corrections allowed Visual & verbal feedback	↔ HR & MAP at start & end ($p>0.05$) ↔ HR increase rate ($p=0.78$) ↑ MAP increase rate ($p=0.03$, $d=0.49$, 0.03%)
Griffith, Yoon & Hunter, 2010 [82]	Young: N=17 (8 m, 9 f) Old: N=12 (7 m, 5 f) 23.6±6.5 yrs 70.0±5.0 yrs MVIC (Nm) young: pre-PIMA: 38.0±10.2 pre-HIMA: 37.4±9.4 MVIC (Nm) old: pre-PIMA: 35.5±8.9 pre-HIMA: 34.9±10.4	Dorsi-flexors PIMA: force transducer HIMA: inertial load, potentiometer, ACC automated beat-by-beat, blood pressure monitor HR MAP	Dorsi-flexion 0°, hip & knee flexion 90° 30% MVIC to failure (1×) HIMA: 561±204s; PIMA: 624±270s* PIMA: >5% force drop for 4s HIMA: >18° angle drop corrections allowed Visual & verbal feedback	↑ HR increase rate young & old ($p=0.02$, $d=0.38$, 41.7%) ↔ MAP & HR at start & end ($p>0.05$) ↑ MAP increase rate young & old ($p=0.001$, $d=0.60$, 44.1%)

Hunter, Rochette, Critchlow & Enoka, 2005 [83]	N=18 (10 m, 8 f) 72±4 yrs MVIC: Pre-PIMA: 180±55 N Pre-HIMA: 178±61 N	Elbow flexors PIMA: force transducers HIMA: inertial load, electrogoniometer, ACC automated beat-by-beat, blood pressure monitor HR MAP	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 20% MVIC to failure (1×) HIMA: 636±366s; PIMA: 1368±546s* PIMA: >10% force drop for >5s HIMA: >26° angle change for >5s corrections allowed Visual & verbal feedback	↔ HR & MAP at start & end ($p>0.05$) ↑ HR increase rate ($p<0.05$, $d=0.81$, 122.2%) ↑ MAP increase rate ($p<0.05$, $d=1.10$, 110.0%)
Hunter, Ryan, Ortega & Enoka, 2002 [84]	N=16 (8 m, 8 f) 27±4 yrs MVIC: Pre-PIMA: 308±151 N Pre-HIMA: 307±152 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC automated beat-by-beat, blood pressure monitor HR MAP	Elbow flexion 90°, forearm horizontal & neutral, slight shoulder abduction 15% MVIC to failure (1×) HIMA: 702±582s; PIMA: 1402±728s* PIMA: >10% force drop for >5s HIMA: >10° angular change for >5s corrections allowed Visual & verbal feedback	↔ HR & MAP at start ($p>0.05$) ↔ HR at end ($p>0.05$) ↑ MAP at end ($p<0.05$, $d=0.65$, 8.3%) ↑ HR increase rate ($p<0.05$, 38.0%) ↑ MAP increase rate ($p<0.05$, 182.5%)
Hunter, Yoon, Farinella, Griffith & Ng, 2008 [85]	N=15 (8 m, 7 f) 21.1±1.4 yrs MVIC: pre-PIMA: 333±71 N pre-HIMA: 334±65 N	Dorsi-flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC automated beat-by-beat, blood pressure monitor HR MAP	Dorsi-flexion 0°, hip & knee flexion 90° 20% MVIC to failure (1×) HIMA: 600±372s; PIMA: 1278±1068s* PIMA: >5% force drop for 4s HIMA: >18° angle drop for 4s corrections allowed Visual & verbal feedback	↔ MAP & HR at start & end ($p>0.05$) ↑ MAP increase rate ($p=0.018$, 195.8%) ↑ HR increase rate ($p=0.014$, 73.5%)
Maluf, Shinohara, Stephenson & Enoka, 2005 [92]	N=20 m (2×n=10) 23±5 yrs MVIC: Low force group (n=10): Pre-PIMA: 34.8±7.5 N Pre-HIMA: 33.8±6.7 N High force group (n=10): Pre-PIMA: 32.5±4.0 N Pre-HIMA: 32.1±3.9 N	First dorsal interosseus (abduction) PIMA: force transducer HIMA: potentiometer, ACC automated blood pressure monitor HR MAP	Index finger abduction 0° 20% or 60% MVIC to failure (1×) 20%: HIMA: 593±212s; PIMA: 938±328s* 60%: HIMA: 86±31s; PIMA: 93±41s PIMA: >1.5% force change for 3s HIMA: >10° angle change for 3s corrections allowed Visual & verbal feedback	↔ HR & MAP at start & end at 20% task ($p\geq 0.504$) ↔ HR & MAP at start & end at 60% task ($p\geq 0.117$) ↔ HR increase rate at 20% task ($p=0.086$) ↔ MAP increase rate at 20% & 60% tasks ($p>0.05$) ↑ HR increase rate at 60% task ($p=0.001$, $d=0.53$, 75.0%)
Mottram, Jakobi, Semmler & Enoka, 2005 [95]	N=15 m 25.6±5.8 yrs MVIC: 265±50 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC automated beat-by-beat, blood pressure monitor HR MAP	Elbow flexion 90°, forearm horizontal & neutral, shoulder abduction 15° 22.2±13.4% MVIC (3.5±2.1% above recruitment threshold) for 161±96s (1×) incl. needle EMG Visual feedback	↑ HR at start, 80s, and 160s ($p\leq 0.03$, $d=0.20$ -0.52, 4.4-8.9%); n.s. at 40s, 120s ↑ MAP at start, 80s, 120s and 160s ($p<0.001$, $d=0.60$, 10.6%); n.s. at 40s ↑ MAP increase rate ($p<0.001$)

Rudroff, Barry, Stone, Barry & Enoka, 2007 [64]	N=20 m 27±5 yrs MVIC horizontal: Pre-PIMA: 309±45 N Pre-HIMA: 307±43 N MVIC vertical: Pre-PIMA: 264±55 N Pre-HIMA: 259±41 N	Elbow flexors in two postures PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC automated beat-by-beat, blood pressure monitor HR MAP	Elbow flexion 90°, forearm vertical or horizontal 20% MVIC to failure (each 1×) horiz.: HIMA: 312±156s, PIMA: 528±216s* vertic.: HIMA: 468±270s; PIMA: 474±246s PIMA: >5% force change for >5s HIMA: >11.5° angle change for >5s corrections allowed Visual & verbal feedback	↔ HR at start & end vertical & horizontal ($p>0.05$) ↔ MAP start & end forearm vertical & horizontal ($p>0.05$) ↑ MAP increase rate forearm horizontal ($p=0.039$)
Rudroff, Justice, Holmes, Matthews & Enoka, 2011 [63]	N=21 (10 m, 11 f) 23±6 yrs MVIC pre-PIMA: 20%: 165±73 N 30%: 169±86 N 45%: 148±68 N 60%: 142±61 N MVIC pre-HIMA: 20%: 181±74 N 30%: 164±88 N 45%: 157±67 N 60%: 152±56 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer automated blood pressure monitor HR MAP	Elbow flexion 90°, forearm horizontal, neutral 20% MVIC (n=10): HIMA: 299±77s; PIMA: 576±80s* 30% MVIC (n=11): HIMA: 168±36s; PIMA: 325±70s* 45% MVIC (n=10): HIMA: 132±29s; PIMA: 178±35s 60% MVIC (n=9): HIMA: 87±14s; PIMA: 86±15s to failure (each 1×) PIMA: >5% force change for >5s HIMA: >11.5° angle change for >5s Visual & verbal feedback	↔ HR at start & end all tasks ($p>0.31$) ↑ HR increase rate at 20% & 30% tasks ($p=0.003$, $d=2.31$, 100%) ↔ HR increase rate at 45% & 60% tasks ($p>0.07$, $d=1.53$, 91.7%) ↔ MAP at start & end all tasks ($p>0.65$) ↑ MAP increase rate at 20% & 30% tasks ($p=0.006$, $d=1.05$, 100%) ↔ MAP increase at 45% & 60% tasks ($p>0.84$)
Rudroff, Justice, Matthews, Zuo & Enoka, 2010 [101]	N=13 (9 m, 4 f) 25±7 yrs MVIC: Pre-PIMA: 189±40 N Pre-HIMA: 179±43 N	Knee extensors PIMA: force transducer HIMA: inertial load, electrogoniometer, force transducer automated beat-by-beat, blood pressure monitor HR MAP	Supine, knee & hip flexion 90° 20% MVIC to failure (1×) HIMA: 110±36s; PIMA: 224±114s* PIMA: >5% force change for 5s HIMA: >10° angle change for 5s Visual & verbal feedback	↔ HR at start & end ($p>0.63$) ↑ HR increase rate ($p=0.03$, $d=1.29$, 137.5%) ↔ MAP at start & end ($p>0.29$) ↑ MAP increase rate ($p=0.03$, $d=0.86$, 50.0%)
Rudroff, Kalliokoski, Block, Gould, Klingensmith III & Enoka, 2013 [33]	<i>Muscle activation experiment:</i> n=12 m 6 young: 26±6 yrs 6 old: 77±6 yrs MVIC: young: 462±77 N old: 354±91 N	Knee extensors PIMA: force transducer HIMA: inertial load, goniometer, force transducer PET & CT scan lower limb (hip to feet); glucose injection to vein SUV: standardized glucose uptake value of agonists (quadriceps femoris), antagonists (hamstrings), hip muscles & lower leg muscles	Supine, knee flexion 45°, trunk-thigh 180° 25% MVIC until 90% of TTF of HIMA (1×) (young: 848±137s; old: 751±83s) PIMA: < target force for 5s HIMA: >10° angular change for 5s Visual feedback	↑ SUV young ($p<0.01$, $d=0.30$, 14.1%) ↑ SUV agonists young ($p<0.01$, $d=0.68$, 32.6%) ↑ SUV antagonists young ($p<0.05$, $d=0.47$, 15.4%) ↑ SUV hip muscles young ($p<0.01$, $d=0.19$, 9.7%) ↔ SUV old ($p>0.05$) ↔ SUV agonists old ($p>0.05$, $d=0.13$, -4.3%) ↔ SUV antagonists old ($p>0.05$, $d=0.26$, -7.2%) ↔ SUV hip muscles old ($p>0.05$, $d=0.25$, -7.7%) ↔ SUV lower leg muscles young & old ($p>0.05$)

Rudroff, Poston, Shin, Bojsen-Møller & Enoka, 2005 [102]	N=8 m 26±5 yrs MVIC: 304±107 N	Elbow flexors PIMA: force transducer HIMA: inertial load, electrogoniometer, ACC automated beat-by-beat, blood pressure monitor HR MAP	Elbow & shoulder flexion 90°, forearm vertical & supinated 20% MVIC to failure (1×) HIMA: 447±276s; PIMA: 609±250s* PIMA: >5% force change for 5s HIMA: >10° angle change for 5s corrections allowed Visual & verbal feedback	↔ HR at start & end ($p>0.05$) ↔ MAP at start & end ($p>0.05$) ↔ MAP increase rate ($p>0.05$)
Abbreviations (alphabetical): °=angular degrees, CT=computed tomography, d =Cohen's d effect size, f=female, HIMA=holding isometric muscle action, HR=heart rate, m=male, MAP=mean arterial pressure, MVIC=maximal voluntary isometric contraction, N=newtons, NIRS=near-infrared spectroscopy, Nm=newton meters, nTHI=normalized index of total hemoglobin, O2C=spectrophotometer, PET=positron emission tomography, PIMA=pushing isometric muscle action, s=seconds, SUV=standardized glucose uptake value, SvO ₂ =capillary venous oxygen saturation, TOI=tissue oxygenation index, TTS=time to leveling off into SvO ₂ steady state, TTF=time to task failure, yrs=years old.				
Numbers are reported as mean±standard deviation. Effect sizes are pairwise. *indicates a significant difference for TTF between HIMA & PIMA.				

Meta-analyses of 11-15 outcomes (7-9 studies) found non-significant and trivial differences in HR ($n=136-170$, $g=-0.11$ to 0.15 , $p=0.12-0.96$) and MAP ($n=136-170$, $g=0.04-0.18$, $p=0.07-0.45$) at 0%, 50% and 100% of TTF (**Figure 8A, 9A**). While not reaching statistical significance, MAP at 100% of TTF was slightly greater during HIMA ($n=170$, $g=0.18$, 95%CI: -0.01 to 0.37 , $p=0.07$). Examining the bubble plots is complicated based on statistical noise and relatively few studies at each intensity or muscle group. However, HR increased by a greater magnitude at 100% TTF under PIMA for larger appendicular muscle groups, whereas the opposite trend was true for the smaller appendicular muscles (**Figure 8C**). However, the opposite appears true when examining MAP during the initial phases (0% and 50% of TTF) (**Figure 9C**). The preceding HR and MAP meta-analyses were almost completely homogeneous (all $I^2 < 0.001$).

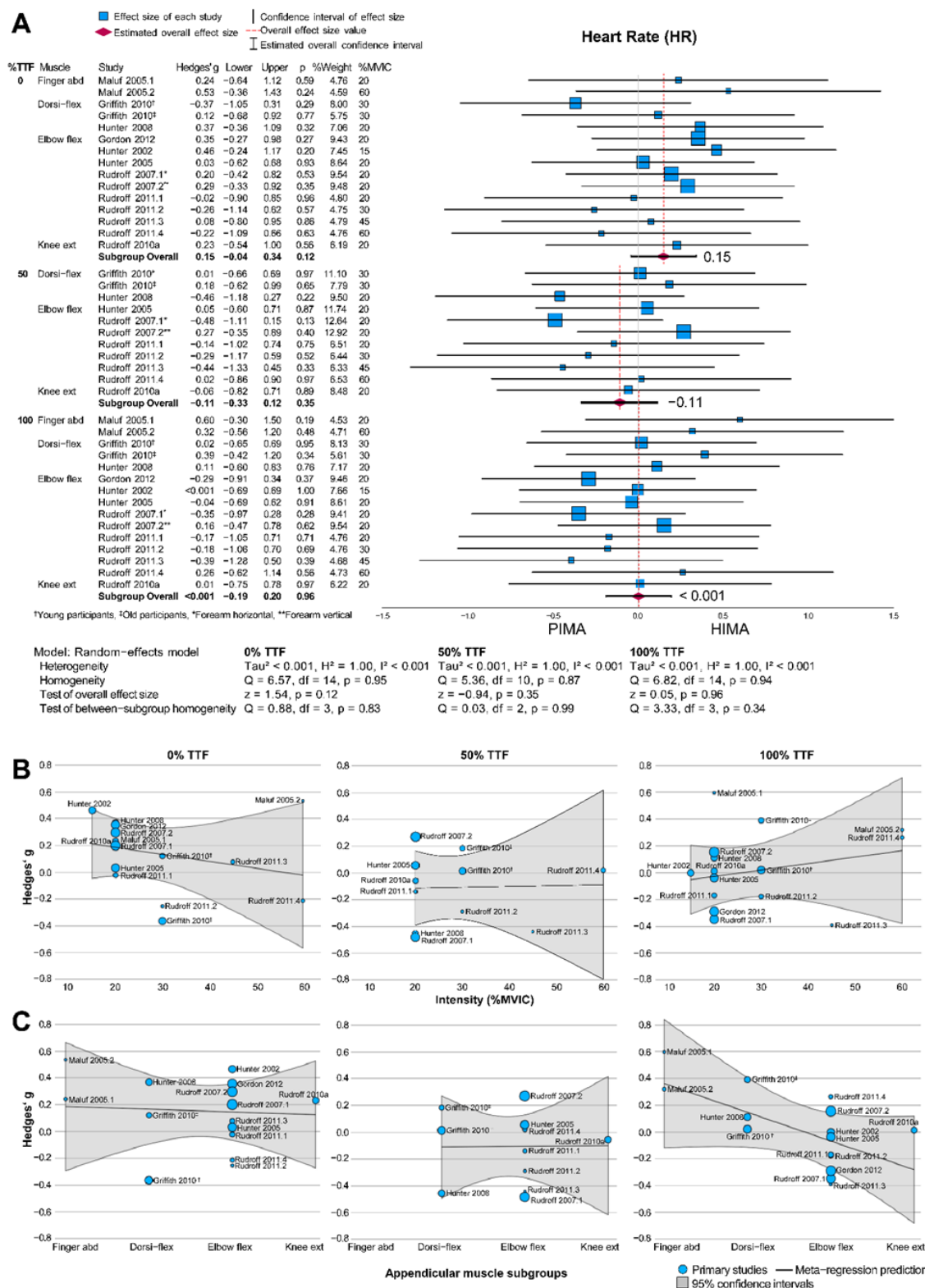


Figure 8. Meta-analysis forest plot (A) comparing heart rate (HR) between PIMA and HIMA with bubble chart regressions for loading intensity (B) and appendicular muscle subgroups (C).

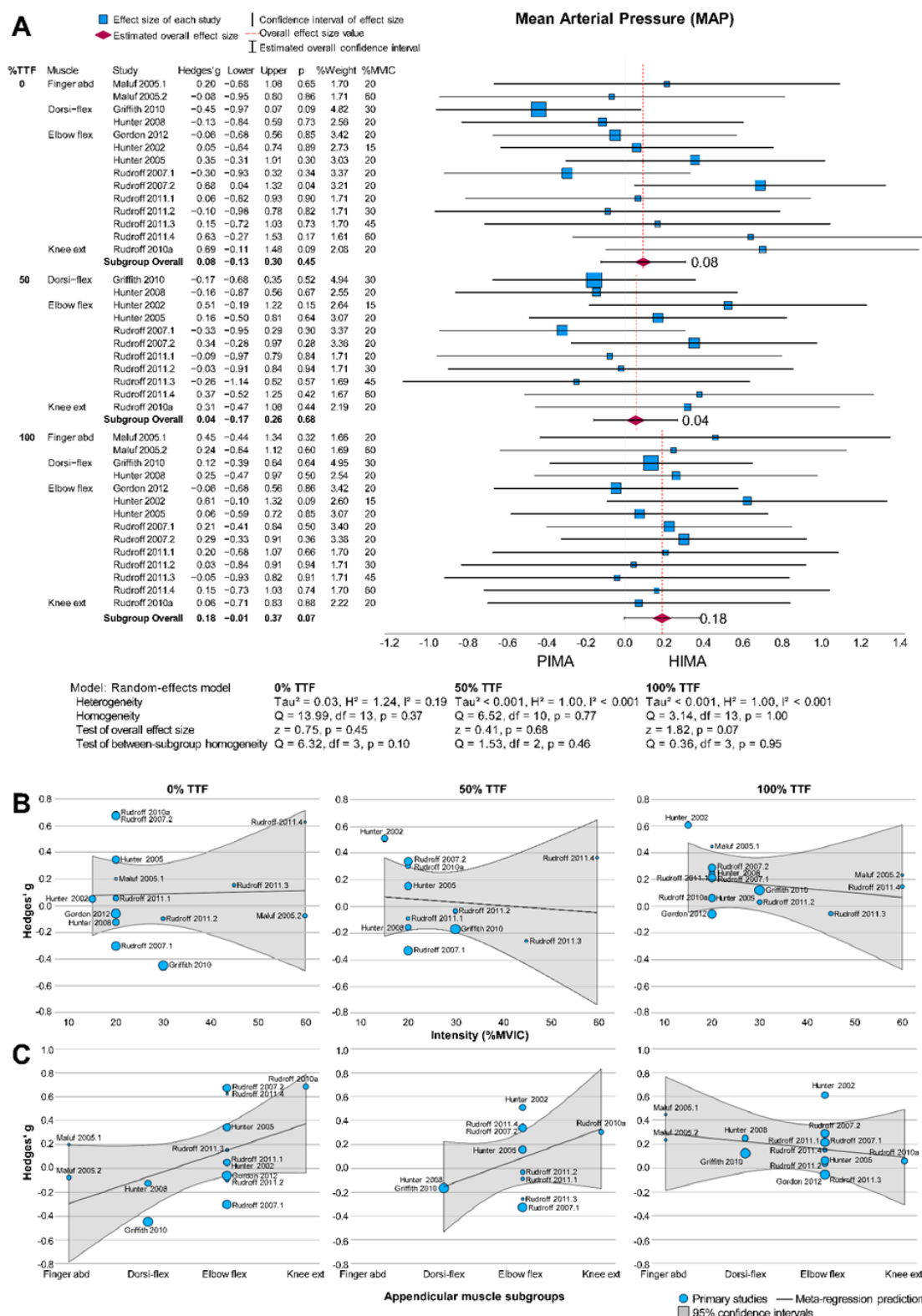


Figure 9. Meta-analysis forest plot (A) comparing mean arterial pressure (MAP) between PIMA and HIMA with bubble chart regressions for loading intensity (B) and appendicular muscle subgroups (C).

The rate of increase of HR and MAP was higher during HIMA for most studies (Figure 10). These data suggest metabolic conditions or consequences contributed to the briefer TTF for HIMA in most studies.

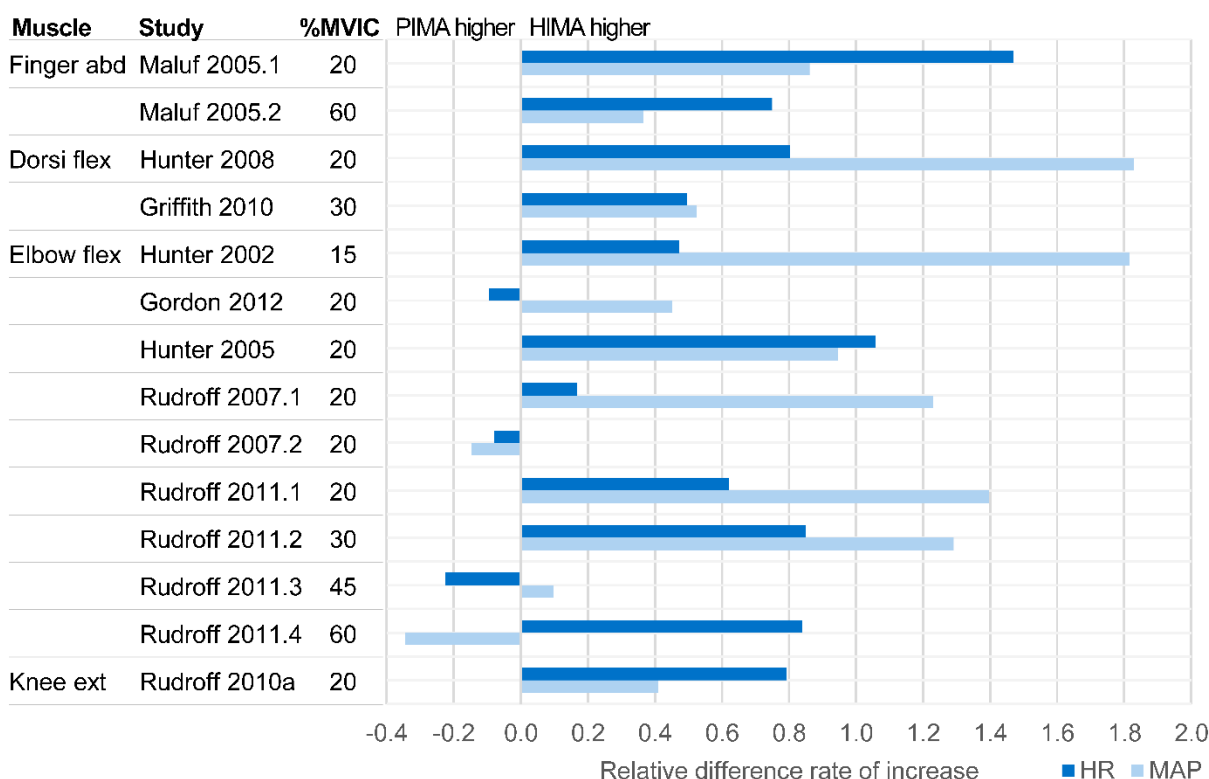


Figure 10. The relative difference between HIMA and PIMA is the rate of increase in heart rate (HR; dark blue) and mean arterial pressure (MAP; light blue). Note: The rate of increase was calculated from the start, end and TTF values for PIMA and HIMA, respectively. Relative differences=(HIMA-PIMA)/PIMA.

4. DISCUSSION

In sports sciences and medicine, it is uncommon to differentiate two types of isometric muscle action. PIMA (the most common isometric muscle action) is usually assessed to quantify muscle strength; however, the focus is increasingly shifting towards HIMA, which is being utilized more and more frequently in various sports medicine and performance practitioners to improve musculotendinous morphology and neuromuscular function [43-47]. Despite a wealth of studies comparing PIMA and HIMA, it is difficult for researchers and practitioners to understand their potential applications due to a wide range of methods and data reporting. Therefore, we aimed to identify and present all studies directly comparing the two types of isometric muscle action while

pooling and interpreting the data together where possible. We also aimed to provide researchers with the most pressing relevant questions and suggestions for future studies. The first question was whether two types of isometric muscle action exist. Based on this review, the answer is yes: HIMA and PIMA can be differentiated by objective measures. When examining the 53 studies comparing both isometric types, our main findings based on the majority of outcomes were that 1) PIMA had longer TTF than HIMA at the same relative intensity, except for the postural muscles of the axial skeleton, larger fluctuations of force, higher discharge rates, greater D1 inhibition and higher peak torques. 2) HIMA showed higher heteronymous facilitation, greater EMG burst rates, interspike interval CV, glucose uptake in muscles of young participants, and faster increases (though nearly identical absolute changes) in force/position fluctuations, EMG amplitude, RPE, HR, and MAP. 3) Although non-significant in most studies, there seems to be a trend for HIMA showing greater MVIC decline and higher coactivation than PIMA. 4) Non-significant or unclear differences (some outcomes revealed higher values for HIMA, some for PIMA) were present for the absolute values of EMG amplitude and frequency, MEP, RPE, HR, MAP and oxygenation. 5) HIMA vs PIMA showed differences in corticocortical and corticomuscular coherence and reacted differently to experimental joint pain.

4.1 Distinguishing holding and pushing isometric muscle actions

4.1.1 Muscle fatigue during HIMA and PIMA

The most frequently investigated parameter which showed the clearest picture was the TTF at the same intensity (force-matched). For appendicular muscles, the TTF was significantly longer for PIMA than HIMA, with large effect in meta-analysis ($g=0.90$). This suggests that maintaining an isometric position (HIMA) factor must be present, reducing the appendicular muscles' endurance ability. However, 12 of 30 outcomes were non-significant, and there was a trend that the phenomenon occurred only for low intensities ($<30\%$ MVIC). Only six studies investigated the TTF with intensities $\geq 50\%$ MVIC [4, 62, 63, 78, 87, 92]. No significant differences between the isometric types were found for five of the six outcomes. Possible reasons might lie in the methods. For example, Schaefer et al. [4] ($\geq 80\%$ MVIC), who found significantly shorter TTF for HIMA, used a pneumatic device to exert external force during the holding task against the participant without permitted position corrections. In contrast, other researchers used a weight whereby the position should be controlled. Three of them stated explicitly that corrections of the

position were allowed. It is suggested that minor muscle shortenings, which must be present by correcting the position, might change neuromuscular control processes in the direction of PIMA. The study of Dech et al. [78] supports this, as they investigated HIMA using a dumbbell without and with short and minor muscle twitches at 60% MVIC. With twitches, TTF was larger than exclusive HIMA and did not differ significantly from PIMA. However, some studies with low intensities also allowed for position corrections during HIMA and found significant differences. It is assumed that the intensity of muscle action might be a decisive factor in whether minor contractions in the direction of muscle shortening can change neuromuscular control. Another strong hint that differences in TTF are present also for high intensities was brought by a study in which the isometric types were investigated during personal interaction [5]. It was determined which partner was no longer able to maintain the task. In most cases (85%), the HIMA performing partner left the position first. As soon as they performed PIMA, they ‘won’ the interaction and their partner, who performed HIMA, yielded first. This was independent of conditional aspects like physical shape or MVIC. Since the setting was equal for HIMA and PIMA (personal interaction just with changing tasks) with the same amount of position stability and possibilities for correction, the differences must result from the different isometric tasks.

For axial muscles, the TTF was significantly longer for HIMA than for PIMA [65, 103], likely reflecting different methods of operation between appendicular and axial muscles. The latter must stabilize the body while standing, walking, or similar, and, thus, are probably more used to holding actions, while appendicular muscles are used to execute arbitrary movements. Although the results for axial muscles are based on only two studies, this might indicate that axial and appendicular muscles should probably be handled differently in assessment and treatment.

The significant differences between HIMA and PIMA regarding TTF suggest that, at least for appendicular muscles, HIMA is more challenging to maintain than PIMA. Hence, factors that contribute to this phenomenon must be present. One factor might lie in the relation of the submaximal intensities to the maximal force, which can be reached in the respective isometric types. In most studies, the intensity was based on the MVIC (max-PIMA). Only three studies directly compared max-HIMA and max-PIMA [18, 20, 23]. It must be considered that the participants had to adapt to an increasing load for HIMA, which was not necessary during PIMA.

Although the physical capacities were identical (same muscle, same participant), they could not generate such high maximum forces during HIMA as during PIMA (~77% of PIMA). This might indicate that the relative force levels in the fatiguing tasks were higher for HIMA than PIMA, although the absolute force level was the same. This could have contributed to the shorter TTF. However, it must be questioned why those restrictions in max torques and maintenance during holding actions occur.

4.1.2 Metabolic, cardiovascular and neurophysiological considerations

In line with the shorter TTF for HIMA compared to PIMA, some parameters (which differed not significantly for absolute values) showed a significantly higher increase during HIMA within one trial (e.g., fluctuation of force/position, RPE, MAP, HR, EMG amplitude and burst rate). The faster increases are strong indications that HIMA is more strenuous than PIMA. Since the perceived effort and MAP are mediated by central processes [82, 108], HIMA is assumed to require greater central processing. Different neural adjustments or motor control strategies are suggested by most researchers [30, 34, 63, 64, 70, 75, 80, 83, 86, 92, 95]. This is underpinned by the absence of differences in muscle oxygenation, which could have been an influencing factor [62, 78]. Spinal processing, especially, was considered relevant, like modulation of motor-unit recruitment and rate coding [63, 64, 74, 81, 84-86, 89, 91, 92, 95, 101], mainly indicated by the faster time course of change regarding different parameters during HIMA. However, central modulations were also suggested primarily based on higher central demands for holding actions (mainly reflected by MAP, HR) [25, 30, 35, 64, 83, 95, 102]. This is supported by the findings of lower D1 inhibition and higher heteronymous facilitation, suggesting a reduction of the presynaptic inhibition of Ia afferents and higher reflex responsiveness when maintaining a position [70-73, 88, 91, 107]. The presynaptic inhibition is primarily controlled by supraspinal centers and sensory feedback [109], reinforcing the relevance of supraspinal pathways for distinguishing both tasks. Accordingly, a strategy for HIMA was suggested “in which supraspinal centers choose to enhance the contribution of muscle afferents to the synaptic input that converges onto spinal motor neurons” [72, 110]. This aligns with the finding that HIMA needs more proprioceptive information [35, 88, 95]. One given explanation was that higher modulations during HIMA are necessary because of the more unstable task, which requires a higher reflex responsiveness [70]. This points to differences in the settings of HIMA and PIMA, which might have influenced the results. Indeed,

45 of 53 studies used an inertial load (43) or a similar approach (2) to realize HIMA (position control) with freedom to move in both directions, muscle shortening and lengthening. The more unstable task (HIMA) might have resulted in stronger neural modulations in those settings. In contrast, four studies used a pneumatic device to prevent muscle shortening [4, 18, 20, 23], and another four used personal interaction [5, 21, 22, 25]. In those studies, the ‘stability’ or ‘freedom to move’ was similar for both tasks. Nevertheless, HIMA and PIMA showed different outcomes. This suggests that, at least for those studies, the differences between HIMA and PIMA do not result from the stability of the position. If the findings of the 45 studies with uneven position stability during both tasks would have also arisen with a more stable position during HIMA remains open. However, there must be other explanations for the differences between the isometric types than the stability of the position.

4.1.3 HIMA and PIMA as ‘restricted’ or ‘stopped’ lengthening or shortening actions

It was previously hypothesized that the two types of isometrics might be a kind of ‘stopped’ or ‘restricted’ lengthening/eccentric (HIMA) or shortening/concentric (PIMA) muscle action [4, 5, 79]. This was mainly based on the execution of the tasks: during PIMA, the participants should push against resistance; if it is not stable, the muscle action will be concentric. For HIMA, the participants should resist a load, and if the participant’s force is reduced, the muscle action shifts to eccentric. This is also indicated by the so-called eccentric quasi-isometric (EQI) muscle actions [13-17]. If the ‘stopped’ muscle action hypothesis is true, it could explain the different outcomes since lengthening muscle actions require unique and more complex control strategies than shortening ones [111, 112]. For the isometric types, different motor control strategies were also suggested [4, 5, 20-22, 25, 27-29, 35, 64, 75]. However, it is difficult to compare the isometric with the anisometric types since the latter includes motion and muscle length changes. Furthermore, it must be considered that assessment of eccentrics using isokinetic devices commonly also includes pushing against the lever while it overcomes the participant’s muscle. This could change the neural control strategies. However, Grabiner and Owings [113] found lower EMG activity during isometric actions (PIMA) preceding maximal eccentric vs concentric actions, indicating a different motor command implemented by the CNS in expectation of an anisometric action. This would contradict the proposal that pushing against the lever might

diminish the neural characteristic of eccentric action and highlight the relevance of the task's expectations.

Despite those considerations, some similarities exist between HIMA and PIMA and the suggested anisometric counterparts. 1) The discharge rate was found to be “systematically lower during lengthening actions” compared to shortening ones [111], which indicates a lower excitability. Similarly, the discharge rate was significantly lower for HIMA vs PIMA in 8 of 12 outcomes [34, 74, 81, 95, 100]. 2) Faster RPE increase for HIMA vs PIMA: Carson et al. [108] reported that the level of force is overestimated during eccentric vs concentric muscle action. The faster RPE increase for HIMA vs PIMA suggests a similar estimation in the studies here. 3) Modulations of neural activation during lengthening actions are assumed to involve supraspinal and spinal mechanisms. However, they were mainly attributed to spinal pre- and postsynaptic inhibitory mechanisms [111, 114, 115]. A similar mechanism was suggested for the modulations during HIMA (see above) [63, 64, 70-74, 81, 84-86, 88, 89, 91, 92, 95, 101, 107]. For lengthening vs shortening muscle actions, a reduced peripheral activity (EMG) accompanied by a higher central activity (EEG) was found [116, 117]. Although Poortvliet et al. [97] found higher beta band power for HIMA vs PIMA, conclusions on those parameters cannot be made for the isometric types based on the scarce literature. Some related aspects are discussed below. 4) The voluntary activation is less for maximal lengthening vs shortening actions, for which insufficient neural commands were suggested to contribute to the lower force capacity [111]. This might support the reported lower max-HIMA achieved in adaptive holding actions [18, 20, 23].

No clear statement can be made when comparing anisometric to isometric muscle action types concerning H-reflex: it is usually depressed during lengthening actions [108] and passive ones [111, 118]. This indicates that the stretch or motion might be a factor for the reflex depression. However, in one study, HIMA led to a decreased H-reflex [89]. Other studies that used the H-reflex as a control reflex to investigate presynaptic inhibition adjusted the reflex to be equal for both tasks [70, 72, 73, 107]. Hence, further research into the behavior of the H-reflex during HIMA and PIMA is necessary.

Findings that might speak against the hypothesis are the MEP and the silent period after transcranial magnetic stimulation. They were found to be reduced and briefer, respectively, in lengthening vs shortening actions [111]. For HIMA vs PIMA, the results are inconsistent or speak against the findings of anisometric actions: three studies found no significant differences in MEP, while one outcome showed a lower increase during HIMA [30, 88, 89]. The silent period was briefer in one [88], non-significant in another [89] and larger in two studies [30, 105]. Despite those aspects, which might contradict the hypothesis that HIMA and PIMA are ‘stopped’ lengthening and shortening muscle actions, respectively, some findings support the assumption that the control strategies of HIMA and PIMA are closer to lengthening and shortening actions. Further research is needed to investigate this suggestion.

4.1.4 Neuromuscular control strategies

Modulations of cortical activity were also attributed to the differences in TTF of HIMA and PIMA. Poortvliet et al. [97] found less cortical network activity (cortico-cortical coherence) for HIMA than PIMA, but higher power in the beta band of the left hemisphere, and a non-significant corticomuscular coherence. They suggested that these findings are attributed to the postural-focused task during HIMA. A case report investigated inter-brain and inter-muscle-brain coherence during personal interaction (n=2) [25]. The inter-brain coherence was similar for both tasks, but the inter-muscle-brain coherence was significantly higher for HIMA than PIMA [25]. This might contradict the non-significant differences in corticomuscular coherence found by Poortvliet et al. [96, 97]. In addition to the small sample size [25], methods differed largely (e.g., EMG vs. MMG, intensity: 10% vs. 80% of MVIC, single-person vs personal interaction). Personal interaction is more complex than performing isometric tasks in a single-person setting. Furthermore, in Poortvliet et al. [96, 97], the stability of the position differed between HIMA and PIMA, which, as mentioned above, could affect the results. Anyway, the conclusions of both research groups seem to contradict each other: Poortvliet et al. [96, 97] consider PIMA to show higher cortical network activity, whereas Schaefer & Bittmann [25] suggested more complex control strategies during HIMA. However, the level of complexity is difficult to estimate using only those approaches. For example, Poortvliet et al. [97] suggested their findings could imply higher subcortical mechanisms during HIMA. This also feeds control mechanisms.

Differences in neural processing and the influence of pain were also suggested [96, 97]. PIMA was found to show higher pain-related modifications than HIMA, contradicting the proposed higher sensitivity to afferents for HIMA based on several studies [64, 84, 85, 92, 98, 99]. Yunoki et al. [35] suggested that sensory input unrelated to movement execution is suppressed during PIMA, especially when processing cutaneous information. During HIMA, “relatively more information regarding the joint angle and muscle length can be provided to the central nervous system” [35]. This would support the suggested higher sensorimotor control during HIMA. The fact that during HIMA, the participant must react to an applied load, while during PIMA, the participant acts against an immovable resistance already indicates that HIMA presumably requires greater responsiveness and sensorimotor modulation. The reaction and adaptation to an external load result in higher processing of proprioceptive inputs and, therefore, in a larger involvement of somatosensory areas. This hypothesis is supported by the higher inter-brain-muscle coherence [25], the more substantial motor-unit activity adjustments [74, 80, 89, 95, 100], the greater modulation of reflexes [70], and the higher interspike interval CV [34, 74, 81, 100] during HIMA. The latter might indicate more regulation since higher variability reflects adequate adaptation and feedback control [5, 119]. However, further research with consistent settings is necessary to conclude the neuromuscular control strategies for HIMA and PIMA.

The research on the Adaptive Force (AF) also speaks for more complex control strategies and a higher vulnerability for HIMA vs PIMA. The AF is based on HIMA where the load is not stationary but increasing. This task involves an even higher amount of reaction and adaptation as constant updates of the sensorimotor system are required [120]. As mentioned above, the max isometric AF (i.e., max-HIMA) was significantly lower than max-PIMA [18, 20, 23]. Moreover, it was significantly reduced in healthy participants in reaction to stimuli like negative vs positive imagery and odors [27-29], and produced slack vs physiologically adjusted muscle spindle [21, 22], and it was found to be significantly lower in patients with knee osteoarthritis [18] and Long-COVID [24]. However, whether the max-PIMA would have reacted similarly was not tested; one aspect speaks against this possibility. The results of Bittmann et al. [22], who assessed the maximal isometric AF (i.e., max-HIMA) during different pre-conditionings of muscle spindles: (1) contraction in lengthened position with passive return to middle (testing) position (provokes a slack of muscle spindles) and (2) contraction in lengthened position with passive return followed

by a second contraction in middle (testing) position. Only the former resulted in a clear and significant reduction of the holding capacity by 47%, while the latter did not. This shows that a voluntary ‘acting’ (not reacting) muscle activation of only ~20% MVIC leads to regaining the max holding capacity after a provoked muscle slack, like MVIC (max-PIMA) and max-HIMA at baseline, respectively. If pre-conditioning had been performed before a max-PIMA task, the execution would have led directly to a voluntary ‘acting’ muscle activation with even higher intensities than 20% MVIC. Hence, it is plausible that max-PIMA would not have reacted to pre-conditioning. It is therefore assumed that the length-tension control during HIMA, especially with the component of adapting to increasing external loads, can be affected by adequate stimuli reflecting a higher sensitivity and more complex control strategies than PIMA. This supports the suggested differences in sensorimotor modulation between both isometric types of other studies, which are essential for diagnostics and therapy. However, further research is required to examine how PIMA reacts to potentially disturbing stimuli and to investigate the underlying neuromuscular control mechanisms.

4.2 Practical applications

4.2.1 Injury prevention

When considering practical applications for the two types of isometric muscle action in injury prevention, two aspects should be considered separately: muscle function assessment and exercise interventions. For both, it is essential to consider injury mechanisms. It is known that non-contact injuries, which account for most injuries in contact sports [121, 122], arise during muscle lengthening under load; when a muscle must hold or decelerate an external force, during shortening contractions, they do not occur [123, 124]. Despite this knowledge, muscle strength is usually assessed by pushing against resistance, where the adaptive holding aspect is not considered. This reinforces the need for different muscle strength or function assessments to uncover potential impairments that might increase the risk of injury. It is suggested that not pushing or shortening but holding and stabilizing muscle actions should be considered due to the nature of injury mechanisms. As motions in sports and daily activities rarely include only one mode of muscle action and frequently involve the reaction and adaptation to external loads, HIMA in response to varying external loads might be a suitable approach to uncover impairments relevant to injury risks. This is proposed not only due to its proximity to motions in daily life or sports, but also

because of the suggested higher sensitivity of HIMA. The finding that the maximal adaptive holding capacity can be reduced immediately by impairing stimuli [18, 21, 22, 24, 27-29], might explain sudden non-contact injuries under normal load during running, direction changes or similar. However, such movements were performed a thousand times without injury. Research is required to investigate if a reduced adaptive holding function leads to a higher risk of injury or, vice versa, if stable holding functions prevent injuries. Additionally, causes of impairments of the adaptive holding functions need to be identified to potentially support injury prevention. The studies on AF indicate that some factors, such as negative emotions, affect most persons' holding capacity or muscular stability [27-29]. Nevertheless, it is proposed that impairing causes must be identified personally since it is very individual (e.g., a perceived mentally stressful situation). Furthermore, from practical experience, nociception, infections, inflammation, and other clinical issues can reduce holding capacity. Since each person brings their backpack of medical history and possible current limiting factors, an individual approach is necessary. In therapeutic practice, manual muscle testing has been used for decades to assess neuromuscular function [125, 126]. However, the execution and interpretation vary broadly. With the knowledge of HIMA and PIMA, it seems necessary to differentiate between holding and pushing actions. Due to the assumed higher sensorimotor control for HIMA, it is suggested that assessments in clinical environments, irrespective of manual or device control, should assess the reaction to external loads in a holding manner. Thereby, not the overall maximal torque should be considered, but the maximal torque reached under static position; thus, force and angle must be evaluated. Moreover, objectification and standardization are urgently needed and recommended for manual testing [127]. Although the rationale behind the approach to use HIMAs for muscle testing seems to be plausible no studies exist to the knowledge of the authors which investigate if stable holding muscle function supports injury prevention. It is hypothesized that 1) adaptive HIMA, in contrast to PIMA, is suitable to uncover functional impairments of neuromuscular control, 2) a reduced holding capacity increases the risk of injury, and 3) such reductions cannot be repealed solely by strength training because it is assumed to be a reflex response. Further research is necessary to investigate those hypotheses and the potential benefits of adaptive HIMA for diagnostics and injury prevention.

The scientific basis of isometric exercise intervention for injury prevention is poor. Two studies suggested additional isometric exercise in training routines reduces injury rates [128, 129].

However, no information was given on the specific nature of the isometric exercise utilized, and the quality of the studies was considered minor. In a review, Ullmann et al. [130] claimed that positive effects of isometric exercise on injury prevention are assumed because of post-activation potentiation. However, a distinction between HIMA and PIMA was not considered. Exercise programs for injury prevention should account for the complex requirements and adaptive nature of motions in sports. For example, the Nordic hamstring exercise may reduce the risk of hamstring injuries by up to 51%, as suggested by two meta-analyses [131, 132]. However, the newest review stated that the evidence supporting the protective effect is inconclusive [133]. Despite this disagreement, it is suggested that such complex and adaptive exercises should be included in injury prevention, respecting the requirements of the sports being performed.

4.2.2 Rehabilitation

The clinical and research interest in using isometric exercise programs for pain syndromes like tendinopathy has increased in recent years [134]. However, researchers used isometric exercise without differentiating between holding and pushing actions. Five of fifteen identified studies investigating the effect of isometric exercise on pain at least used the word “hold” in describing the program [38, 41, 135-137]. However, only one study examined pure HIMA [135]. The others assessed a combination of HIMA and PIMA, a phase of concentric or eccentric action before the holding phase or did not specify the execution of isometrics [38, 41, 136-147]. Van der Vlist et al. [138] compared the effect of four interventions (isometric dorsiflexed, isometric plantarflexed (tiptoes), isotonic, rest) on immediate improvement of Achilles tendinopathy symptoms. The dorsiflexed isometric intervention, which is assumed to be closest to HIMA, showed a significantly lower VISA-A score (pain, activity, function) than the isotonic and rest groups. However, the stretched position of the Achilles tendon might have contributed to this effect. Overall, no firm evidence exists that isometric exercise is superior to isotonic or other treatments for pain relief [134]. Since no study directly compared HIMA and PIMA exercise programs, no statement can be made on their efficacy in reducing pain or other symptoms in rehabilitation.

However, some suggestions should be made based on the findings of both isometric types. Based on the longer TTF, PIMA would be recommendable if prolonged activation is required. It

was suggested, additionally, to improve cutaneous information processing [35]. HIMA should be preferred if a higher sensorimotor activation and stability control is desired. Furthermore, due to the higher demands on neural control strategies that have been suggested, it could be beneficial for neurological rehabilitation. Following Bauer et al. [75], the two isometric types in rehabilitation should be incorporated and differentiated to fit the individual patient's needs. Based on the findings on the maximal holding capacity and the considerations regarding injury prevention, it is suggested that the focus in rehabilitation should be on muscle stability in adaptation to external loads rather than on muscle strength.

4.2.3 Sports performance

Isometric strength training for sports performance has gained popularity recently as more evidence of its efficacy in improving various sports performance indicators became available [1, 39, 40, 148-151]. However, most of these studies performed PIMA [148-151], with only a limited number of studies employing HIMA [39, 40]. Additionally, no study compared the efficacy of PIMA and HIMA in improving sports performance. Hence, it may be difficult for practitioners to decide which method to use at different phases of the training cycle. Nevertheless, information from the current review may provide some insights to guide practitioners on when best to include either HIMA or PIMA in their athletes' training. Firstly, during the general preparation phase, where the training objective is to increase the overall strength to endure the demands of the subsequent phase and to reduce the risk of injury as training intensity increases, practitioners may include HIMA. This is because HIMA results in greater synergist muscle activation, which will be important in maintaining joint integrity during intensive movement, thus lowering injury risks [20]. Furthermore, appendicular muscles can sustain PIMA longer than HIMA at a given intensity, while the opposite finding was true for the trunk muscles. During the general preparation phase, coaches plan for a higher training volume, meaning that contraction duration would be longer if isometric training is included. If PIMA were to be used during an exercise such as the isometric squat, the adaptation of the lower limb may be limited by the duration at which the trunk muscle could sustain a given force. However, if HIMA were to be used, the trunk muscles would likely be able to maintain the given force for as long as, or longer, than the lower limb. Hence, the adaptation to the lower limb would less likely be limited by the trunk muscles' ability to sustain the force. Secondly, during the training phase, where improving maximum force development is

the main objective, it is likely that PIMA would be preferred as it involves exerting force against an immovable object, where an athlete could exert maximal force. In contrast, HIMA typically involves exerting submaximal force. Similarly, several studies [18, 20, 23] demonstrated that PIMA versus HIMA makes greater maximal torques possible. Hence, high-intensity PIMA would likely result in greater strength adaptations, at least without considering near-maximal HIMA followed by eccentric muscle action [13].

The application of either PIMA or HIMA may also be considered concerning the SAID principle (specific adaptations to imposed demands) [152, 153]. The SAID principle states that the human body will adapt specifically to the needs that are placed on it [152]. While there is rarely only one mode of muscle contraction in sports, brief or extended moments of isometric action are typical in many sports movements. Hence, the decision to use PIMA or HIMA in training may also rely on the specific sporting action(s) that must be developed and prepared for. For example, sporting actions against high external loads involving relatively static pushing, like a rugby union scrum or grappling/wrestling maneuver, may best be trained using PIMA. Sporting actions that require more of a holding action, like the static position held during a springboard dive or different static positions held during gymnastic maneuvers with the rings, may best be prepared for using HIMA.

Finally, if improving the rate of force development is the objective, both PIMA and HIMA with rapid contraction can be utilized, as it has been previously reported that rapid isometric contractions are effective [153, 154]. While it is relatively simple to perform PIMA with rapid contraction, practitioners may not have insights into how to perform HIMA with rapid contraction, including rapid landings or untraditional exercises such as rapid decelerations, ‘hamstring switches,’ or oscillatory training, almost all of which are lacking in the scientific literature. Only one approach was found using pneumatics to induce rapid HIMAs [26]. Furthermore, while several examples of multi-joint HIMA exist [39, 40, 155], no studies have compared PIMA and HIMA muscle actions via closed-chain movements, presenting several exciting research opportunities.

4.3 Limitations and future research directions

While the primary aims were accomplished, there are numerous limitations. First, we included only studies that directly compared PIMA and HIMA. While it would have been exceptionally time-consuming, we could have collected and extracted data from all studies employing PIMA or HIMA. While enough studies utilized TTF as a dependent variable for confident meta-analyses, far fewer studies reported HR, MAP, or RPE; therefore, readers should understand the somewhat limited authority of these analyses. Similarly, the between-study variability in neurological (e.g., EEG, CT) and neuromuscular (e.g., sEMG, intramuscular EMG) assessment methodology prevented us from performing further meta-analyses. Moreover, the small number of studies focusing on the same objective (e.g., brain activity, tendon oscillations, max torque) makes final statements difficult. Other methods may uncover further insights, including fMRI, fascicle tracking, echo intensity, shear wave elastography, and tensiomyography. The ‘stability of position’ factor was impossible to include in the interpretation of data (i.e., HIMA is typically more unstable than PIMA). Thus, aligning the position stability during both functions would be advisable. While the risk-of-bias was modest for all studies, our quality scoring criteria are somewhat subjective. As such, other researchers may have different means of assessing quality. Many included studies did not provide effect size statistics, leaving us to calculate them based on text, tables, or figure extractions, potentially (minimally) affecting some of the meta-analyses. The mean contraction intensity of the included studies was far lower than nearly all but the most conservative rehabilitation or sports performance guidelines. While these low loads allowed for greater contraction durations and, thus, more time intervals for comparison, future studies should examine more practically relevant intensities. An even more explicit limitation is the complete absence of studies comparing PIMA and HIMA during multi-joint movements (e.g., squat, leg press, pull-up, bench press). Researchers could also compare the two types of isometric muscle action over multiple days to examine the time course of fatigue and recovery. Moreover, investigations on how external and internal stimuli might influence HIMA and PIMA could be relevant for gathering further information on underlying control mechanisms and for diagnostics, injury prevention and rehabilitation. Most importantly, longitudinal, randomized control trials are required to ascertain if PIMA or HIMA have any measurable advantage over the other for use in diagnostics and for inducing various rehabilitation or sports performance adaptations.

5. CONCLUSIONS

Objective measures can distinguish two types of isometric muscle action with the synthesized findings suggest more complex control strategies for HIMA than PIMA. It is hypothesized that the neuromuscular processing of HIMA and PIMA is close to those of muscle lengthening and shortening, respectively, indicating they might be ‘stopped’ anisometric actions. Due to their excellent responsiveness and sensorimotor modulation, HIMAs could be beneficial for diagnostics, injury prevention and rehabilitation, especially by including varying external loads where the neuromuscular system can be challenged further, showing immediate responses to impairing and helpful stimuli that could be used for therapeutic purposes. When integrated with relevant physiological, biomechanical, and psychological research, HIMAs could provide a time-efficient approach for inducing musculoskeletal, neural, and cardiovascular adaptations in rehabilitation settings. PIMAs may be more effective for prolonged activation and for promoting agonist neuromuscular adaptations, primarily due to their greater force capacity. The wide variation in methods across studies precludes additional meta-analyses. Consistent methodology and improved raw data reporting are needed. Randomized controlled trials are necessary to validate the use of HIMA vs PIMA in clinical or sports performance contexts. It is recommended that the existence and knowledge of both isometric types be implemented in research and education in sports, health sciences, and medicine.

Author contributions

DJO conceptualized the review topic and formed the research team. DJO and ARN fine-tuned the research question and review methods. DJO and ARN performed the literature search and initial screenings. DJO, LVS, DL, AON and ARN assessed inclusion and quality/bias studies. DJO and LVS extracted data, while ARN and FNB checked the extracted data for accuracy. DJO, LVS and ARN performed the statistical analyses and created the figures. DJO wrote the first draft of the title, abstract, introduction and methods. DJO and LVS wrote the first draft of the results. All authors contributed to the discussion's first draft and approved the manuscript's final version.

REFERENCES

1. Oranchuk DJ, Storey AG, Nelson AR, Cronin JB. Isometric training and long-term adaptations; effects of muscle length, intensity, and intent: A systematic review. *Scand J Med Sci Sports*. 2019;29(4):484-503.

- 1131 2. Douglas J, Pearson S, Ross A, McGuigan M. Chronic adaptations to eccentric training: A
1132 systematic review. *Sports Med.* 2017;47(5):917-41.
- 1133 3. Douglas J, Pearson S, Ross A, McGuigan M. Eccentric exercise: Physiological
1134 characteristics and acute responses. *Sports Med.* 2017;47(4):663-75.
- 1135 4. Schaefer LV, Bittmann FN. Are there two forms of isometric muscle action? Results of the
1136 experimental study support a distinction between a holding and a pushing isometric muscle
1137 function. *BMC Sports Sci Med Rehabil.* 2017;9(11):1-13.
- 1138 5. Schaefer LV, Bittmann FN. Paired personal interaction reveals objective differences
1139 between pushing and holding isometric muscle action. *PLoS One.* 2021;16(5):e0238331.
- 1140 6. Sharifnezhad A, Marzilger R, Arampatzis A. Effects of load magnitude, muscle length and
1141 velocity during eccentric chronic loading on the longitudinal growth of the vastus lateralis muscle.
1142 *J Exp Biol.* 2014;217(Pt 15):2726-33.
- 1143 7. Kassiano W, Costa B, Kunevaliki G, Soares D, Zacarias G, Manske I, et al. Greater
1144 gastrocnemius muscle hypertrophy after partial range of motion training carried out at long muscle
1145 lengths. *J Strength Cond Res.* 2023;37(9):1746-53.
- 1146 8. Pedrosa GF, Lima FV, Schoenfeld BJ, Lacerda LT, Simões MG, Pereira MR, et al. Partial
1147 range of motion training elicits favorable improvements in muscular adaptations when carried out
1148 at long muscle lengths. *Eur J Sport Sci.* 2022;22(8):1250-60.
- 1149 9. Chapman D, Newton M, Sacco P, Nosaka K. Greater muscle damage induced by fast
1150 versus slow velocity eccentric exercise. *Int J Sports Med.* 2006;27(8):591-8.
- 1151 10. Sterling W. Outlines of practical physiology. 2nd ed. Philadelphia: P. Blakiston, Son &
1152 Co; 1895.
- 1153 11. Kordi M, Evans M, Howatson G. Quasi-isometric cycling: A case study investigation of a
1154 novel method to augment peak power output in sprint cycling. *Int J Sports Physiol Perform.*
1155 2021;16(3):452-5.
- 1156 12. Spurway NC. Hiking physiology and the "quasi-isometric" concept. *J Sports Sci.*
1157 2007;25(10):1081-93.
- 1158 13. Oranchuk DJ, Nelson AR, Storey AG, Cronin JB. Scientific basis of eccentric quasi-
1159 isometric resistance training: A narrative review. *J Strength Cond Res.* 2019;33(10):2846-59.
- 1160 14. Oranchuk DJ, Nelson AR, Storey AG, Diewald SN, Cronin JB. Short-term neuromuscular,
1161 morphological, and architectural responses to eccentric quasi-isometric muscle actions. *Eur J Appl*
1162 *Physiol.* 2021;121(1):141-58.
- 1163 15. Henderson ZJ, Wang S, Cornish SM, Scribbans TD. Exploring the acute muscle fatigue
1164 response in resistance trained individuals during eccentric quasi-isometric elbow flexions-A cross-
1165 sectional comparison of repetition and sex. *Sports Biomech.* 2023;Ahead of print.

- 1166 16. Oranchuk DJ, Diwald SN, McGrath JW, Nelson AR, Storey AG, Cronin JB. Kinetic and
1167 kinematic profile of eccentric quasi-isometric loading. *Sports Biomech.* 2024;23(6):758-71.
- 1168 17. Ličen U, Oranchuk DJ, Kozinc Z. Exploring the biomechanics and fatigue patterns of
1169 eccentric quasi-isometric muscle actions in the knee extensors and flexors. *Eur J Appl Physiol.*
1170 2024;Ahead of print.
- 1171 18. Schaefer LV, Dech S, Carnarius F, Ronnert F, Bittmann FN, Becker R. Adaptive force of
1172 hamstring muscles in reduced in patients with knee osteoarthritis compared to asymptomatic
1173 controls. *BMC Musculoskeletal Disorders.* 2024;25(34).
- 1174 19. Hoff M, Schaefer L, Heinke N, Bittmann F. Report on adaptive force, a specific
1175 neuromuscular function. *European Journal of Translational Myology.* 2015;25(3):183-9.
- 1176 20. Schaefer LV, Carnarius F, Dech S, Bittmann FN. Repeated measurements of adaptive
1177 force: Maximal holding capacity differs from other maximal strength parameters and preliminary
1178 characteristics for non-professional strength vs. endurance athletes. *Front Physiol.*
1179 2023;14(1020954):1-19.
- 1180 21. Bittmann FN, Dech S, Schaefer LV. Another way to confuse motor control: Manual
1181 technique supposed to shorten muscle spindles reduces the muscular holding stability in the sense
1182 of adaptive force in male soccer players. *Brain Sciences.* 2023;13(7):1105.
- 1183 22. Bittmann FN, Dech S, Schaefer LV. How to confuse motor control: Passive muscle
1184 shortening after contraction in lengthened position reduces the muscular holding stability in the
1185 sense of adaptive force. *Life.* 2023;13(911):1-18.
- 1186 23. Dech S, Bittmann FN, Schaefer LV. Assessment of the adaptive force of elbow extensions
1187 in healthy subjects quantified by a novel pneumatically driven measurement system with
1188 considerations of its quality criteria. *Diagnostics.* 2021;11(923):1-23.
- 1189 24. Schaefer LV, Bittmann FN. The adaptive force as a potential biomechanical parameter in
1190 the recovery process of patients with long COVID. *Diagnostics.* 2023;13(5):882.
- 1191 25. Schaefer LV, Bittmann FN. Case study: Intra- and interpersonal coherence of muscle and
1192 brain activity of two coupled persons during pushing and holding isometric muscle actions. *Brain*
1193 *Sciences.* 2022;12(6):703.
- 1194 26. Schaefer LV, Bittmann FN. Muscular pre-activation can boost the maximal explosive
1195 eccentric adaptive force. *Front Physiol.* 2019;10(910).
- 1196 27. Schaefer LV, Dech S, Aehle M, Bittmann FN. Disgusting odours affect the characteristics
1197 of the Adaptive Force in contrast to neutral and pleasant odours. *Sci Rep.* 2021;11(16410).
- 1198 28. Schaefer LV, Dech S, Bittmann FN. Adaptive Force and emotionally related imaginations
1199 – preliminary results suggest a reduction of the maximal holding capacity as reaction to disgusting
1200 food imagination. *Heliyon.* 2021;7(8):e07827.

- 1201 29. Schaefer LV, Dech S, Wolff LL, Bittmann FN. Emotional imagery influences the Adaptive
1202 Force in young women: Unpleasant imagery reduces instantaneously the muscular holding
1203 capacity. *Brain Sciences*. 2022;12(10):1318.
- 1204 30. Mathis J, de Quervain D, Hess CW. Task-dependent effects on motor-evoked potentials
1205 and on the following silent period. *J Clin Neurophysiol*. 1999;16(6):556-65.
- 1206 31. Dietz C, Peterson B. Triphasic training: A systematic approach to elite speed and explosive
1207 strength performance. Hudson, WI: Dietz Sport Enterprise; 2012.
- 1208 32. Verkhoshansky V, Verkhoshansky N. Special strength training manual for coaches. Rome,
1209 Italy: Verkhoshansky SSTM; 2011.
- 1210 33. Rudroff T, Kalliokoski KK, Block DE, Gould JR, Klingensmith WC, Enoka RM. PET/CT
1211 imaging of age- and task-associated differences in muscle activity during fatiguing contractions. *J*
1212 *Appl Physiol*. 2013;114(9):1211-9.
- 1213 34. Pascoe MA, Gould JR, Enoka RM. Motor unit activity when young and old adults perform
1214 steady contractions while supporting an inertial load. *J Neurophysiol*. 2013;109(4):1055-64.
- 1215 35. Yunoki K, Watanabe T, Matsumoto T, Kuwabara T, Horinouchi T, Ito K, et al. Cutaneous
1216 information processing differs with load type during isometric finger abduction. *PLoS One*.
1217 2022;17(12):e0279477.
- 1218 36. Oranchuk DJ, Gullett LK, Kicia M, Thome B, Game A. The relationship of anthropometric
1219 and physical performance characteristics on competitive success in amateur, elite, and professional
1220 rodeo athletes. *J Strength Cond Res*. 2023;37(7):1495-506.
- 1221 37. Juneau CM, Diewald SN, Cronin JB, Neville J, Oranchuk DJ. Intersession variability of
1222 knee extension kinetics using a strain gauge device with differing clinically practical physical
1223 constraints. *J Sport Rehabil*. 2023;31(1):96-101.
- 1224 38. Bradford B, Rio E, Murphy M, Wells J, Khondoker M, Clarke C, et al. Immediate effects
1225 of two isometric calf muscle exercise on mid-portion achilles tendon pain. *Int J Sports Med*.
1226 2021;42(12):1122-7.
- 1227 39. Allegue H, Khmiri A, Oranchuk DJ, Chelly MS. The effect of combined isometric and
1228 plyometric- versus contrast strength training on athletic performance in male junior handball
1229 players. *Applied Sciences*. 2023;13(16):9069.
- 1230 40. Lee B, McGill S. The effect of core training on distal limb performance during ballistic
1231 strike manoeuvres. *J Sports Sci*. 2017;35(18):1768-80.
- 1232 41. Rio E, van Ark M, Docking S, Moseley GL, Kidgell D, Gaida JE, et al. Isometric
1233 contractions are more analgesic than isotonic contractions for patellar tendon pain: An in-season
1234 randomized clinical trial. *Clin J Sport Med*. 2017 May;27(3):253-9.

1235 42. Le Flao E, Pichardo AW, Ganpatt S, Oranchuk DJ. An accessible, 16-week neck strength
1236 training program improves head kinematics following chest perturbation in young soccer athletes.
1237 J Sport Rehabil. 2021;30(8):1158-65.

1238 43. CeJ. Hold on, it's Isometric Time. 2024 [cited; Available from:
1239 <https://unionfitness.com/hold-on-its-isometric-time/>

1240 44. Natera A. Modern Speed Training: Q&A with Alex Natera on Isometrics for Running
1241 Performance. 2017 [cited; Available from: [https://www.just-fly-sports.com/modern-speed-](https://www.just-fly-sports.com/modern-speed-training-alex-natera/)
1242 [training-alex-natera/](https://www.just-fly-sports.com/modern-speed-training-alex-natera/)

1243 45. Natera A. All isometric training is not the same: Differences in the execution, prescription
1244 and use of pushing and holding isometrics. 2021 [cited; Available from:
1245 [https://www.sportsmith.co/articles/all-isometric-training-is-not-the-same-differences-in-the-](https://www.sportsmith.co/articles/all-isometric-training-is-not-the-same-differences-in-the-execution-prescription-and-use-of-pushing-and-holding-isometrics/)
1246 [execution-prescription-and-use-of-pushing-and-holding-isometrics/](https://www.sportsmith.co/articles/all-isometric-training-is-not-the-same-differences-in-the-execution-prescription-and-use-of-pushing-and-holding-isometrics/)

1247 46. Plaza JV. Isometric Training for Building Muscle. 2023 [cited; Available from:
1248 <https://vitruve.fit/blog/isometric-training-for-building-muscle/>

1249 47. Simpson P. Two Types of Isometrics!? 2022 [cited; Available from:
1250 <https://www.airspaceperth.com.au/blog/two-types-of-isometrics>

1251 48. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The
1252 PRISMA 2020 statement: An updated guideline for reporting systematic reviews. BMJ.
1253 2021;372(n71).

1254 49. Ouzzani M, Hammady H, Fedorowics Z, Elmagarmid A. Rayyan - A web and mobile app
1255 for systematic reviews. Systematic Reviews. 2016;5(210).

1256 50. H DS, Black N. The feasibility of creating a checklist for the assessment of the
1257 methodological quality both of randomised and non-randomised studies of health care
1258 interventions. J Epidemiol Community Health. 1998;52(6):377-84.

1259 51. Sterne JA, Hernan MA, Reeves BC, Savovic J, Berkman ND, Viswanathan M, et al.
1260 ROBINS-I: A tool for assessing risk of bias in non-randomised studies of interventions. BMJ.
1261 2016;355(i4919).

1262 52. Sterne JAC, Savovic J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a
1263 revised tool for assessing risk of bias in randomised trials. BMJ. 2019;366(I4898).

1264 53. Manjali JJ, Gupta T. Critical appraisal of a clinical research paper: What one needs to
1265 know. Cancer Research, Statistics, and Trends. 2020;3(3):545-51.

1266 54. Medicine CfE-B. Study designs. 2024 [cited; Available from:
1267 <https://www.cebm.ox.ac.uk/resources/ebm-tools/study-designs>

- 1268 55. Barker TH, Habibi N, Aromataris E, Stone JC, Leonardi-Bee J, Sears K, et al. The revised
1269 JBI critical appraisal tool for the assessment of risk of bias for quasi-experimental studies. JBI
1270 Evidence Synthesis. 2024;22(3):378-88.
- 1271 56. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in
1272 sports medicine and exercise science. Med Sci Sports Exerc. 2009;41(1):3-13.
- 1273 57. McKay AKA, Stellingwerff T, Smith ES, Martin DT, Mujika I, Goosey-Tolfrey VL, et al.
1274 Defining training and performance caliber: A participant classification framework. Int J Sports
1275 Physiol Perform. 2022;17(2):317-31.
- 1276 58. McGuinness LA, Higgins JPT. Risk-of-bias VISualization (robvis): An R package and
1277 Shiny web app for visualizing risk-of-bias assessments. Research Synthesis Methods.
1278 2020;12(1):55-61.
- 1279 59. Cohen J. Statistical Power Analysis for the Behavioural Sciences. Hillsdale, NJ: Lawrence
1280 Erlbaum Associates; 1988.
- 1281 60. Park S, Beretvas SN. Synthesizing effects for multiple outcomes per study using robust
1282 variance estimation versus the three-level model. Behavior Research Methods. 2019;51(1):152-
1283 71.
- 1284 61. Becker BJ. Multivariate meta-analysis: Academic Press; 2000.
- 1285 62. Booghs C, Baudry S, Enoka RM, Duchateau J. Influence of neural adjustments and muscle
1286 oxygenation on task failure during sustained isometric contractions with elbow flexor muscles.
1287 Exp Physiol. 2012;97(8):918-29.
- 1288 63. Rudroff T, Justice JN, Holmes MR, Matthews SD, Enoka RM. Muscle activity and time to
1289 task failure differ with load compliance and target force for elbow flexor muscles. J Appl Physiol.
1290 2011;110(1):125-36.
- 1291 64. Rudroff T, Barry BK, Stone AL, Barry CJ, Enoka RM. Accessory muscle activity
1292 contributes to the variation in time to task failure for different arm postures and loads. J Appl
1293 Physiol. 2007;102(3):1000-6.
- 1294 65. Thomas JS, Ross AJ, Russ DW, Clark BC. Time to task failure of trunk extensor muscles
1295 differs with load type. J Motor Behav. 2010;43(1):27-9.
- 1296 66. Holzbaur KRS, Murry WM, Gold GE, Delp SL. Upper limb muscle volumes in adult
1297 subjects. J Biomech. 2007;40(4):742-9.
- 1298 67. Ito J, Moriyama H, Inokuchi S, Goto N. Human lower limb muscles: An evaluation of
1299 weight and fiber size. Okajimas Folia Anat Jpn. 2003;80(2-3):47-55.
- 1300 68. Higgins JPT, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-
1301 analyses. BMJ. 2003;327(7414):557-60.

- 1302 69. Sterne JAC, Sutton AJ, Ioannidis JPA, Terrin N, Jones DR, Lau J, et al. Recommendations
1303 for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled
1304 trials. *BMJ*. 2011;343(d4002).
- 1305 70. Baudry S, Enoka RM. Influence of load type on presynaptic modulation of Ia afferent input
1306 onto two synergist muscles. *Exp Brain Res*. 2009;199(1):83-8.
- 1307 71. Baudry S, Jordan K, Enoka RM. Heteronymous reflex responses in a hand muscle when
1308 maintaining constant finger force or position at different contraction intensities. *Clin*
1309 *Neurophysiol*. 2009;120(1):210-7.
- 1310 72. Baudry S, Maerz AH, Enoka RM. Presynaptic modulation of Ia afferents in young and old
1311 adults when performing force and position control. *J Neurophysiol*. 2010;103(2):623-31.
- 1312 73. Baudry S, Maerz AH, Gould JR, Enoka RM. Task- and time-dependent modulation of Ia
1313 presynaptic inhibition during fatiguing contractions performed by humans. *J Neurophysiol*.
1314 2011;106(1):265-73.
- 1315 74. Baudry S, Rudroff T, Pierpoint LA, Enoka RM. Load type influences motor unit
1316 recruitment in biceps brachii during a sustained contraction. *J Neurophysiol*. 2009;102(3):1725-
1317 35.
- 1318 75. Bauer P, Gomes JS, Oliveira J, Santos P, Pezarat-Correia P, Vaz JR. Torque regulation is
1319 influenced by the nature of the isometric contraction. *Sensors*. 2023;23(2):726.
- 1320 76. Bojsen-Møller J, Schwartz S, Magnusson SP. Limited difference in time to failure between
1321 sustained force and position control contractions with the knee extensors. *Scand J Med Sci Sports*.
1322 2011;21(6):e48-55.
- 1323 77. Coletta NA, Mallette MM, Gabriel DA, Tyler CJ, Cheung SS. Core and skin temperature
1324 influences on the surface electromyographic responses to an isometric force and position task.
1325 *PLoS One*. 2018;13(3):e0195219.
- 1326 78. Dech S, Bittmann FN, Schaefer LV. Muscle oxygenation and time to task failure of
1327 submaximal holding and pulling isometric muscle actions and influence of intermittent voluntary
1328 muscle twitches. *BMC Sports Sci Med Rehabil*. 2022;14(55).
- 1329 79. Garner JC, Blackburn T, Wiemar W, Campbell B. Comparison of electromyographic
1330 activity during eccentrically versus concentrically loaded isometric contractions. *J Electromyogr*
1331 *Kinesiol*. 2008;18(3):466-71.
- 1332 80. Gordon NM, Rudroff T, Enoka JA, Enoka RM. Handedness but not dominance influences
1333 variability in endurance time for sustained, submaximal contractions. *J Neurophysiol*.
1334 2012;108(5):1501-10.
- 1335 81. Gould JR, Cleland BT, Mani D, Amiridis IG, Enoka RM. Motor unit activity in biceps
1336 brachii of left-handed humans during sustained contractions with two load types. *J Neurophysiol*.
1337 2016;116(3):1358-65.

- 1338 82. Griffith EE, Yoon T, Hunter SK. Age and load compliance alter time to task failure for a
1339 submaximal fatiguing contraction with the lower leg. *J Appl Physiol*. 2010;108(6):1510-9.
- 1340 83. Hunter SK, Rochette L, Critchlow A, Enoka RM. Time to task failure differs with load
1341 type when old adults perform a submaximal fatiguing contraction. *Muscle Nerve*. 2005;31(6):730-
1342 40.
- 1343 84. Hunter SK, Ryan DL, Ortega JD, Enoka RM. Task differences with the same load torque
1344 alter the endurance time of submaximal fatiguing contractions in humans. *J Neurophysiol*.
1345 2002;88(6):3087-96.
- 1346 85. Hunter SK, Yoon T, Farinella J, Griffith EE, Ng AV. Time to task failure and muscle
1347 activation vary with load type for a submaximal fatiguing contraction with the lower leg. *J Appl*
1348 *Physiol*. 2008;105(2):463-72.
- 1349 86. Jeon S, Miller WM, Ye X. A comparison of motor unit control strategies between two
1350 different isometric tasks. *International Journal of Environmental Research and Public Health*.
1351 2020;17(8):2799.
- 1352 87. Jeon S, Ye X, Miller WM. Sex comparisons of agonist and antagonist muscle
1353 electromyographic parameters during two different submaximal isometric fatiguing tasks. *Physiol*
1354 *Rep*. 2019;7(5):e14022.
- 1355 88. Kirimoto H, Tamaki H, Suzuki M, Matsumoto T, Sugawara K, Kojima S, et al.
1356 Sensorimotor modulation differs with load type during constant finger force or position. *PLoS*
1357 *One*. 2014;9(9):e108058.
- 1358 89. Klass M, Levenez M, Enoka RM, Duchateau J. Spinal mechanisms contribute to
1359 differences in the time to failure of submaximal fatiguing contractions performed with different
1360 loads. *J Neurophysiol*. 2008;99(3):1096-104.
- 1361 90. Kunugi S, Holobar A, Kodera T, Toyoda H, Watanabe K. Motor unit firing patterns on
1362 increasing force during force and position tasks. *J Neurophysiol*. 2021;126(5):1653-9.
- 1363 91. Maluf KS, Barry BK, Riley ZA, Enoka RM. Reflex responsiveness of a human hand
1364 muscle when controlling isometric force and joint position. *Clin Neurophysiol*. 2007;118(9):2063-
1365 71.
- 1366 92. Maluf KS, Shinohara M, Stephenson JL, Enoka RM. Muscle activation and time to task
1367 failure differ with load type and contraction intensity for a human hand muscle. *Exp Brain Res*.
1368 2005;167(2):165-77.
- 1369 93. Marion R, Power GA. Residual force enhancement due to active muscle lengthening allows
1370 similar reductions in neuromuscular activation during position and force control tasks. *J Sport*
1371 *Health Sci*. 2020;9(6):670-6.

- 1372 94. Mottram CJ, Christou EA, Meyer FG, Enoka RM. Frequency modulation of motor unit
1373 discharge has task-dependent effects on fluctuations in motor output. *J Neurophysiol.*
1374 2005;94(4):2878-87.
- 1375 95. Mottram CJ, Jakobi JM, Semmler JG, Enoka RM. Motor-unit activity differs with load
1376 type during a fatiguing contraction. *J Neurophysiol.* 2005;93(3):1381-92.
- 1377 96. Poortvliet PC, Tucker KJ, Finnigan S, Scott D, Hodges PW. Experimental pain decreases
1378 corticomuscular coherence in a force- but not a position-control task. *J Pain.* 2019;20(2):192-200.
- 1379 97. Poortvliet PC, Tucker KJ, Finnigan S, Scott D, Sowman P, Hodges PW. Cortical activity
1380 differs between position- and force-control knee extension tasks. *Exp Brain Res.*
1381 2015;233(12):3447-57.
- 1382 98. Poortvliet PC, Tucker KJ, Hodges PW. Changes in constraint of proximal segments effects
1383 time to task failure and activity of proximal muscles in knee position-control tasks. *Clin*
1384 *Neurophysiol.* 2013;124(4):732-9.
- 1385 99. Poortvliet PC, Tucker KJ, Hodges PW. Experimental pain has a greater effect on single
1386 motor unit discharge during force-control than position-control tasks. *Clin Neurophysiol.*
1387 2015;126(7):1378-86.
- 1388 100. Rudroff T, Jordan K, Enoka JA, Matthews SD, Baudry S, Enoka RM. Discharge of biceps
1389 brachii motor units is modulated by load compliance and forearm posture. *Exp Brain Res.*
1390 2010;202(1):111-20.
- 1391 101. Rudroff T, Justice JN, Matthews S, Zuo R, Enoka RM. Muscle activity differs with load
1392 compliance during fatiguing contractions with the knee extensor muscles. *Exp Brain Res.*
1393 2010;203(2):307-16.
- 1394 102. Rudroff T, Poston B, Shin I-S, Bojsen-Møller F, Enoka RM. Net excitation of the motor
1395 unit pool varies with load type during fatiguing contractions. *Muscle Nerve.* 2005;31(1):78-87.
- 1396 103. Russ DW, Ross AJ, Clark BC, Thomas JS. The effects of task type on time to task failure
1397 during fatigue: A modified Sørensen test. *Journal of Motor Behavior.* 2018;50(1):96-103.
- 1398 104. Semmler JG, Kornatz KW, Dinunno DV, Shi Z, Enoka RM. Motor unit synchronisation is
1399 enhanced during slow lengthening contractions of a hand muscle. *The Journal of Physiology.*
1400 2002;545(2):681-95.
- 1401 105. Williams PS, Hoffman RL, Clark BC. Cortical and spinal mechanisms of task failure of
1402 sustained submaximal fatiguing contractions. *PLoS One.* 2014;9(3):e93284.
- 1403 106. Buchanan TS, Lloyd DG. Muscle activity is different for humans performing static tasks
1404 which require force control and position control. *Neurosci Lett.* 1995;194(1-2):61-4.

- 1405 107. Magalhães FHE, L A, da Silva CR, de Lima FF, de Toledo DR, Kohn AF. D1 and D2
1406 inhibitions of the soleus H-reflex are differentially modulated during plantarflexion force and
1407 position tasks. PLoS One. 2015;10(11):e0143862.
- 1408 108. Carson RG, Riek S, Shahbazzpour N. Central and peripheral mediation of human force
1409 sensation following eccentric or concentric contractions. J Physiol. 2002;539(Pt 3):913-25.
- 1410 109. Stein RB. Presynaptic inhibition in humans. Prog Neurobiol. 1995;47(6):533-44.
- 1411 110. Nielsen JB. Sensorimotor integration at spinal level as a basis for muscle coordination
1412 during voluntary movement in humans. J Appl Physiol. 2004;96(5):1961-7.
- 1413 111. Duchateau J, Enoka RM. Neural control of lengthening contractions. J Exp Biol.
1414 2016;219(2):197-204.
- 1415 112. Enoka RM. Eccentric contractions require unique activation strategies by the nervous
1416 system. J Appl Physiol. 1996;81(6):2339-46.
- 1417 113. Grabiner MD, Owings TM. EMG differences between concentric and eccentric maximum
1418 voluntary contractions are evident prior to movement onset. Exp Brain Res. 2002;145(4):505-11.
- 1419 114. Duchateau J, Enoka RM. Neural control of shortening and lengthening contractions:
1420 Influence of task constraints. J Physiol. 2008;586(24):5853-64.
- 1421 115. Duclay J, Pasquet B, Martin A, Duchateau J. Specific modulation of corticospinal and
1422 spinal excitabilities during maximal voluntary isometric, shortening and lengthening contractions
1423 in synergistic muscles. J Physiol. 2011;589(11):2657-916.
- 1424 116. Fang Y, Siemionow V, Sahgal V, Xiong F, Yue GH. Greater movement-related cortical
1425 potential during human eccentric versus concentric muscle contractions. J Neurophysiol.
1426 2001;86(4):1764-72.
- 1427 117. Perrey S. Brain activation associated with eccentric movement: A narrative review of the
1428 literature. Eur J Sport Sci. 2017;18(1):75-82.
- 1429 118. Budini F, Tilp M. Changes in H-reflex amplitude to muscle stretch and lengthening in
1430 humans. Rev Neurosci. 2016;27(5):511-22.
- 1431 119. Todorov E. Optimality principles in sensorimotor control. Nat Neurosci. 2004;7(9):907-
1432 15.
- 1433 120. Bays PM, Wolpert DM. Computational principles of sensorimotor control that minimize
1434 uncertainty and variability. J Physiol. 2007;578(Pt 2):387-96.
- 1435 121. Della Villa F, Buckthorpe MG, A, Nabiuzzi A, Tosarelli F, Zaffagnini S, Della Villa S.
1436 Systematic video analysis of ACL injuries in professional male football (soccer): injury
1437 mechanisms, situational patterns and biomechanics study on 134 consecutive cases. Br J Sports
1438 Med. 2020;54(23):1423-32.

- 1439 122. Krosshaug T, Nakamae A, Boden BP, Engebretsen L, Smith G, Slauterbeck JR, et al.
1440 Mechanisms of anterior cruciate ligament injury in basketball: Video analysis of 39 cases. *Am J*
1441 *Sports Med.* 2017;35(3):359-67.
- 1442 123. Fridén J, Lieber RL. Eccentric exercise-induced injuries to contractile and cytoskeletal
1443 muscle fibre components. *Acta Physiol Scand.* 2008;171(3):321-6.
- 1444 124. Garrett WEJ. Muscle strain injuries: Clinical and basic aspects. *Med Sci Sports Exerc.*
1445 1990;22(4):436-43.
- 1446 125. Cuthbert SC, Goodheart Jr GJ. On the reliability and validity of manual muscle testing: A
1447 literature review. *Chiropractic and Manual Therapies.* 2007;15(4).
- 1448 126. Conable KM, Rosner AL. A narrative review of manual muscle testing and implications
1449 for muscle testing research. *Journal of Chiropractic Medicine.* 2011;10(3):157-65.
- 1450 127. Bittmann FN, Dech S, Aehle M, Schaefer LV. Manual muscle testing - Force profiles and
1451 their reproducibility. *Diagnostics.* 2020;10(12):996.
- 1452 128. Bolotin A, Bakayev V. Efficacy of using isometric exercises to prevent basketball injuries
1453 *Journal of Physical Education and Sport.* 2016;16(4):1177-85.
- 1454 129. Brynzak S, Putrov S, Olena O, Kostenko M, Myroshnichenko V. Analysis of the use of
1455 isometric exercises to prevent injuries in beach soccer players. *Journal of Physical Education and*
1456 *Sport.* 2023;23(2):440-8.
- 1457 130. Ullman ZJ, Fernandez MB, Klein M. Effects of isometric exercises versus static stretching
1458 in warm-up regimens for running sport athletes: A systematic review. *International Journal of*
1459 *Exercise Science.* 2021;14(6):1204-18.
- 1460 131. van Dyk N, Behan FP, Whiteley R. Including the Nordic hamstring exercise in injury
1461 prevention programmes halves the rate of hamstring injuries: a systematic review and meta-
1462 analysis of 8459 athletes. *Br J Sports Med.* 2019;53(21):1362-70.
- 1463 132. Al Attar WSA, Soomro N, Sinclair PJ, Pappas E, Sanders RH. Effect of injury prevention
1464 programs that include the Nordic hamstring exercise on hamstring injury rates in soccer players:
1465 A systematic review and meta-analysis. *Sports Med.* 2016;47(5):907-16.
- 1466 133. Impellizzeri FM, McCall A, van Smeden M. Why methods matter in a meta-analysis: A
1467 reappraisal showed inconclusive injury preventive effect of Nordic hamstring exercise. *J Clin*
1468 *Epidemiol.* 2021;140:111-24.
- 1469 134. Clifford C, Challoumas DC, Paul L, Syme G, Millar NL. Effectiveness of isometric
1470 exercise in the management of tendinopathy: A systematic review and meta-analysis of
1471 randomised trials. *BMJ Open Sport & Exercise Medicine.* 2020;6(1):e000760.

- 1472 135. Vuvan V, Vicenzino B, Mellor R, Heales LJ, Coombes BK. Unsupervised isometric
1473 exercise versus wait-and-see for lateral elbow tendonopathy. *Med Sci Sports Exerc.*
1474 2020;52(2):287-95.
- 1475 136. Clifford C, Paul L, Syme G, Millar NL. Isometric versus isotonic exercise for greater
1476 trochanteric pain syndrome: a randomised controlled pilot study. *BMJ Open Sport & Exercise*
1477 *Medicine.* 2019;5(1):e000558.
- 1478 137. Hagberg M, Harms-Ringdahl K, Nisell R, Hjelm EW. Rehabilitation of neck-shoulder pain
1479 in women industrial workers: A randomized trial comparing isometric shoulder endurance training
1480 with isometric shoulder strength training. *Arch Phys Med Rehabil.* 2000;81(8):1051-8.
- 1481 138. van der Vlist AC, van Veldhoven PLJ, van Oosterom RF, Verhaar JAN, de Vos RJ.
1482 Isometric exercises do not provide immediate pain relief in Achilles tendinopathy: A quasi-
1483 randomized clinical trial. *Scand J Med Sci Sports.* 2020;30(9):1712-21.
- 1484 139. Gatz M, Betsch M, Dirrichs T, Schrading S, Tingart M, Michalik R, et al. Eccentric and
1485 isometric exercise in Achilles tendinopathy evaluated by the VISA-A score and shear wave
1486 elastography. *Sports Health.* 2020;12(4):373-81.
- 1487 140. Dupuis F, Barrett E, Dube MO, McCreesh KM, Lewis JS, Roy JS. Cryotherapy or gradual
1488 reloading exercises in acute presentations of rotator cuff tendinopathy: A randomised controlled
1489 trial. *BMJ Open Sport & Exercise Medicine.* 2018;4(1):e000477.
- 1490 141. Holden S, Lyng K, Graven-Nielsen T, Riel H, Olesen JL, Larsen LH, et al. Isometric
1491 exercise and pain in patellar tendinopathy: A randomized crossover trial. *J Sci Med Sport.*
1492 2020;23(3):208-14.
- 1493 142. Parle PJ, Riddiford-Harland DL, Howitt CD, Lewis JS. Acute rotator cuff tendinopathy:
1494 does ice, low load isometric exercise, or a combination of the two produce an analgaesic effect?
1495 *Br J Sports Med.* 2017;51(3):208-9.
- 1496 143. Rio E, Kidgell D, Purdam C, Gaida J, Moseley GL, Pearce AJ, et al. Isometric exercise
1497 induces analgesia and reduces inhibition in patellar tendinopathy. *Br J Sports Med.*
1498 2015;49(19):1277-83.
- 1499 144. Stasinopoulos D, Stasinopoulos I. Comparison of effects of eccentric training, eccentric-
1500 concentric training, and eccentric-concentric training combined with isometric contraction in the
1501 treatment of lateral elbow tendinopathy. *J Hand Ther.* 2017;30(1):13-9.
- 1502 145. van Ark M, Cook JL, Docking SI, Zwerver J, Gaida JE, van den Akker-Scheek I, et al. Do
1503 isometric and isotonic exercise programs reduce pain in athletes with patellar tendinopathy in-
1504 season? A randomised clinical trial. *J Sci Med Sport.* 2016;19(9):702-6.
- 1505 146. Rio E, Purdam C, Girdwood M, Cook J. Isometric exercise to reduce pain in patellar
1506 tendinopathy in-season: Is it effective "on the road"? *Clin J Sport Med.* 2019;29(3):188-92.

1507 147. Topp R, Woolley S, Hornyak J, Khuder S, Kahaleh B. The effect of dynamic versus
1508 isometric resistance training on pain and functioning among adults with osteoarthritis of the knee.
1509 Arch Phys Med Rehabil. 2002;83(9):1187-95.

1510 148. Lum D, Barbosa TM, Balasekaran G. Sprint kayaking performance enhancement by
1511 isometric strength training inclusion: A randomized controlled trial. Sci Period Res Tech Sport.
1512 2021;9(2):16.

1513 149. Lum D, Comfort P, Barbosa TM, Balasekaran G. Comparing the effects of plyometric and
1514 isometric strength training on dynamic and isometric force-time characteristics. Biol Sport.
1515 2022;39(1):189-97.

1516 150. Lum D, Barbosa TM, Aziz AR, Balasekaran G. Effects of isometric strength and
1517 plyometric training on running performance: A randomized controlled study. Res Q Exerc Sport.
1518 2023;94(1):263-71.

1519 151. Lum D, Joseph R, Ong KY, Tang JM, Suchomel TJ. Comparing the effects of long-term
1520 vs periodic inclusion of isometric strength training on strength and dynamic performances. J
1521 Strength Cond Res. 2023;37(2):305-14.

1522 152. Pearson D, Faigenbaum A, Conley M, Kraemer WJ. The National Strength and
1523 Conditioning Association's basic guidelines for the resistance training of athletes. Strength Cond
1524 J. 2000;22(4):14-27.

1525 153. Tillin NA, Folland JP. Maximal and explosive strength training elicit distinct
1526 neuromuscular adaptations, specific to the training stimulus. Eur J Appl Physiol. 2014;14(2):365-
1527 74.

1528 154. Massey GJ, Balshaw TG, Maden-Wilkinson TM, Tillin NA, Folland JP. Tendinous tissue
1529 adaptation to explosive- vs. sustained-contraction strength training. Front Physiol. 2018
1530 Sep;9(1170).

1531 155. Kozlenia D, Domaradzki J. The effectiveness of isometric protocols using an external load
1532 or voluntary effort on jump height enhancement in trained females. Sci Rep. 2023;13(13535):1-
1533 10.

1534